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CHAPTER 4

SPECIAL DETAILED REQUIREMENTS BASED ON USE AND OCCUPANCY

SECTION BC 401 SCOPE

401.1 Detailed use and occupancy requirements. In addition to the occupancy and construction requirements in this code, the provisions of this chapter apply to the special uses and occupancies described herein.

SECTION BC 402 COVERED MALL AND OPEN MALL BUILDINGS

402.1 Applicability. The provisions of this section shall apply to buildings or structures defined herein as covered or open mall buildings not exceeding three floor levels at any point nor more than three stories above grade plane. Except as specifically required by this section, covered and open mall buildings shall meet applicable provisions of this code.

Exceptions:

1. Foyers and lobbies of Groups B, R-1 and R-2 are not required to comply with this section.
2. Buildings need not comply with the provisions of this section where they fully comply with other applicable provisions of this code.

402.1.1 Occupancy classification. A mall in compliance with the provisions of this section shall be classified as occupancy Group M.

402.1.2 Open space. A covered mall building and attached anchor buildings and parking garages shall be surrounded on all sides by a permanent open space of not less than 60 feet (18 288 mm). An open mall building and anchor buildings and parking garages adjoining the perimeter line shall be surrounded on all sides by a permanent open space of not less than 60 feet (18 288 mm).

Exception: The permanent open space of 60 feet (18 288 mm) shall be permitted to be reduced to not less than 40 feet (12 192 mm), provided the following requirements are met:

1. The reduced open space shall not be allowed for more than 75 percent of the perimeter of the covered mall building and anchor buildings or of the perimeter line of the open mall building and anchor buildings;
2. The exterior wall facing the reduced open space shall have a fire-resistance rating of not less than three hours;
3. Openings in the exterior wall facing the reduced open space shall have opening protectives with a fire protection rating of not less than three hours; and
4. Group E, H, I or R occupancies are not located within the covered or open mall building or anchor buildings.

402.1.3 Open mall building perimeter line. For the purpose of this code, a perimeter line shall be established. The perimeter line shall encircle all buildings and structures that comprise the open mall building and shall encompass any open-air interior walkways, open-air courtyards or similar open-air spaces. The perimeter line shall define the extent of the open mall building. Anchor buildings and parking structures shall be outside of the perimeter line and are not considered as part of the open mall building.

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402.2 Definitions. This section contains terms defined elsewhere in this code, and terms with definitions that are specific to this section.

402.2.1 Terms defined elsewhere in this code. The following terms are defined in Chapter 2:

ANCHOR BUILDING.

COVERED MALL BUILDING.

Mall.

Open mall.

Open mall building.

402.2.2 Definitions specific to this section. The following terms shall, for the purposes of this section, have the meanings shown herein:

FOOD COURT. A public seating area located in the mall that serves adjacent food preparation tenant spaces.

GROSS LEASABLE AREA. The total floor area designed for tenant occupancy and exclusive use. The area of tenant occupancy is measured from the centerlines of joint partitions to the outside of the tenant walls. All tenant areas, including areas used for storage, shall be included in calculating gross leasable area.

402.3 Fire protection plan. A fire protection plan shall be provided to the Fire Department pursuant to Article 109 of Title 28.

402.4 Construction. The construction of covered and open mall buildings, anchor buildings and parking garages associated with a mall building shall comply with Sections 402.4.1 through 402.4.3.

402.4.1 Area and types of construction. The building area and type of construction of covered mall or open mall buildings, anchor buildings and parking garages shall comply with this section.

402.4.1.1 Covered and open mall buildings. Covered and open mall buildings shall be of Type I, II or IV construction. The height of covered mall buildings of Type IIB construction shall be limited to one story. The areas of covered mall buildings or open mall buildings shall not be limited provided that the covered mall building or open mall building and attached anchor buildings and parking garages are surrounded on all sides by a permanent open space in accordance with Section 402.1.2.

402.4.1.2 Anchor buildings. Anchor buildings shall be of Type I, II or IV construction. The height of anchor buildings, of Type IIB construction, shall be limited to one story. The allowable height and area of anchor buildings greater than three stories in height shall comply with Section 503, as modified by Sections 504 and 506.

402.4.1.3 Parking garage. The building area and building height of any parking garage, open or enclosed, shall be based on the type of construction as required by Sections 406.5 and 406.6, respectively.

402.4.2 Fire-resistance-rated separation. Fire-resistance-rated separation is not required between tenant spaces and the mall. Fire-resistance-rated separation is not required between a food court and adjacent tenant spaces or the mall.

402.4.2.1 Tenant separations. Each tenant space shall be separated from other tenant spaces by a fire partition complying with Section 708. A tenant separation wall is not required between any tenant space and the mall.

402.4.2.2 Anchor building separation. An anchor building shall be separated from the covered or open mall building by fire walls complying with Section 706.

Exceptions:

1. Anchor buildings of not more than three stories above grade that have an occupancy classification of the same uses permitted as tenants of the covered mall building shall be separated by 2-hour fire-resistance-rated fire barriers complying with Section 707. Openings between such buildings and the mall need not be protected.
2. The exterior walls of anchor buildings separated from an open mall building by an open mall shall comply with Table 602.

402.4.2.3 Parking garages. An attached garage for the storage of passenger vehicles having a capacity of not more than nine persons and open parking garages shall be considered as a separate building where it is separated from the covered or open mall building or anchor building by not less than 2-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.

Parking garages, open or enclosed, which are separated from covered mall buildings, open mall buildings or anchor buildings, shall comply with the provisions of Table 602.

Pedestrian walkways and tunnels that connect garages to the mall buildings or anchor buildings shall be constructed in accordance with Section 3104.

402.4.3 Open mall construction. Floor assemblies in, and roof assemblies over, the open mall of an open mall building shall be open to the atmosphere for not less than 20 feet (9096 mm), measured perpendicular from the face of the tenant spaces on the lowest level, from edge of balcony to edge of balcony on upper floors and from edge of roof line to edge of roof line. The openings within, or the unroofed area of, an open mall shall extend from the lowest/grade level of the open mall through the entire roof assembly. Balconies on upper levels of the mall shall not project into the required width of the opening.

402.4.3.1 Pedestrian walkways. Pedestrian walkways connecting balconies in an open mall shall be located not less than 20 feet (9096 mm) from any other pedestrian walkway.

402.5 Automatic sprinkler system. Covered and open mall buildings and buildings connected shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1, which shall comply with all of the following:

1. The automatic sprinkler system shall be complete and operative throughout occupied space in the mall building prior to occupancy of any of the tenant spaces. Unoccupied tenant spaces shall be similarly protected unless provided with approved alternative protection.
2. Sprinkler protection for the mall of a covered mall building shall be controlled independently from that provided for tenant spaces or anchor buildings.
3. Sprinkler protection for the tenant spaces of an open mall building shall be controlled independently from that provided for anchor buildings.
4. Sprinkler protection shall be provided beneath exterior circulation balconies located adjacent to an open mall.
5. Where tenant spaces are supplied by the same system, they shall be independently controlled.

Exception: An automatic sprinkler system shall not be required in spaces or areas of open parking garages separated from the covered or open mall building in accordance with Section 402.4.2.3 and constructed in accordance with Section 406.5.

402.6 Interior finishes and features. Interior finishes within the mall and installations within the mall shall comply with Sections 402.6.1 through 402.6.4.

402.6.1 Interior finish. Interior wall and ceiling finishes within the mall of a covered mall building and within the exits of covered or open mall buildings shall have a minimum flame spread index and smoke-developed index of Class B in accordance with Chapter 8. Interior floor finishes shall meet the requirements of Section 804.

402.6.2 Kiosks. Kiosks and similar structures (temporary or permanent) located within the mall of a covered mall building or within the perimeter line of an open mall building shall meet the following requirements:

1. Combustible kiosks or other similar structures shall not be located within a covered or open mall unless constructed of any of the following materials:
 - 1.1. Fire-retardant-treated wood complying with Section 2303.2.
 - 1.2. Foam plastics having a maximum heat release rate not greater than 100 kW (105 Btu/h) when tested in accordance with the exhibit booth protocol in UL 1975 or when tested in accordance with NFPA 289 using the 20 kW ignition source.
 - 1.3. Aluminum composite material (ACM) having a flame spread index of not more than 25 and a smoke-developed index of not more than 450 when tested as an assembly in the maximum thickness intended for use in accordance with ASTM E 84 or UL 723.

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2. Covered kiosks or similar covered structures located within the covered mall building shall be provided with an approved automatic sprinkler system and detection devices where required by the *New York City Fire Code* and Section 903.3.3 of this code.
3. The horizontal separation between kiosks or groupings thereof and other similar structures within the mall shall be not less than 20 feet (6096 mm).
4. Each kiosk or similar structure or groupings thereof shall have an area not greater than 300 square feet (28 m²).

402.6.3 Children's play structures. Children's play structures located within the mall of a covered mall building or within the perimeter line of an open mall building shall comply with Section 424. The horizontal separation between children's play structures, kiosks and similar structures within the mall shall be not less than 20 feet (6096 mm).

402.6.4 Plastic signs. Plastic signs affixed to the storefront of any tenant space facing a mall or open mall shall be limited as specified in Sections 402.6.4.1 through 402.6.4.5.

402.6.4.1 Area. Plastic signs shall be not more than 20 percent of the wall area facing the mall.

402.6.4.2 Height and width. Plastic signs shall be not greater than 36 inches (914 mm) in height, except that where the sign is vertical, the height shall be not greater than 96 inches (2438 mm) and the width shall be not greater than 36 inches (914 mm).

402.6.4.3 Location. Plastic signs shall be located not less than 18 inches (457 mm) from adjacent tenants.

402.6.4.4 Plastics other than foam plastics. Plastics other than foam plastics used in signs shall be lighttransmitting plastics complying with Section 2606.4 or shall have a self-ignition temperature of 650°F (343°C) or greater when tested in accordance with ASTM D 1929, and a flame spread index not greater than 75 and smoke-developed index not greater than 450 when tested in the manner intended for use in accordance with ASTM E 84 or UL 723 or meet the acceptance criteria of Section 803.1.2.1 when tested in accordance with NFPA 286.

402.6.4.4.1 Encasement. Edges and backs of plastic signs in the mall shall be fully encased in metal.

402.6.4.5 Foam plastics. Foam plastics used in signs shall have flame-retardant characteristics such that the sign has a maximum heat-release rate of 150 kilowatts when tested in accordance with UL 1975 or when tested in accordance with NFPA 289 using the 20 kW ignition source, and the foam plastics shall have the physical characteristics specified in this section. Foam plastics used in signs installed in accordance with Section 402.6.4 shall not be required to comply with the flame spread and smoke-developed indices specified in Section 2603.3.

402.6.4.5.1 Density. The density of foam plastics used in signs shall be not less than 20 pounds per cubic foot (pcf) (320 kg/m³).

402.6.4.5.2 Thickness. The thickness of foam plastic signs shall be not greater than 1/2 inch (12.7 mm).

402.7 Emergency systems. Covered and open mall buildings, anchor buildings and associated parking garages shall be provided with emergency systems complying with Sections 402.7.1 through 402.7.5.

402.7.1 Standpipe system. Covered and open mall buildings shall be equipped throughout with a standpipe system as required by Section 905.3.3.

402.7.2 Smoke control. Where a covered mall building contains an atrium, a smoke control system shall be provided in accordance with Section 404.5.

402.7.3 Emergency power. Covered mall buildings greater than 50,000 square feet (4645 m²) in area and open mall buildings greater than 50,000 square feet (4645 m²) within the established perimeter line shall be provided with emergency power that is capable of operating the emergency voice/alarm communication system in accordance with Section 2702.

402.7.4 Emergency voice/alarm communication system. Where the total floor area is greater than 50,000 square feet (4645 m²) within either a covered mall building or within the perimeter line of an open mall building, an emergency voice/alarm communication system shall be provided.

Emergency voice/alarm communication systems serving a mall, required or otherwise, shall be accessible to the fire department. The systems shall be provided in accordance with Section 907.5.2.2.

402.7.5 Fire Department access to equipment. Rooms or areas containing controls for air-conditioning systems, automatic fire-extinguishing systems, automatic sprinkler systems or other detection, suppression or control elements shall be identified for use by the Fire Department.

402.8 Means of egress. Covered mall buildings, open mall buildings and each tenant space within a mall building shall be provided with means of egress as required by this section and this code. Where there is a conflict between the requirements of this code and the requirements of Sections 402.8.1 through 402.8.8, the requirements of Sections 402.8.1 through 402.8.8 shall apply.

402.8.1 Mall width. For the purpose of providing required egress, malls are permitted to be considered as corridors but need not comply with the requirements of Section 1005.1 of this code where the width of the mall is as specified in this section.

402.8.1.1 Minimum width. The aggregate clear egress width of the mall in either a covered or open mall building shall be not less than 20 feet (6096 mm). The mall width shall be sufficient to accommodate the occupant load served. No portion of the minimum required aggregate egress width shall be less than 10 feet (3048 mm) measured to a height of 8 feet (2438 mm) between any projection of a tenant space bordering the mall and the nearest kiosk, vending machine, bench, display opening, food court or other obstruction to means of egress travel.

402.8.2 Occupant load. The calculated occupant load used to determine the required number of means of egress shall be in accordance with this section.

402.8.2.1 Occupant load of tenant spaces. The occupant load permitted in any individual tenant space in a covered mall building shall be determined in accordance with Chapter 10. Means of egress requirements for individual tenant spaces shall be based on the occupant load thus determined.

402.8.2.2 Occupant load of mall. In determining required means of egress of the mall, the total occupant load in the mall shall include the occupant load attributed to the mall as determined by Section 402.8.2.3, plus the occupant load of the food court as determined by Section 402.8.2.5, if any.

402.8.2.3 Occupant load formula. To determine the occupant load attributed to the mall, the gross leasable area (square feet), excluding any food court and anchor buildings, shall be divided by the occupant load factor (OLF) value determined by Equation 4-1.

$$OLF = (0.00007)(GLA) + 25 \quad \text{(Equation 4-1)}$$

where:

OLF = The occupant load factor (square feet per person).

GLA = The gross leasable area (square feet).

Exception: Tenant spaces attached to a covered mall building but with a means of egress system that is totally independent of the covered mall building shall not be considered as gross leasable area for determining the required means of egress for the covered mall building.

402.8.2.4 OLF range. The occupant load factor (OLF) is not required to be less than 30 and shall not exceed 50.

402.8.2.5 Food courts. The occupant load of a food court shall be determined in accordance with Section 1004 and shall be added to the occupant load of the mall.

402.8.2.6 Anchor buildings. The occupant load of anchor buildings opening into the mall shall not be included in computing the total number of occupants for the covered mall building.

402.8.3 Number of means of egress. The required number of means of egress shall be determined in accordance with this section.

402.8.3.1 Number of means of egress within tenant spaces. Wherever the distance of travel to the mall from any location within a tenant space used by persons other than employees exceeds 75 feet (22 860 mm) or the tenant space exceeds an occupant load of 74, no fewer than two means of egress shall be provided.

402.8.3.2 Number of means of egress from the mall. The number of means of egress from the mall shall be determined in accordance with Chapter 10 and shall be based upon an occupant load calculated in accordance with Section 402.8.2.

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402.8.4 Arrangements of means of egress. Means of egress shall be arranged in accordance with this section.

402.8.4.1 Anchor building means of egress. Required means of egress for anchor buildings shall be provided independently from the mall means of egress system. The occupant load of anchor buildings opening into the mall shall not be included in determining means of egress requirements for the mall. The path of egress travel of malls shall not exit through anchor buildings. Malls terminating at an anchor building where no other means of egress has been provided shall be considered as a dead-end mall.

402.8.4.2 Tenant spaces requiring more than one means of egress. Where more than one means of egress is required from an individual tenant space as determined by Chapter 10, not more than 50 percent of the occupant load shall discharge into the mall.

402.8.4.3 Large assembly occupancy. Assembly occupancies with an occupant load of 500 or more shall be so located in the covered mall building such that their entrance will be immediately adjacent to a principal entrance to the mall and shall have not less than one-half of their required means of egress opening directly to the exterior of the covered mall building.

402.8.5 Distance to exits. Within each individual tenant space in a covered or open mall building, the distance of travel from any point to an exit or entrance to the mall shall be not greater than 200 feet (60 960 mm).

The distance of travel from any point within a mall of a covered mall building to an exit shall be not greater than 200 feet (60 960 mm). The maximum distance of travel from any point within an open mall to the perimeter line of the open mall building shall be not greater than 200 feet (60 960 mm).

402.8.6 Access to exits. Where more than one exit is required, they shall be so arranged that it is possible to travel in either direction from any point in a mall of a covered mall building to separate exits or from any point in an open mall of an open mall building to two separate locations on the perimeter line, provided neither location is an exterior wall of an anchor building or parking garage. The width of an exit passageway or corridor from a mall shall be not less than 66 inches (1676 mm).

Exception: Access to exits is permitted by way of a dead-end mall that does not exceed a length equal to twice the width of the mall measured at the narrowest location within the dead-end portion of the mall.

402.8.6.1 Exit passageway enclosures. Where exit passageway enclosures provide a secondary means of egress from a tenant space, doors to the exit passageway enclosures shall be minimum 1-hour fire doors with panic hardware. Such doors shall be self-closing and be so maintained or shall be automatic closing by smoke detection.

402.8.7 Service areas fronting on exit passageways. Mechanical rooms, electrical rooms, building service areas and service elevators are permitted to open directly into exit passageways provided that the exit passageway is separated from such rooms by fire barriers providing the same fire-resistance rating as required for the exit passageway. Door swings from these rooms shall not project into the minimum width of such exit passageways.

402.8.8 Security grilles and doors. Horizontal sliding or vertical security grilles or doors that are a part of a required means of egress shall conform to the following:

1. Doors and grilles shall remain in the full open position during the period of occupancy by the general public.
2. Doors or grilles shall not be brought to the closed position when there are 10 or more persons occupying spaces served by a single exit or 50 or more persons occupying spaces served by more than one exit.
3. The doors or grilles shall be openable from within without the use of any special knowledge or effort where the space is occupied.
4. Where two or more exits are required, not more than one-half of the exits shall be permitted to include either a horizontal sliding or vertical rolling grille or door.

SECTION BC 403 HIGH-RISE BUILDINGS

403.1 Applicability. High-rise buildings shall comply with Sections 403.2 through 403.8.

Exception: The provisions of Sections 403.2 through 403.8 shall not apply to the following buildings and structures:

1. Air traffic control towers in accordance with Section 412.3.
2. Buildings occupied entirely by Group R-3 classification.
3. Open parking garages in accordance with Section 406.5.
4. The portion of a building containing a Group A-5 occupancy in accordance with Section 303.6.
5. Special industrial occupancies in accordance with Section 503.1.1.
6. Buildings with:
 - 6.1. A Group H-1 occupancy;
 - 6.2. A Group H-2 occupancy in accordance with Section 415.8, 415.9.2, 415.9.3 or 426.1; or
 - 6.3. A Group H-3 occupancy in accordance with Section 415.8.

403.2 Construction. The construction of high-rise buildings shall comply with the provisions of Sections 403.2.1 through 403.2.4.

403.2.1 Types of construction. The following modifications to the minimum fire-resistance rating of the building elements in Table 601 shall be as follows:

1. High rise buildings 420 feet (128 m) or greater in building height shall be constructed of Type IA construction.
2. For high rise buildings not greater than 420 feet (128 m) in building height, and constructed to meet the fire-resistance rating requirements of Type IB or IIA construction, the required fire-resistance rating of columns supporting floors shall be constructed to meet Type IA construction.

403.2.2 Seismic considerations. For seismic considerations, see Chapter 16.

403.2.3 Structural integrity of interior exit stairways and elevator hoistway enclosures. For all high-rise buildings, exit enclosures for interior exit stairways and elevator hoistway enclosures shall comply with Sections 403.2.3.1 through 403.2.3.4.

Exception: Where fire-resistance rating is not required by this code for the enclosure or portions thereof.

403.2.3.1 Wall assembly. The wall assemblies making up the enclosures for interior exit stairways and elevator hoistway enclosures shall meet or exceed Soft Body Impact Classification Level 2 as measured by the test method described in ASTM C 1629/C 1629M.

403.2.3.2 Wall assembly materials. The face of the wall assemblies making up the enclosures for interior exit stairways and elevator hoistway enclosures that are not exposed to the interior of the enclosures for interior exit stairways or elevator hoistway enclosure shall be constructed in accordance with one of the following methods:

1. The wall assembly shall incorporate no fewer than two layers of impact-resistant construction board each of which meets or exceeds Hard Body Impact Classification Level 2 as measured by the test method described in ASTM C 1629/C 1629M.
2. The wall assembly shall incorporate no fewer than one layer of impact-resistant construction material that meets or exceeds Hard Body Impact Classification Level 3 as measured by the test method described in ASTM C 1629/C 1629M.
3. The wall assembly incorporates multiple layers of any material, tested in tandem, that meets or exceeds Hard Body Impact Classification Level 3 as measured by the test method described in ASTM C 1629/C 1629M.

403.2.3.3 Concrete and masonry walls. Concrete or masonry walls shall be deemed to satisfy the requirements of Sections 403.2.3.1 and 403.2.3.2.

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403.2.3.4 Other wall assemblies. Any other wall assembly that provides impact resistance equivalent to that required by Sections 403.2.3.1 and 403.2.3.2 for Hard Body Impact Classification Level 3, as measured by the test method described in ASTM C 1629/C 1629M, shall be permitted.

403.2.4 Sprayed fire-resistant materials (SFRM). The bond strength of the SFRM installed throughout the building shall be in accordance with Table 403.2.4.

**TABLE 403.2.4
MINIMUM BOND STRENGTH**

HEIGHT OF BUILDING ^a	SFRM MINIMUM BOND STRENGTH
Up to 420 feet	430 psf
Greater than 420 feet	1,000 psf

For SI: 1 foot = 304.8 mm, 1 pound per square foot (psf) = 0.0479 kW/m².

a. Above the lowest level of fire department vehicle access.

403.3 Automatic sprinkler system. Buildings and structures shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 and a secondary water supply where required by Section 403.3.3.

403.3.1 Number of sprinkler risers and system design. Each sprinkler system zone in buildings that are more than 300 feet (91 440 mm) in building height shall be supplied by no fewer than two risers. Each riser shall supply sprinklers on alternate floors. If more than two risers are provided for a zone, sprinklers on adjacent floors shall not be supplied from the same riser.

403.3.1.1 Riser location. Sprinkler risers shall be placed in interior exit stairways and ramps that are remotely located in accordance with Section 1007.1.

403.3.2 Water supply to required fire pumps. Required fire pumps shall be provided with water supplies in accordance with Section 905.2 and NFPA 14 as modified by Appendix Q of this code.

403.3.3 Secondary on-site water supply. A secondary on-site water supply equal to the hydraulically calculated sprinkler demand, including the hose stream requirement, shall be provided for high-rise buildings in Seismic Design Category C or D as determined by this code, and in any high-rise building with occupied floors located more than 500 feet (152 m) above the lowest level of fire department vehicle access. The secondary on-site water supply shall have a duration of not less than 30 minutes as determined by the occupancy hazard classification in accordance with NFPA 13 as modified by Appendix Q of this code.

403.3.4 Fire pump room. Fire pumps shall be located in rooms protected in accordance with Section 913.2.1.

403.4 Emergency systems. The detection, alarm and emergency systems of high-rise buildings shall comply with Sections 403.4.1 through 403.4.8.

403.4.1 Smoke detection. Smoke detection shall be provided in accordance with Section 907.2.13.1.

403.4.2 Fire alarms systems. A fire alarm system shall be provided in accordance with Section 907.2.13.

403.4.3 Standpipe system. A high-rise building shall be equipped with a standpipe system as required by Section 905.3.

403.4.4 Emergency voice/alarm communication system. An emergency voice/alarm communication system shall be provided in accordance with Section 907.5.2.2.

403.4.5 Emergency responder radio coverage. Emergency responder radio coverage shall be provided in accordance with the *New York City Fire Code* and Section 907.2.13.2 of this code.

403.4.6 Fire command. A fire command center complying with Section 911 shall be provided in a location approved by the Fire Department.

403.4.7 Post-fire smoke purge. A post-fire smoke purge system shall be installed in accordance with Section 917.

403.4.8 Standby and emergency power. A standby power system complying with Section 2702 and Section 3003 shall be provided for the standby power loads specified in Section 403.4.8.3. An emergency power system complying with Section 2702 shall be provided for the emergency power loads specified in Section 403.4.8.4.

403.4.8.1 Equipment room. If the standby or emergency power system includes a generator set inside a building, the system shall be located in a separate room in accordance with Section 2702.1.2.2.

403.4.8.2 Fuel line piping protection. Fuel lines supplying a generator set inside a building shall comply with Section 1305.9 of the *New York City Mechanical Code*.

403.4.8.3 Standby power systems. A standby power system complying with Section 2702 shall be provided for standby power loads specified in Sections 403.4.8.3.1 and 403.4.8.3.2. Fuel sources for generators shall be in accordance with Section 2702.1.1.

403.4.8.3.1 Standby power loads in occupancies other than Group R-2. In buildings of any occupancy group other than Group R-2, the following are classified as standby power loads:

1. Power and lighting for fire command center required by Section 403.4.6;
2. Ventilation and automatic fire detection equipment for smokeproof enclosures;
3. Elevators, in accordance with Section 3003; and
4. Stair pressurization systems when provided.

403.4.8.3.2 Standby power loads in Group R-2 occupancies. Group R-2 occupancies in buildings greater than 125 feet (38 100 mm) in height shall be required to provide a standby power system to support the following loads:

1. Power and lighting for fire command center required by Section 403.4.6;
2. Ventilation and automatic fire detection equipment for smokeproof enclosures;
3. At least one elevator serving all floors, or one elevator per bank where different banks serve different portions of the building; and
4. Stair pressurization systems when provided.

403.4.8.4 Emergency power systems. An emergency power system complying with Section 2702 shall be provided for emergency power loads specified in Sections 403.4.8.4.1 and 403.4.8.4.3. Fuel sources for generators shall be in accordance with Section 2702.1.1.

403.4.8.4.1 Emergency power loads in occupancies other than Group R-2. In buildings of any occupancy group other than Group R-2, the following are classified as emergency power loads:

1. Exit signs and means of egress illumination required by Chapter 10;
2. Elevator car lighting;
3. Emergency voice/alarm communications systems, including Fire Department in-building Auxiliary Radio Communication systems (ARCs);
4. Automatic fire detection systems;
5. Fire alarm systems; and
6. Electrically powered fire pumps, including manual fire pumps, automatic fire pumps, and sprinkler booster pumps.

403.4.8.4.2 Emergency power loads in Group R-2 occupancies. Group R-2 occupancies in buildings greater than 125 feet (38 100 mm) in height shall be required to provide an emergency power system to support the following loads:

1. Exit signs and means of egress illumination required by Chapter 10;
2. Emergency voice communications systems, including Fire Department in-building Auxiliary Radio Communication systems (ARCs); and
3. Electrically powered fire pumps, unless electrical power to the motor is taken ahead of the main from the street side of the house service switch.

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403.4.8.4.3 Emergency power loads in Group R-2 occupancies 125 feet or less in height. Group R-2 occupancies in buildings 125 feet (38 100 mm) or less in height shall be required to provide an emergency power system to support the following loads:

1. Emergency voice communications systems in buildings containing Group R-2 occupancies in accordance with Section 907.5.2.2 of this code, or where otherwise provided. Batteries in accordance with the *New York City Electrical Code* are permitted to serve as the secondary power supply for such systems.
2. Fire Department in-building Auxiliary Radio Communication systems (ARCs) in buildings containing Group R-2 occupancies in accordance with Section 916.3 of this code, or where otherwise provided. Batteries in accordance with the *New York City Electrical Code* are permitted to serve as the secondary power supply for such systems.

403.5 Means of egress and evacuation. The means of egress in high-rise buildings shall comply with Sections 403.5.1 through 403.5.6.

403.5.1 Remoteness of interior exit stairways. Required interior exit stairways shall be separated by a distance not less than 30 feet (9144 mm) or not less than $\frac{1}{4}$ of the length of the maximum overall diagonal dimension of the building or area to be served, whichever is less. The distance shall be measured in a straight line between the nearest points of the enclosure surrounding the interior exit stairways. In buildings with three or more interior exit stairways, no fewer than two of the interior exit stairways shall comply with this section. Stairs sharing any common wall, floors, ceilings, scissor stair assemblies, or other enclosures shall be counted as one interior exit stairway.

Exception: Group R-2 occupancies.

403.5.2 Additional exit stairway. For buildings other than Group R-2 that are more than 420 feet (128 m) in building height, one additional exit stairway meeting the requirements of Sections 1011 and 1023 shall be provided in addition to the minimum number of exits required by Section 1006.3. The total width of any combination of remaining exit stairways with one exit stairway removed shall not be less than the total width required by Section 1005.1. Stairs sharing any common wall, floors, ceilings, scissor stair assemblies, or other enclosures shall not be considered the additional exit stairway required by this section.

Exceptions: An additional exit stairway shall not be required to be installed in any of the following instances:

1. In buildings where all passenger elevators for general public use and all other elevators used for occupant self-evacuation comply with Sections 3008.1 through 3008.11;
2. In buildings where all of the following conditions are met:
 - 2.1. The commissioner has approved a timed egress analysis establishing the egress time it would take for a full building evacuation, utilizing the stairs required by Section 1006 and the additional stair that would have been required pursuant to Section 403.5.2;
 - 2.2. The commissioner has approved a timed egress analysis establishing the egress time it would take for a full building evacuation, utilizing the combination of the stairs required by Section 1006 and elevators used for occupant self-evacuation as follows:
 - 2.2.1. Such analysis need only utilize a number of designated elevators designed for occupant self-evacuation necessary to demonstrate an evacuation time that is less than the time established in Exception 2.1; and
 - 2.2.2. Every bank of passenger elevators for general public use shall be served by at least one such designated elevator;
 - 2.3. All elevators in the building that either are passenger elevators for general public use or are used for occupant self-evacuation shall comply with Sections 3008.1 through 3008.11. However, the standby power generating equipment required by Section 3008.9 need only be sized to satisfy the loads required to simultaneously operate those elevators identified in the timed egress analysis described in Exception 2.2.

The minimum number of exits required by Section 1006 are all at least 25 percent wider than the exit width required by Chapter 10. The increase in the stair width shall not be utilized in the timed egress analyses required by Exceptions 2.1 and 2.2.

403.5.3 Stairway door operation. Doors opening into interior stair enclosures shall not be locked from either side. However, a door locked from the stair side may be permitted provided that such door is equipped with an automatic fail safe system for opening in the event of the activation of any automatic fire detection system, or when any elevator recall is activated, or when any signal is received from the fire command center. Such door shall be deemed as openable from the stair side. Stair reentry signs shall be posted throughout the stairway indicating that reentry is provided only during fire emergencies. Such signs shall be in accordance with Section 1031.4.2.

Exception: In schools classified as Group E occupancies that require “lockdown drills” as per Section 807 of the *New York State Education Law*, stairway egress doors from occupied spaces are permitted to be locked from the side opposite the egress side for the duration of the lockdown drill, provided they are openable from the occupied side for egress and are unlocked upon directions from the school administrative personnel following the completion of the lockdown drill.

403.5.3.1 Stairway communication system. A telephone or other two-way voice[‡] communications system connected to an approved constantly attended station shall be provided at not less than every fifth floor in each stairway where the doors to the stairway are locked in accordance with Section 403.5.3.

403.5.4 Smokeproof enclosures. Every required interior exit stairway serving floors more than 75 feet (22 860 mm) above the lowest level of fire department vehicle access shall comply with Sections 909.20 and 1023.11.

Exception: Smokeproof enclosures are not required in occupancy Group R-2 unless provided pursuant to Exception 2 of Section 917.1.

403.5.5 Luminous egress path markings. Luminous egress path markings shall be provided in accordance with Section 1025 and Appendix S.

Exception: Egress paths serving Group R-2.

403.5.6 Emergency escape and rescue. Emergency escape and rescue openings specified in Section 1030 are not required.

403.6 Elevators. Elevator installation and operation in high-rise buildings shall comply with Chapter 30 and Sections 403.6.1 and 403.6.2.

403.6.1 Fire service access elevator. In buildings with an occupied floor more than 120 feet (36 576 mm) above the lowest level of fire department vehicle access, a minimum of one fire service access elevator shall be provided in accordance with Section 3007.

403.6.2 Occupant evacuation elevators. Where installed in accordance with Section 3008, passenger elevators for general public use shall be permitted to be used for occupant self-evacuation.

403.6.3 Elevator lobbies. Elevator lobbies shall be provided in accordance with Sections 3006.1.1 and 3006.1.2.

403.7 Outdoor air intakes. For high-rise buildings, outdoor air intakes serving spaces above the second story and serving spaces greater than 10,000 square feet (929 m²) of floor area shall be located in accordance with Section 401.5 of the *New York City Mechanical Code*.

Exception: Group R-2 occupancy.

403.8 Open web steel joists. The use of open web steel joists shall be prohibited in high-rise buildings until the commissioner promulgates rules establishing minimum acceptable fireproofing methods.

SECTION BC 404 ATRIUMS

404.1 General. In other than Group H occupancies, and where permitted by Section 712.1.7, the provisions of Sections 404.1 through 404.10 shall apply to buildings or structures containing vertical openings defined as “atriums.”

404.1.1 Definition.[‡] The following term is defined in Chapter 2:

ATRIUM.

SPECIAL DETAILED REQUIREMENTS BASED ON USE AND OCCUPANCY

404.2 Use. The floor of the atrium shall not be used for other than low fire hazard uses and only approved materials and decorations in accordance with the *New York City Fire Code* shall be used in the atrium space.

Exception: The atrium floor area is permitted to be used for any approved use where the individual space is provided with an automatic sprinkler system in accordance with Section 903.3.1.1.

404.3 Automatic sprinkler protection. An approved automatic sprinkler system shall be installed throughout the entire building.

Exceptions:

1. That area of a building adjacent to or above the atrium need not be sprinklered provided that portion of the building is separated from the atrium portion by not less than 2-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.
2. Where the ceiling of the atrium is more than 55 feet (16 764 mm) above the floor, sprinkler protection at the ceiling of the atrium is not required.

404.4 Fire alarm system. A fire alarm system shall be provided in accordance with Section 907.2.14.

404.5 Smoke control. A smoke control system shall be installed in accordance with Section 909.

Exception: In other than Groups I-1 and I-2, and covered mall buildings containing an atrium pursuant to Section 402.7.2, smoke control is not required for atriums that connect only two stories.

404.6 Enclosure of atriums. Atrium spaces shall be separated from adjacent spaces by a 1-hour fire barrier constructed in accordance with Section 707 or a horizontal assembly constructed in accordance with Section 711, or both.

Exceptions:

1. A fire barrier is not required where a glass or glass-block wall forming a smoke partition is provided. The glass or glass-block wall shall comply with all of the following:
 - 1.1. Automatic sprinklers are provided along both sides of the separation wall and doors, or on the room side only if there is not a walkway on the atrium side. The sprinklers shall be located between 4 inches and 12 inches (102 mm and 305 mm) away from the glass and at intervals along the glass not greater than 6 feet (1829 mm). The sprinkler system shall be designed so that the entire surface of the glass is wet upon activation of the sprinkler system without obstruction;
 - 1.2. The glass wall shall be installed in a gasketed frame in a manner that the framing system deflects without breaking (loading) the glass before the sprinkler system operates; and
 - 1.3. Where glass doors are provided in the glass wall, they shall be either self-closing or automatic-closing.
2. The adjacent spaces of any three floors of the atrium shall not be required to be separated from the atrium where such spaces are accounted for in the design of the smoke control system.

404.7 Standby power. Equipment required to provide smoke control shall be provided with standby power in accordance with Section 909.11.

404.8 Interior finish. The interior finish of walls and ceilings of the atrium shall be not less than Class B with no reduction in class for sprinkler protection.

404.9 Exit access travel distance. Exit access travel distance for areas open to an atrium shall comply with the requirements of this section.

404.9.1 Egress not through the atrium. Where required access to the exits is not through the atrium, exit access travel distance shall comply with Section 1017.

404.9.2 Exit access travel distance at the level of exit discharge. Where the path of egress travel is through an atrium space, exit access travel distance at the level of exit discharge shall be determined in accordance with Section 1017.

404.9.3 Exit access travel distance at other than the level of exit discharge. Where the path of egress travel is not at the level of exit discharge from the atrium, that portion of the total permitted exit access travel distance that occurs within the atrium shall be not greater than 200 feet (60 960 mm).

404.10 Interior exit stairways. A maximum of 50 percent of interior exit stairways are permitted to egress through an atrium on the level of exit discharge in accordance with Section 1028.

SECTION BC 405 UNDERGROUND BUILDINGS AND SPACES

405.1 General. The provisions of Sections 405.2 through 405.9 apply to building spaces having a floor level used for human occupancy more than 30 feet (9144 mm) below the finished floor of the lowest level of exit discharge.

Exceptions: The provisions of Section 405 are not applicable to the following buildings or portions of buildings:

1. One- and two-family dwellings, sprinklered in accordance with Section 903.3.1.3.
2. Parking garages provided with automatic sprinkler systems in compliance with Section 405.3.
3. Fixed guideway transit systems.
4. Grandstands, bleachers, stadiums, arenas and similar facilities.
5. Where the lowest story is the only story that would qualify the building as an underground building and has an area not greater than 1,500 square feet (139 m²) and has an occupant load less than 10.
6. Pumping stations and other similar mechanical spaces intended only for limited periodic use by service or maintenance personnel.

405.2 Construction requirements. The underground portion of the building shall be of Type I construction.

405.3 Automatic sprinkler system. The highest level of exit discharge serving the underground portions of the building and all levels below shall be equipped with an automatic sprinkler system installed in accordance with Section 903.3.1.1. Water-flow switches and control valves shall be supervised in accordance with Section 903.4.

405.4 Compartmentation. Compartmentation shall be in accordance with Sections 405.4.1 through 405.4.3.

405.4.1 Number of compartments. A building having a floor level more than 60 feet (18 288 mm) below the finished floor of the lowest level of exit discharge shall be divided into no fewer than two compartments of approximately equal size. Such compartmentation shall extend through the highest level of exit discharge serving the underground portions of the building and all levels below.

Exception: The lowest story need not be compartmented where the area is not greater than 1,500 square feet (139 m²) and has an occupant load of less than 10.

405.4.2 Compartment separation. The separation between the two compartments shall be of minimum 2-hour fire barrier wall construction that shall extend from floor slab to floor deck above. Penetrations between the two compartments shall be limited to plumbing and electrical piping and conduit that are firestopped in accordance with Section 714. Doorways shall be protected by fire door assemblies that are automatic-closing by smoke detection in accordance with Section 716.5.9.3 and are installed in accordance with NFPA 105 and Section 716.5.3 of this code. Where provided, each compartment shall have an air supply and an exhaust system independent of the other compartments.

405.4.3 Elevators. Where elevators are provided, each compartment shall have direct access to an elevator. Where an elevator serves more than one compartment, an elevator lobby shall be provided and shall be separated from each compartment by a 2-hour fire barrier. Doors shall be gasketed, have a drop sill, and be automatic-closing by smoke detection in accordance with Section 716.5.9.3.

405.5 Smoke control system. A smoke control system shall be provided in accordance with Sections 405.5.1 and 405.5.2.

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405.5.1 Control system. A smoke control system is required to control the migration of products of combustion in accordance with Section 909 and the provisions of this section. Smoke control shall restrict movement of smoke to the general area of fire origin and maintain means of egress in a usable condition.

405.5.2 Compartment smoke control system. Where compartmentation is required, each compartment shall have an independent smoke control system. The system shall be automatically activated and capable of manual operation in accordance with Sections 907.2.18 and 907.2.19.

405.6 Fire alarm systems. A fire alarm system shall be provided where required by Sections 907.2.18 and 907.2.19.

405.7 Means of egress. Means of egress shall be in accordance with Sections 405.7.1 and 405.7.2.

405.7.1 Number of exits. Each floor level shall be provided with no fewer than two exits. Where compartmentation is required by Section 405.4, each compartment shall have no fewer than one exit and shall also have no fewer than one exit access doorway into the adjoining compartment.

405.7.2 Smokeproof enclosure. Every required stairway serving floor levels more than 30 feet (9144 mm) below its level of exit discharge shall comply with the requirements for a smokeproof enclosure as provided in Section 1023.11.

405.8 Standby and emergency power. A standby power system complying with Section 2702 shall be provided for the standby power loads specified in Section 405.8.1. An emergency power system complying with Section 2702 shall be provided for the emergency power loads specified in Section 405.8.2.

405.8.1 Standby power loads. The following loads are classified as standby power loads:

1. Smoke control system.
2. Ventilation and automatic fire detection equipment for smokeproof enclosures.
3. Fire pumps.
4. Elevators, as required in Section 3003; and
5. Stair pressurization systems when provided.

405.8.2 Emergency power loads. The following loads are classified as emergency power loads:

1. Emergency voice/alarm communications systems, including Fire Department in-building Auxiliary Radio Communication systems (ARCs) provided where required or installed voluntarily in accordance with Section 916;
2. Fire alarm systems;
3. Automatic fire detection systems;
4. Elevator car lighting; and
5. Means of egress and exit sign illumination as required by Chapter 10.

405.9 Standpipe system. The underground building shall be equipped throughout with a standpipe system in accordance with Section 905.

SECTION BC 406 MOTOR-VEHICLE-RELATED OCCUPANCIES

406.1 General. Motor-vehicle-related occupancies shall comply with Sections 406.1 through 406.8.

406.2 Definitions. The following terms are defined in Chapter 2:

CARPORT.

MECHANICAL-ACCESS OPEN PARKING GARAGE.

OPEN PARKING GARAGE.

PRIVATE GARAGE.

RAMP-ACCESS OPEN PARKING GARAGE.

406.3 Private garages and carports. Private garages and carports shall comply with Sections 406.3.1 through 406.3.5.

406.3.1 Classification. Private garages and carports shall be classified as Group U occupancy.

406.3.2 Clear height. In private garages and carports, the clear height in vehicle and pedestrian traffic areas shall be not less than 7 feet (2134 mm). Vehicle and pedestrian areas accommodating van-accessible parking shall comply with Section 1106.5.

406.3.3 Floor construction. Private garage and carport floors shall be of concrete or equivalent noncombustible material that will not absorb flammable liquids. The area of floor used for parking of vehicles shall be sloped to facilitate the movement of liquids to a drain. The sills of all door openings connecting a dwelling unit to a private garage shall be raised at least 4 inches (102 mm) above the floor level of the garage.

Exception: Asphalt surfaces shall be permitted at ground level in carports.

406.3.3.1 Ramps. Ramps shall have a gradient not exceeding one in seven, with nonslip surfaces, and shall be located entirely within the property line. However, upon application to the commissioner, steeper gradients may be accepted, but in no event greater than one in three.

406.3.4 Separation. Separations shall comply with the following:

1. The private garage shall be separated from the dwelling unit and its attic area by means of a minimum 1-hour fire-rated construction. Garages beneath habitable rooms shall be separated from all habitable rooms above by not less than 1-hour fire-rated construction. Openings from a private garage directly into a room used for sleeping purposes shall not be permitted.
2. No air used for heating, cooling or ventilation shall be circulated through private garages to dwelling areas.
3. A separation is not required between a Group R-3 occupancy and a carport, provided the carport does not have any enclosed areas above.
4. For free standing private garages where the fire separation distance from the nearest exterior wall is less than 5 feet (1524 mm), such exterior walls shall have a fire-resistance rating not less than one hour.

406.3.5 Automatic garage door openers. Automatic garage door openers, where provided, shall be listed in accordance with UL 325.

406.4 Public parking garages. Parking garages, other than private parking garages, shall be classified as public parking garages and shall comply with the provisions of Sections 406.4.2 through 406.4.10 and shall be classified as either an open parking garage or an enclosed parking garage. Open parking garages shall also comply with Section 406.5. Enclosed parking garages shall also comply with Section 406.6. See Section 510 for special provisions for parking garages.

406.4.1 Clear height. The clear height of each floor level in vehicle and pedestrian traffic areas shall be not less than 7 feet (2134 mm). Vehicle and pedestrian areas accommodating van-accessible parking shall comply with Section 1106.5.

406.4.2 Guards. Guards shall be provided in accordance with Section 1015. Guards serving as vehicle barrier systems shall comply with Sections 406.4.3 and 1015.

406.4.3 Vehicle barrier systems. Vehicle barriers not less than 2 feet 9 inches (835 mm) in height shall be placed where the vertical distance from the floor of a drive lane or parking space to the ground or surface directly below is greater than 1 foot (305 mm). Vehicle barriers shall comply with the loading requirements of Section 1607.9.

Exception: Vehicle barriers are not required in vehicle storage compartments in a mechanical access parking garage.

SPECIAL DETAILED REQUIREMENTS BASED ON USE AND OCCUPANCY

406.4.4 Ramps. Ramps shall have a gradient not exceeding one in seven, with nonslip surfaces. A level landing having a minimum dimension of 20 feet (6096 mm) shall be provided at the discharge point of all ramps at the street level, within the property line. Ramps used for the movement of motor vehicles need not be enclosed when serving tiers above grade. Vehicle ramps may serve as part of a means of egress in accordance with Section 1006.4.

406.4.5 Floor surface. Parking surfaces shall be of concrete or similar noncombustible and nonabsorbent materials.

The area of floor used for parking of vehicles shall be sloped to facilitate the movement of liquids to a drain.

Exceptions:

1. Asphalt parking surfaces shall be permitted at ground level.
2. Floors of Group S-2 parking garages shall not be required to have a sloped surface.

406.4.6 Separation. Parking garages shall be separated from other occupancies in accordance with Section 508.1.

406.4.7 Special hazards. Connection of a parking garage with any room in which there is a fuel-fired appliance shall be by means of a vestibule providing a two-doorway separation. Such vestibule and doorway shall be minimum 1-hour rated construction.

Exception: A single door shall be allowed provided the sources of ignition in the appliance are not less than 18 inches (457 mm) above the floor.

406.4.8 Attached to rooms. Openings from a parking garage directly into a room used for sleeping purposes shall not be permitted.

406.4.9 Fire-fighter aisles. In any parking garage or lot, one or more aisles, at least 24 inches (610 mm) in width, shall be provided to permit access by fire-fighting personnel to all parts of the garage or lot. There shall not be more than three rows of parked motor vehicles between aisles.

406.4.10 Electric vehicle charging stations. Parking garages shall be capable of supporting electrical vehicle charging stations in accordance with this section. Electrical raceway to the electrical supply panel serving the garage shall be capable of providing a minimum of 3.1 kW of electrical capacity to at least 20 percent of the parking spaces of the garage. The electrical room supplying the garage must have the physical space for an electrical supply panel sufficient to provide 3.1kW of electrical capacity to at least 20 percent of the parking spaces of the garage. Such raceway and all components and work appurtenant thereto shall be in accordance with the *New York City Electrical Code*.

Exceptions:

1. The provisions of this section shall not apply to parking garages for buildings of Occupancy Group M (Mercantile).
2. The commissioner may waive compliance with this section if the commissioner determines that the parking garage is a temporary facility that will be in service no longer than three years.
3. The provisions of this section shall not apply to parking garages for buildings in which not less than 50 percent of the residential units are for households earning up to 60 percent of the area median income as determined by the United States Department of Housing and Urban Development.

406.5 Open parking garages. Open parking garages shall comply with Sections 406.5.1 through 406.5.11.

406.5.1 Construction. Open parking garages shall be of Type I, II or IV construction. Open parking garages shall meet the design requirements of Chapter 16. For vehicle barriers, see Section 406.4.3.

406.5.2 Openings. For natural ventilation purposes, the exterior side of the structure shall have uniformly distributed openings on two or more sides. The area of such openings in exterior walls on a tier shall be not less than 20 percent of the total perimeter wall area of each tier. The aggregate length of the openings considered to be providing natural ventilation shall be not less than 40 percent of the perimeter of the tier. Interior walls shall be not less than 20 percent open with uniformly distributed openings.

Exception: Openings are not required to be distributed over 40 percent of the building perimeter where the required openings are uniformly distributed over two opposing sides of the building.

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406.5.2.1 Openings below grade. Where openings below grade provide required natural ventilation, the outside horizontal clear space shall be 1½ times the depth of the opening. The width of the horizontal clear space shall be maintained from grade down to the bottom of the lowest required opening.

406.5.3 Uses. Mixed uses shall be allowed in the same building as an open parking garage subject to the provisions of Sections 402.4.2.3, 406.5.11, 508.1, 510.3, 510.4 and 510.7.

406.5.4 Area and height. Area and height of open parking garages shall be limited as set forth in Chapter 5 for Group S-2 occupancies.

**TABLE 406.5.4
OPEN PARKING GARAGES AREA AND HEIGHT**

TYPE OF CONSTRUCTION	AREA PER TIER (square feet)	HEIGHT (in tiers)		
		Ramp access	Mechanical access	
			Automatic sprinkler system	
			No	Yes
IA	Unlimited	Unlimited	Unlimited	Unlimited
IB	Unlimited	12 tiers	12 tiers	18 tiers
IIA	50,000	10 tiers	10 tiers	15 tiers
IIB	50,000	8 tiers	8 tiers	12 tiers
IV	50,000	4 tiers	4 tiers	4 tiers

For SI: 1 square foot = 0.0929 m².

406.5.4.1 Single use. Where the open parking garage is used exclusively for the parking or storage of private motor vehicles, with no other uses in the building, the area and height shall be permitted to comply with Table 406.5.4.

Exception: The grade-level tier is permitted to contain an office, waiting and toilet rooms having a total combined area of not more than 1,000 square feet (93 m²). Such area need not be separated from the open parking garage.

In open parking garages having a spiral or sloping floor, the horizontal projection of the structure at any cross section shall not exceed the allowable area per parking tier. In the case of an open parking garage having a continuous spiral floor, each 9 feet 6 inches (2896 mm) of height, or portion thereof, shall be considered a tier.

The clear height of a parking tier shall be not less than 7 feet (2134 mm), except that a lower clear height is permitted in mechanical-access open parking garages where approved by the commissioner.

406.5.5 Area and height increases. The allowable area and height of open parking garages shall be increased in accordance with the provisions of this section. Garages with sides open on three-fourths of the building's perimeter are permitted to be increased by 25 percent in area and one tier in height. Garages with sides open around the entire building's perimeter are permitted to be increased by 50 percent in area and one tier in height. For a side to be considered open under the above provisions, the total area of openings along the side shall not be less than 50 percent of the interior area of the side at each tier and such openings shall be equally distributed along the length of the tier. For purposes of calculating the interior area of the side, the height shall not exceed 7 feet (2134 mm).

Allowable tier areas in Table 406.5.4 shall be increased for open parking garages constructed to heights less than the table maximum. The gross tier area of the garage shall not exceed that permitted for the higher structure. No fewer than three sides of each such larger tier shall have continuous horizontal openings not less than 30 inches (762 mm) in clear height extending for not less than 80 percent of the length of the sides and no part of such larger tier shall be more than 200 feet (60 960 mm) horizontally from such an opening. In addition, each such opening shall face a street or yard accessible to a street with a width of not less than 30 feet (9144 mm) for the full length of the opening, and standpipes shall be provided in each such tier.

Open parking garages of Type II construction, with all sides open, shall be unlimited in allowable area where the building height does not exceed 75 feet (22 860 mm). For a side to be considered open, the total area of openings

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along the side shall be not [be] less than 50 percent of the interior area of the side at each tier and such openings shall be equally distributed along the length of the tier. For purposes of calculating the interior area of the side, the height shall not exceed 7 feet (2134 mm). All portions of tiers shall be within 200 feet (60 960 mm) horizontally from such openings or other natural ventilation openings as defined in Section 406.5.2. These openings shall be permitted to be provided in courts with a minimum dimension of 20 feet (6096 mm) for the full width of the openings.

406.5.6 Fire separation distance. Exterior walls and openings in exterior walls shall comply with Tables 601 and 602. The distance from an adjacent property line shall be determined in accordance with Table 602 and Section 705.

406.5.7 Means of egress. Open parking garages shall meet the means of egress requirements of Chapter 10. However, where no persons other than parking attendants are permitted, required exit stairways shall not be less than 36 inches (914 mm) wide.

406.5.8 Standpipes. Standpipes shall be installed where required by the provisions of Section 905.

406.5.9 Enclosure of vertical openings. Enclosure shall not be required for vertical openings.

406.5.10 Ventilation. Openings specified in Section 406.5.2 shall satisfy natural ventilation requirements. Additional mechanical ventilation shall not be required.

406.5.11 Prohibitions. The following uses and alterations are not permitted:

1. Vehicle repair work.
2. Parking of commercial vehicles, buses, trucks and similar vehicles.
3. Partial or complete closing of required openings in exterior walls by tarpaulins or any other means.
4. Dispensing of fuel.

406.6 Enclosed parking garages. Parking garages and portions thereof that do not meet the definition of “Open parking garages” shall be classified as an enclosed parking garage and shall meet the requirements of this section.

406.6.1 Heights and areas. Enclosed parking garages shall be limited to the allowable heights and areas specified in Chapter 5. Permitted roof parking shall not be considered as a separate story.

406.6.2 Ventilation. A mechanical ventilation system shall be provided in accordance with the *New York City Mechanical Code*.

406.6.3 Automatic sprinkler system. Automatic sprinkler systems and standpipes shall be installed in accordance with the provisions of Section 903. An automatic dry sprinkler system may be installed in unheated garages.

406.6.4 Means of egress. Enclosed parking garages shall meet the means of egress requirements of Chapter 10.

406.7 Motor fuel-dispensing facilities. Motor fuel-dispensing facilities shall be designed and constructed in accordance with the *New York City Fire Code* and Sections 406.7.1 through 406.7.3 of this code.

406.7.1 Vehicle fueling pad. The vehicle shall be fueled on noncoated concrete or other approved paving material having a resistance not exceeding 1 megohm as determined by the methodology in CEN 1081.

406.7.2 Canopies. Canopies under which fuels are dispensed shall have a clear, unobstructed height of not less than 14 feet (4267 mm) to the lowest projecting element in the vehicle drive-through area. Canopies and their supports over pumps shall be of noncombustible materials. However, panels constructed of light-transmitting plastic materials shall be permitted to be installed in canopies erected over motor vehicle fuel-dispensing station fuel dispensers, provided the panels are located not less than 10 feet (3048 mm) from any building on the same property and face yards or streets not less than 40 feet (12 192 mm) in width on the other sides. The aggregate areas of plastics shall be not greater than 1,000 square feet (93 m²). The maximum area of any individual panel shall be not greater than 100 square feet (9.3 m²).

406.7.2.1 Canopies used to support gaseous hydrogen systems. Where flammable compressed gases are permitted by the *New York City Fire Code* to be located on the roof of a canopy that is used to shelter dispensing operations, such canopy shall be in accordance with the following:

1. The canopy shall meet or exceed Type I construction requirements.
2. Operations located under canopies shall be limited to refueling only.
3. The canopy shall be constructed in a manner that prevents the accumulation of hydrogen gas.

406.7.3 Storage tanks. Motor fuel storage tanks shall be installed below ground, except as authorized by the rules of the Fire Department. The installation and venting of storage tanks shall be in accordance with the rules of the Fire Department and the requirements of the *New York City Fire Code*. Storage tanks installed below ground shall comply with the following:

1. The top of the storage tanks shall be at least 24 inches (610 mm) below finished grade and at least 24 inches (610 mm) below the level of any cellar or basement floor within 10 feet (3048 mm) of the tanks.
2. Storage tanks shall be located so that the forces from any building foundation and support loads are not transmitted to the tanks. The distance from any part of a storage tank to the nearest wall of any basement, pit or cellar, or from any property line that may be built upon, shall not be less than 36 inches (914 mm).
3. Storage tanks shall be covered with a structurally supported reinforced concrete slab at least 8 inches (203 mm) thick, extending at least 12 inches (305 mm) beyond the horizontal outlines of the storage tanks, and placed over a cover of suitable clean backfill material. Such slab shall be designed and reinforced in accordance with ACI 318. All concrete shall have a minimum compressive strength of 4,000 pounds per square inch (27 580 kPa) at 28 days.
4. Storage tanks shall be placed on a 12-inch (305 mm) thick concrete base slab or installed in such other manner, secured against flotation, and approved by the department.

406.8 Repair garages. Repair garages shall be constructed in accordance with the *New York City Fire Code* and Sections 406.8.1 through 406.8.6 of this code. This occupancy shall not include motor fuel-dispensing facilities, as regulated in Section 406.7.

406.8.1 Mixed uses. Mixed uses shall be allowed in the same building as a repair garage subject to the provisions of Section 508.1.

406.8.2 Ventilation. Repair garages shall be mechanically ventilated in accordance with the *New York City Mechanical Code*. The ventilation system shall be controlled at the entrance to the garage.

406.8.3 Floor surface. Repair garage floors shall be of concrete or similar noncombustible and nonabsorbent materials.

Exception: Slip-resistant, nonabsorbent, interior floor finishes having a critical radiant flux not more than 0.45 W/cm², as determined by NFPA 253, shall be permitted.

406.8.4 Heating equipment. Heating equipment shall be installed in accordance with the *New York City Mechanical Code*.

406.8.5 Gas detection system. Repair garages used for the repair of vehicles fueled by CNG, LNG, or hydrogen, shall be provided with an approved flammable gas-detection system.

406.8.5.1 System design. The flammable gas-detection system shall be listed or approved and shall be calibrated to the types of fuels or gases used by vehicles to be repaired. The gas detection system shall be designed to activate when the level of flammable gas exceeds 25 percent of the lower flammable limit (LFL). Gas detection shall be provided in lubrication or chassis service pits of repair garages used for repairing nonodorized LNG-fueled vehicles.

406.8.5.1.1 Gas detection system components. Gas detection system control units shall be listed and labeled in accordance with UL 864 or UL 2017. Gas detectors shall be listed and labeled in accordance with UL 2075 for use with the gases and vapors being detected.

406.8.5.2 Operation. Activation of the gas detection system shall result in all of the following:

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1. Initiation of distinct audible and visual alarm signals in the repair garage.
2. Deactivation of all heating systems located in the repair garage.
3. Activation of the mechanical ventilation and exhaust system, where the system is interlocked with gas detection.

406.8.5.3 Failure of the gas detection system. Failure of the gas detection system shall result in the deactivation of the heating system, activation of the mechanical ventilation system where the system is interlocked with the gas detection system and cause a trouble signal to sound in an approved location.

406.8.6 Automatic sprinkler system. A repair garage shall be equipped with an automatic sprinkler system in accordance with Section 903.2.4.2.

406.9 Open parking lots. The provisions of this section shall govern the construction of open parking lots and to all such existing premises hereafter enlarged or changed in location.

406.9.1 Definition.‡ The following term is defined in Chapter 2:

OPEN PARKING LOT.

406.9.2 Permit required. Any premise intended to be occupied for the storage or sale of motor vehicles on an open parking lot shall require an application to be filed with the department in order to obtain a Certificate of Occupancy. Such Certificate of Occupancy shall indicate the maximum number of vehicles to be accommodated and the type of vehicle, whether private passenger or commercial, to be stored. An application for or including an open parking lot shall be accompanied by a plan exhibiting the following:

1. Dimensions of the lot and its location in relation to adjoining streets;
2. Any structure(s) existing or to be erected on the plot;
3. The relative elevations of the parking area, curbs and adjoining yards or courts;
4. Structures, retaining walls, and open spaces on adjoining premises;
5. Retaining walls to be built;
6. Location and dimensions of curb cuts, driveways, and enclosures;
7. Drainage diagram;
8. Specification of surfacing material;
9. Parking stall layout with dimensions;
10. Analysis exhibiting compliance with the *New York City Zoning Resolution*; and
11. Other information as may be requested by the commissioner.

A copy of the plan or diagram approved by the department shall be kept on the premises. Certified, reduced size, legible copies may be used for this purpose. When an attendant's shelter is provided on the parking lot, the Certificate of Occupancy issued shall be posted and maintained under glass in the shelter.

406.9.3 Locations prohibited. All vehicular activities associated with the operation of open parking lots shall be entirely within the property lines of the premises. Vehicles shall not be permitted to encroach upon the sidewalks. No motor vehicle may be stored or parked in any location where it would obstruct a required window or required exit of any adjacent building.

406.9.4 Surfacing. All driveways, parking stalls, and open spaces used for the parking or storage of motor vehicles shall be surfaced with concrete asphalt, or equivalent durable, dustless material.

406.9.4.1 Drainage. Open parking lots shall be graded and maintained such that no drainage will flow onto abutting sidewalks and adjoining properties.

406.9.4.2 Nonporous surfaces. Where the surface paving of an open parking lot is nonporous, such lot shall be drained as required by the *New York City Plumbing Code*.

406.9.4.3 Resurfacing. Resurfacing of existing open parking lots shall comply with this section.

406.9.5 Curbs and bumpers. Open parking lots shall be completely separated from adjoining premises and public sidewalks by curbs or bumpers of concrete, masonry, steel, heavy timber, or other similar and equally substantial

materials, and shall be securely anchored so as to stop motor vehicles. Curbs and bumpers shall be at least 8 inches (203 mm) high and 8 inches (203 mm) wide.

406.9.5.1 Openings. Openings in required perimeter curbs and bumpers shall be permitted only for drainage, for motor vehicle entrances and exits, and for pedestrian entrances and exits. The width of an opening for motor vehicle access shall not exceed the length of the curb cut, not including the lengths of the splays. When an opening for pedestrian access is adjacent to a motor vehicle access opening, the two openings shall be separated by a permanent and substantial post. In no case shall any pedestrian entrance or exit exceed 5 feet (1524 mm) in width.

406.9.6 Accessory uses and occupancies. Parking lot offices, attendant shelters, storage facilities, and similar structures used in conjunction with open parking lots may be provided as an accessory use and shall comply with the applicable code provisions. Such accessory structures may be constructed of combustible materials subject to the limitations of Section D105.1 of Appendix D.

406.9.6.1 Motor fuel-dispensing facilities. Motor fuel-dispensing facilities, accessory to open parking lots, shall comply with the requirements of the *New York City Fire Code* and Section 406.8 of this code. Motor fuel pumps shall be located at least 30 feet (9144 mm) from any parking space or interior lot line.

406.9.7 Special provisions applicable to large open parking lots. Open parking lots storing 10 or more motor vehicles shall also be subject to the special provisions of this section.

406.9.7.1 Curbs and bumpers. Bumpers shall be situated not less than 1 foot (305 mm) from the perimeter edge of the open parking lot where vehicles are parked parallel to such perimeter edge, and not less than 4 feet (1219 mm) from the perimeter edge where vehicles are parked perpendicular to such perimeter edge. Openings shall be limited as provided for in Section 406.9.5.1.

Exception: A steel guardrail or other substantial barrier designed in accordance with the provisions of Section 1607.9, that will prevent any part of a vehicle from extending across a property line, may be accepted in lieu of bumpers.

406.9.7.2 Screen enclosures. Open parking lots shall provide a perimeter screen enclosure as per this section unless otherwise required by the *New York City Zoning Resolution*. Such screen enclosure may be constructed as a masonry wall, woven wire fence, iron picket fence, or uniformly painted fence of fire-resistant material at least 4 feet (1219 mm) high, but not more than 8 feet (2438 mm) above finished grade, subject to the limitations of Section 3111. Such enclosures shall completely separate the lot from adjoining premises and public sidewalks. Openings shall be limited as provided for in Section 406.9.5.1.

Exception: Screening shall not be required for any portion of the lot adjacent to an existing wall located at the lot line.

406.9.7.3 Illumination. Open parking lots operating between 6 p.m. and 6 a.m. shall be adequately illuminated to a minimum of 1 footcandle (10.8 lux) measured at grade level, distributed over the entire area. Illumination provided with reflectors or floodlights shall be arranged such that the illumination is directed downward and away from adjacent premises.

406.9.7.4 Fire-fighter aisles. In any open parking lot, one or more aisles, at least 24 inches (610 mm) in width, shall be provided to permit access by fire-fighting personnel to all parts of the garage or lot. There shall not be more than three rows of parked motor vehicles between aisles.

406.9.8 Electric vehicle charging stations. Open parking lots shall be capable of supporting electric vehicle charging stations in accordance with this section. A minimum of 20 percent of the parking spaces in an open parking lot shall be equipped with electrical raceway capable of providing a minimum supply of 11.5 kVA to an electrical vehicle supply equipment (EVSE) from an electrical supply panel. The raceway shall be no smaller than 1 inch (25 mm). The electrical supply panel serving such parking spaces must have at least 3.1 kW of available capacity for each stall connected to it with raceway. Such raceway and all components and work appurtenant thereto shall be in accordance with the *New York City Electrical Code*.

Exceptions:

1. The provisions of this section shall not apply to open parking lots for buildings of Occupancy Group M (Mercantile).

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2. The commissioner may waive compliance with this section if the commissioner determines that the open parking lot is a temporary facility that will be in service no longer than 3 years.
3. The provisions of this section shall not apply to open parking lots for buildings in which not less than fifty percent of the residential units are for households earning up to sixty percent of the area median income as determined by the United States Department of Housing and Urban Development.

406.10 Curb cuts and driveways. Except where specifically provided for in the *New York City Zoning Resolution*, all curb cuts and driveways shall be constructed in accordance with Sections 406.10.1 and 406.10.2. All sidewalks and driveways or portions thereof that are structurally supported shall be designed for loads prescribed in Chapter 16.

406.10.1 Curb cuts. For the purpose of this section, a curb cut shall include all splays when determining the total length of cut curb. All curb cuts shall comply with the following requirements:

1. The aggregate length of curb cuts shall not exceed 60 percent of any street frontage 100 feet (30 480 mm) or less in length. No single curb cut shall exceed 30 feet (9144 mm) in length, and there shall not be more than two curb cuts on any street frontage 100 feet (30 480 mm) or less in length. The minimum distance between two curb cuts shall be 5 feet (1524 mm).
2. For each 50 feet (15 240 mm) of street frontage length over 100 feet (30 480 mm), an additional curb cut no greater than 30 feet (9144 mm) in length may be permitted.
3. No curb cut shall commence within 8 feet (2438 mm) of a side lot line.

Exceptions:

1. For open parking lots on corner lots and on lots with street frontage length of 50 feet (15 240 mm) or less, the curb cut may commence no less than 30 inches (762 mm) from the side lot line.
2. In open or enclosed parking garages where a level landing having a minimum dimension of 20 feet (6096 mm) is provided at the discharge point of all ramps at the street level within the property line, the curb cut may commence no less than 30 inches (762 mm) from a side lot line.
4. No curb cut, including splays, shall be less than 10 feet (3048 mm) in length.

406.10.2 Driveways. Driveways serving passenger vehicles shall be a minimum of 8 feet (2438 mm) in width and shall not exceed the length of the curb cut, not including the lengths of the splays. For all other motor vehicles, the minimum width of driveways shall be 10 feet (3048 mm). Portions of the driveway located between the curb line and the lot line shall be paved in accordance with the requirements of the Department of Transportation for the construction of sidewalks.

406.11 Battery-powered industrial trucks, equipment and mobility devices. Buildings or portions thereof that contain or store battery-powered industrial trucks, equipment and mobility devices shall be designed and constructed in accordance with Section 309.3 of the *New York City Fire Code*.

SECTION BC 407 GROUP I-2

407.1 General. Occupancies in Group I-2 shall comply with the provisions of Sections 407.1 through 407.10 and other applicable provisions of this code.

407.2 Corridors continuity and separation. Corridors in occupancies in Group I-2 shall be continuous to the exits and shall be separated from other areas in accordance with Section 407.3 except spaces conforming to Sections 407.2.1 through 407.2.6.

407.2.1 Waiting and similar areas. Waiting areas and similar spaces constructed as required for corridors shall be permitted to be open to a corridor, only where all of the following criteria are met:

1. The spaces are not occupied as care recipient's sleeping rooms, treatment rooms, incidental uses in accordance with Section 509, or hazardous uses.
2. The open space is protected by an automatic fire detection system installed in accordance with Section 907.

3. The corridors onto which the spaces open, in the same smoke compartment, are protected by an automatic fire detection system installed in accordance with Section 907, or the smoke compartment in which the spaces are located is equipped throughout with quick-response sprinklers in accordance with Section 903.3.2.
4. The space is arranged so as not to obstruct access to the required exits.

407.2.2 Care providers' stations. Spaces for care providers', supervisory staff, doctors' and nurses' charting, communications and related clerical areas shall be permitted to be open to the corridor, where such spaces are constructed as required for corridors.

407.2.3 Psychiatric treatment areas. Areas wherein psychiatric care recipients who are not capable of self-preservation are housed, or group meeting or multipurpose therapeutic spaces other than incidental uses in accordance with Section 509, under continuous supervision by facility staff, shall be permitted to be open to the corridor, where the following criteria are met:

1. Each area does not exceed 1,500 square feet (140 m²).
2. The area is located to permit supervision by the facility staff.
3. The area is arranged so as not to obstruct any access to the required exits.
4. The area is equipped with an automatic fire detection system installed in accordance with Section 907.2.
5. Not more than one such space is permitted in any one smoke compartment.
6. The walls and ceilings of the space are constructed as required for corridors.

407.2.4 Gift shops. Gift shops and associated storage that are less than 500 square feet (46 m²) in area shall be permitted to be open to the corridor where such spaces are constructed as required for corridors.

407.2.5 Nursing home housing units. In Group I-2 occupancies, in areas where nursing home residents are housed, shared living spaces, group meeting or multipurpose therapeutic spaces shall be permitted to be open to the corridor, where all of the following criteria are met:

1. The walls and ceilings of the space are constructed as required for corridors.
2. The spaces are not occupied as resident sleeping rooms, treatment rooms, incidental uses in accordance with Section 509, or hazardous uses.
3. The open space is protected by an automatic fire detection system installed in accordance with Section 907.
4. The corridors onto which the spaces open, in the same smoke compartment, are protected by an automatic fire detection system installed in accordance with Section 907, or the smoke compartment in which the spaces are located is equipped throughout with quick-response sprinklers in accordance with Section 903.3.2.
5. The space is arranged so as not to obstruct access to the required exits.

407.2.6 Nursing home cooking facilities. In Group I-2 occupancies, rooms or spaces that contain a cooking facility with domestic cooking appliances shall be permitted to be open to the corridor where all of the following criteria are met:

1. The number of care recipients housed in the smoke compartment is not greater than 30.
2. The number of care recipients served by the cooking facility is not greater than 30.
3. Only one cooking facility area is permitted in a smoke compartment.
4. The types of domestic cooking appliances permitted are limited to ovens, cooktops, ranges, warmers and microwaves.
5. The corridor is a clearly identified space delineated by construction or floor pattern, material or color.
6. The space containing the domestic cooking facility shall be arranged so as not to obstruct access to the required exit.
7. A domestic cooking hood installed and constructed in accordance with Section 505 of the *New York City Mechanical Code* is provided over the cooktop or range.

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8. The domestic cooking hood provided over the cooktop or range shall be equipped with an automatic fire-extinguishing system of a type recognized for protection of domestic cooking equipment. Preengineered automatic extinguishing systems shall be tested in accordance with UL 300A and listed and labeled for the intended application. The system shall be installed in accordance with this code, its listing and the manufacturer's instructions.
9. A manual actuation device for the hood suppression system shall be installed in accordance with Sections 904.12.1 and 904.12.2.
10. An interlock device shall be provided such that upon activation of the hood suppression system, the power or fuel supply to the cooktop or range will be turned off.
11. A shut-off for the fuel and electrical power supply to the cooking equipment shall be provided in a location that is accessible only to staff.
12. A timer shall be provided that automatically deactivates the cooking appliances within a period of not more than 120 minutes.
13. A portable fire extinguisher shall be installed in accordance with Section 906 of the *New York City Fire Code*.

407.3 Corridor wall construction. Corridor walls shall be constructed as smoke partitions in accordance with Section 710.

407.3.1 Corridor doors. Corridor doors, other than those in a wall required to be rated by Section 509.4 or for the enclosure of a vertical opening or an exit, shall not have a required fire protection rating and shall not be required to be equipped with self-closing or automatic-closing devices, but shall provide an effective barrier to limit the transfer of smoke and shall be equipped with positive latching. Roller latches are not permitted. Other doors shall conform to Section 716.5.

407.4 Means of egress. Group I-2 occupancies shall be provided with means of egress complying with Chapter 10 and Sections 407.4.1 through 407.4.4. The fire and emergency preparedness plans provided in accordance with Section 1001.4 of this code shall identify the building components necessary to support a defend-in-place emergency response in accordance with the *New York City Fire Code*.

407.4.1 Direct access to a corridor. Habitable rooms in Group I-2 occupancies shall have an exit access door leading directly to a corridor.

Exceptions:

1. Rooms with exit doors opening directly to the outside at ground level.
2. Rooms arranged as care suites complying with Section 407.4.4.

407.4.1.1 Locking devices. Locking devices that restrict access to a care recipient's room from the corridor and that are operable only by staff from the corridor side shall not restrict the means of egress from the care recipient's room.

Exceptions:

1. This section shall not apply to rooms in psychiatric treatment and similar care areas.
2. Locking arrangements in accordance with Section 1010.1.9.6.

407.4.2 Distance of travel. The distance of travel between any point in a Group I-2 occupancy sleeping room, not located in a care suite, and an exit access door in that room shall be not greater than 50 feet (15 240 mm).

407.4.3 Projections in nursing home corridors. In nursing homes of Group I-2 occupancies, where the corridor width is a minimum of 96 inches (2440 mm), projections shall be permitted for furniture where all of the following criteria are met:

1. The furniture is attached to the floor or to the wall.
2. The furniture does not reduce the clear width of the corridor to less than 72 inches (1830 mm) except where other encroachments are permitted in accordance with Section 1005.7.
3. The furniture is positioned on only one side of the corridor.

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4. Each arrangement of furniture is 50 square feet (4.6 m²) maximum in area.
5. Furniture arrangements are separated by 10 feet (3048 mm) minimum.
6. Placement of furniture is considered as part of the fire and emergency preparedness plans in accordance with Section 1001.4.

407.4.4 Group I-2 care suites. Care suites in Group I-2 shall comply with Sections 407.4.4.1 through 407.4.4.4 and either Section 407.4.4.5 or 407.4.4.6.

407.4.4.1 Exit access through care suites. Exit access from all other portions of a building not classified as a care suite shall not pass through a care suite. In a care suite required to have more than one exit, one exit access is permitted to pass through an adjacent care suite provided all of the other requirements of Sections 407.4 and 1016.2 are satisfied.

407.4.4.2 Separation. Care suites shall be separated from other portions of the building, including other care suites, by a smoke partition complying with Section 710.

††† 407.4.4.3 Access to corridor. Movement from habitable rooms shall not require passage through more than three doors and 100 feet (30 480 mm) distance of travel within the suite.

Exception: The distance of travel shall be permitted to be increased to 125 feet (38 100 mm) where an automatic smoke detection system is provided throughout the care suite and installed in accordance with NFPA 72 as modified by Appendix Q of this code.

407.4.4.4 Doors within care suites. Doors in care suites serving habitable rooms shall be permitted to comply with one of the following:

1. Manually operated horizontal sliding doors permitted in accordance with Exception 9 to Section 1010.1.2.1.
2. Power-operated doors permitted in accordance with Exception 7 to Section 1010.1.2.1.
3. Means of egress doors complying with Section 1010.

407.4.4.5 Care suites containing sleeping room areas. Sleeping rooms shall be permitted to be grouped into care suites where one of the following criteria is met:

1. The care suite is not used as an exit access for more than eight care recipient beds.
2. The arrangement of the care suite allows for direct and constant visual supervision into the sleeping rooms by care providers.
3. An automatic smoke detection system is provided in the sleeping rooms and installed in accordance with NFPA 72 as modified by Appendix Q of this code.

407.4.4.5.1 Area. Care suites containing sleeping rooms shall be not greater than 7,500 square feet (696 m²) in area.

Exception: Care suites containing sleeping rooms shall be permitted to be not greater than 10,000 square feet (929 m²) in area where an automatic smoke detection system is provided throughout the care suite and installed in accordance with NFPA 72 as modified by Appendix Q of this code.

407.4.4.5.2 Exit access. Any sleeping room, or any care suite that contains sleeping rooms, of more than 1,000 square feet (93 m²) shall have no fewer than two exit access doors from the care suite located in accordance with Section 1007.

407.4.4.6 Care suites not containing sleeping rooms. Areas not containing sleeping rooms, but only treatment areas and the associated rooms, spaces or circulation space, shall be permitted to be grouped into care suites and shall conform to the limitations in Sections 407.4.4.6.1 and 407.4.4.6.2.

407.4.4.6.1 Area. Care suites of rooms, other than sleeping rooms, shall have an area not greater than 12,500 square feet (1161 m²).

Exception: Care suites not containing sleeping rooms shall be permitted to be not greater than 15,000 square feet (1394 m²) in area where an automatic smoke detection system is provided throughout the care suite in accordance with Section 907.

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407.4.4.6.2 Exit access. Care suites, other than sleeping rooms, with an area of more than 2,500 square feet (232 m²) shall have no fewer than two exit access doors from the care suite located in accordance with Section 1007.1.

407.5 Smoke barriers. Smoke barriers shall be provided to subdivide every story used by persons receiving care, treatment or sleeping and to divide other stories with an occupant load of 50 or more persons, into no fewer than two smoke compartments. Such stories shall be divided into smoke compartments with an area of not more than 22,500 square feet (2090 m²) in nursing homes in Group I-2, and not more than 40,000 square feet (3716 m²) in detoxification facilities, hospitals and psychiatric centers in Group I-2, and the distance of travel from any point in a smoke compartment to a smoke barrier door shall be not greater than 200 feet (60 960 mm). The smoke barrier shall be in accordance with Section 709.

407.5.1 Refuge area. Refuge areas shall be provided within each smoke compartment. The size of the refuge area shall accommodate the occupants and care recipients from the adjoining smoke compartment. Where a smoke compartment is adjoined by two or more smoke compartments, the minimum area of the refuge area shall accommodate the largest occupant load of the adjoining compartments. The size of the refuge area shall provide the following:

1. Not less than 30 net square feet (2.8 m²) for each care recipient confined to bed or stretcher.
2. Not less than 6 square feet (0.56 m²) for each ambulatory care recipient not confined to bed or stretcher and for other occupants.

Areas or spaces permitted to be included in the calculation of refuge area are corridors, sleeping areas, treatment rooms, lounge or dining areas and other low-hazard areas.

407.5.2 Independent egress. A means of egress shall be provided from each smoke compartment created by smoke barriers without having to return through the smoke compartment from which means of egress originated.

407.5.3 Horizontal assemblies. Horizontal assemblies supporting smoke barriers required by this section shall be designed to resist the movement of smoke. Elevator lobbies shall be in accordance with Section 3006.1.

407.6 Automatic sprinkler system. Smoke compartments containing sleeping rooms shall be equipped throughout with an automatic fire sprinkler system in accordance with Sections 903.3.1.1 and 903.3.2.

407.7 Fire alarm system. A fire alarm system shall be provided in accordance with Section 907.2.6.

407.8 Automatic fire detection. Corridors in nursing homes (both intermediate-care and skilled nursing facilities) and spaces permitted to be open to the corridors by Section 407.2 shall be equipped with an automatic fire detection system installed in accordance with Section 907. Detoxification facilities, hospitals and psychiatric centers shall be equipped with smoke detection as required in Section 407.2.

Exceptions:

1. Corridor smoke detection is not required where sleeping rooms are provided with smoke detectors that comply with UL 268. Such detectors shall provide a visual display on the corridor side of each sleeping room and an audible and visual alarm at the care provider's station attending each unit.
2. Corridor smoke detection is not required where sleeping room doors are equipped with automatic door-closing devices with integral smoke detectors on the unit sides installed in accordance with their listing, provided that the integral detectors perform the required alerting function.

407.9 Secured yards. Grounds are permitted to be fenced and gates therein are permitted to be equipped with locks, provided that safe dispersal areas having 30 net square feet (2.8 m²) for bed and stretcher care recipients and 6 net square feet (0.56 m²) for ambulatory care recipients and other occupants are located between the building and the fence. Such provided safe dispersal areas shall be located not less than 50 feet (15 240 mm) from the building they serve.

407.10 Electrical systems. In Group I-2 occupancies, the essential electrical system for electrical components, equipment and systems shall be designed and constructed in accordance with the provisions of NFPA 99 and Chapter 27 of this code.

**SECTION BC 408
GROUP I-3**

408.1 General. Occupancies in Group I-3 shall comply with the provisions of Sections 408.1 through 408.11 and other applicable provisions of this code (see Section 308.5).

408.1.1 Definitions specific to this section. The following terms shall, for the purposes of this section, have the meanings shown herein:

CELL. A room within a housing unit in a detention or correctional facility used to confine inmates or prisoners.

CELL TIER. Levels of cells vertically stacked above one another within a housing unit.

HOUSING UNIT. A dormitory or a group of cells with a common dayroom in Group I-3.

SALLYPORT. A security vestibule with two or more doors or gates where the intended purpose is to prevent continuous and unobstructed passage by allowing the release of only one door or gate at a time.

408.2 Other occupancies. Buildings or portions of buildings in Group I-3 occupancies where security operations necessitate the locking of required means of egress shall be permitted to be classified as a different occupancy. Occupancies classified as other than Group I-3 shall meet the applicable requirements of this code for that occupancy where provisions are made for the release of occupants at all times.

Means of egress from detention and correctional occupancies that traverse other use areas shall, as a minimum, conform to requirements for detention and correctional occupancies.

Exception: It is permissible to exit through a horizontal exit into other contiguous occupancies that do not conform to detention and correctional occupancy egress provisions but that do comply with requirements set forth in the appropriate occupancy, as long as the occupancy is not a Group H use.

408.3 Means of egress. Except as modified or as provided for in this section, the means of egress provisions of Chapter 10 shall apply.

408.3.1 Door width. Doors to resident sleeping units shall have a clear width of not less than 28 inches (711 mm).

408.3.2 Sliding doors. Where doors in a means of egress are of the horizontal-sliding type, the force to slide the door to its fully open position shall be not greater than 50 pounds (220 N) with a perpendicular force against the door of 50 pounds (220 N).

408.3.3 Guard tower doors. A hatch or trap door not less than 16 square feet (1.49 m²) in area through the floor and having dimensions of not less than 2 feet (610 mm) in any direction shall be permitted to be used as a portion of the means of egress from guard towers.

408.3.4 Spiral stairways. Spiral stairways that conform to the requirements of Section 1011.10 are permitted for access to and between staff locations.

408.3.5 Ship ladders. Ship ladders shall be permitted for egress from control rooms or elevated facility observation rooms in accordance with Section 1011.15.

408.3.6 Exit discharge. Exits are permitted to discharge into a fenced or walled courtyard. Enclosed yards or courts shall be of a size to accommodate all occupants, be located not less than 50 feet (15 240 mm) from the building and have an area of not less than 15 square feet (1.4 m²) per person.

408.3.7 Sallyports. A sallyport shall be permitted in a means of egress where there are provisions for continuous and unobstructed passage through the sallyport during an emergency egress condition.

408.3.8 Interior exit stairway and ramp construction. One interior exit stairway or ramp in each building shall be permitted to have glazing installed in doors and interior walls at each landing level providing access to the interior exit stairway or ramp, provided that the following conditions are met:

1. The interior exit stairway or ramp shall not serve more than four floor levels.
2. Exit doors shall be not less than ³/₄ hour fire door assemblies complying with Section 716.5.

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3. The total area of glazing at each floor level shall not exceed 5,000 square inches (3.2 m²) and individual panels of glazing shall not exceed 1,296 square inches (0.84 m²).
4. The glazing shall be protected on both sides by an automatic sprinkler system. The sprinkler system shall be designed to wet completely the entire surface of any glazing affected by fire when actuated.
5. The glazing shall be in a gasketed frame and installed in such a manner that the framing system will deflect without breaking (loading) the glass before the sprinkler system operates.
6. Obstructions, such as curtain rods, drapery traverse rods, curtains, drapes or similar materials shall not be installed between the automatic sprinklers and the glazing.

408.4 Locks. Egress doors are permitted to be locked in accordance with the applicable use condition. Doors from a refuge area to the outside are permitted to be locked with a key in lieu of locking methods described in Section 408.4.1. The keys to unlock the exterior doors shall be available at all times and the locks shall be operable from both sides of the door.

408.4.1 Remote release. Remote release of locks on doors in a means of egress shall be provided with reliable means of operation, remote from the resident living areas, to release locks on all required doors. In Occupancy Condition 3 or 4, the arrangement, accessibility and security of the release mechanisms required for egress shall be such that with the minimum available staff at any time, the lock mechanisms are capable of being released within 2 minutes.

Exception: Provisions for remote locking and unlocking of occupied rooms in Occupancy Condition 4 are not required provided that not more than 10 locks are necessary to be unlocked in order to move occupants from one smoke compartment to a refuge area within 3 minutes. The opening of necessary locks shall be accomplished with not more than two separate keys.

408.4.2 Power-operated doors and locks. Power-operated sliding doors or power-operated locks for swinging doors shall be operable by a manual release mechanism at the door. Emergency power shall be provided for the doors and locks in accordance with Section 2702.

Exceptions:

1. Emergency power is not required in facilities with 10 or fewer locks complying with the exception to Section 408.4.1.
2. Emergency power is not required where remote mechanical operating releases are provided.

408.4.3 Redundant operation. Remote release, mechanically operated sliding doors or remote release, mechanically operated locks shall be provided with a mechanically operated release mechanism at each door, or shall be provided with a redundant remote release control.

408.4.4 Relock capability. Doors remotely unlocked under emergency conditions shall not automatically relock when closed unless specific action is taken at the remote location to enable doors to relock.

408.5 Protection of vertical openings. Any vertical opening shall be protected by a shaft enclosure in accordance with Section 713, or shall be in accordance with Sections 408.5.1 and 408.5.2.

408.5.1 Floor openings. Openings in floors within a housing unit are permitted without a shaft enclosure, provided all of the following conditions are met:

1. The entire normally occupied areas so interconnected are open and unobstructed so as to enable observation of the areas by supervisory personnel;
2. Means of egress capacity is sufficient for all occupants from all interconnected cell tiers and areas;
3. The height difference between the floor levels of the highest and lowest cell tiers shall not exceed 23 feet (7010 mm); and
4. Egress from any portion of the cell tier to an exit or exit access door shall not require travel on more than one additional floor level within the housing unit.

408.5.2 Shaft openings in communicating floor levels. Where a floor opening is permitted between communicating floor levels of a housing unit in accordance with Section 408.5.1, plumbing chases serving vertically stacked individual cells contained within the housing unit shall be permitted without a shaft enclosure.

408.6 Smoke barrier. Occupancies in Group I-3 shall have smoke barriers complying with Sections 408.8 and 709 to divide every story occupied by residents for sleeping, or any other story having an occupant load of 50 or more persons, into no fewer than two smoke compartments.

Exception: Spaces having a direct exit to one of the following, provided that the locking arrangement of the doors involved complies with the requirements for doors at the smoke barrier for the use condition involved:

1. A public way.
2. A building separated from the resident housing area by a 2-hour fire-resistance-rated assembly or 50 feet (15 240 mm) of open space.
3. A secured yard or court having a holding space 50 feet (15 240 mm) from the housing area that provides 6 square feet (0.56 m²) or more of refuge area per occupant, including residents, staff and visitors.

408.6.1 Smoke compartments. The number of residents in any smoke compartment shall be not more than 200. The distance of travel to a door in a smoke barrier from any room door required as exit access shall be not greater than 150 feet (45 720 mm). The distance of travel to a door in a smoke barrier from any point in a room shall be not greater than 200 feet (60 960 mm).

408.6.2 Refuge area. Not less than 6 net square feet (0.56 m²) per occupant shall be provided on each side of each smoke barrier for the total number of occupants in adjoining smoke compartments. This space shall be readily available wherever the occupants are moved across the smoke barrier in a fire emergency.

408.6.3 Independent egress. A means of egress shall be provided from each smoke compartment created by smoke barriers without having to return through the smoke compartment from which means of egress originates.

408.7 Security glazing. In occupancies in Group I-3, windows and doors in 1-hour fire barriers constructed in accordance with Section 707, fire partitions constructed in accordance with Section 708 and smoke barriers constructed in accordance with Section 709 shall be permitted to have security glazing installed provided that the following conditions are met. Such glazing shall be subject to the requirements in Chapter 24.

1. Individual panels of glazing shall not exceed 1,296 square inches (0.84 m²).
2. The glazing shall be protected on both sides by an automatic sprinkler system. The sprinkler system shall be designed to, when actuated, wet completely the entire surface of any glazing affected by fire.
3. The glazing shall be in a gasketed frame and installed in such a manner that the framing system will deflect without breaking (loading) the glass before the sprinkler system operates.
4. Obstructions, such as curtain rods, drapery traverse rods, curtains, drapes or similar materials shall not be installed between the automatic sprinklers and the glazing.

408.8 Subdivision of resident housing areas. Sleeping areas and any contiguous day room, group activity space or other common spaces where residents are housed shall be separated from other spaces in accordance with Sections 408.8.1 through 408.8.4.

408.8.1 Occupancy Conditions 3 and 4. Each sleeping area in Occupancy Conditions 3 and 4 shall be separated from the adjacent common spaces by a smoke-tight partition where the distance of travel from the sleeping area through the common space to the corridor exceeds 50 feet (15 240 mm).

408.8.2 Occupancy Condition 5. Each sleeping area in Occupancy Condition 5 shall be separated from adjacent sleeping areas, corridors and common spaces by a smoke-tight partition. Additionally, common spaces shall be separated from the corridor by a smoke-tight partition.

408.8.3 Openings in room face. The aggregate area of openings in a solid sleeping room face in Occupancy Conditions 2, 3, 4 and 5 shall not exceed 120 square inches (0.077 m²). The aggregate area shall include all openings including door undercuts, food passes and grilles. Openings shall be not more than 36 inches (914 mm) above the floor. In Occupancy Condition 5, the openings shall be closeable from the room side.

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408.8.4 Smoke-tight doors. Doors in openings in partitions required to be smoke tight by Section 408.8 shall be substantial doors, of construction that will resist the passage of smoke. Latches and door closures are not required on cell doors.

408.9 Windowless buildings. For the purposes of this section, a windowless building or portion of a building is one with nonopenable windows, windows not readily breakable or without windows. Windowless buildings shall be provided with an engineered smoke control system to provide a tenable environment for exiting from the smoke compartment in the area of fire origin in accordance with Section 909 for each windowless smoke compartment.

408.10 Fire alarm system. A fire alarm system shall be provided in accordance with Section 907.2.6.3.

408.11 Automatic sprinkler system. Group I-3 occupancies shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.2.6.

SECTION BC 409 MOTION PICTURE PROJECTION ROOMS

409.1 General. The provisions of Sections 409.1 through 409.5 of this code shall apply to rooms in which ribbon-type cellulose acetate or other safety film is utilized in conjunction with electric arc, xenon or other light-source projection equipment that develops hazardous gases, dust or radiation. The projection, use or storage of film having a nitrocellulose base (commonly known as nitrate film) shall not be permitted except under conditions specified in special permits when issued by the Fire Department. Safety film meeting the specification and test standards of NFPA 40 may be projected, used or stored.

409.1.1 Projection room required. Every motion picture machine projecting film as mentioned within the scope of this section shall be enclosed in a projection room. Appurtenant electrical equipment, such as rheostats, transformers and generators, shall be within the projection room or in an adjacent room of equivalent construction.

409.2 Construction of projection rooms. Every projection room shall be of permanent construction consistent with the construction requirements for the type of building in which the projection room is located. Openings are not required to be protected.

The room shall have a floor area of not less than 80 square feet (7.44 m²) for a single machine and not less than 40 square feet (3.7 m²) for each additional machine. Each motion picture projector, floodlight, spotlight or similar piece of equipment shall have a clear working space of not less than 30 inches by 30 inches (762 mm by 762 mm) on each side and at the rear thereof, but only one such space shall be required between two adjacent projectors. The projection room and the rooms appurtenant thereto shall have a ceiling height of not less than 7 feet 6 inches (2286 mm). The aggregate of openings for projection equipment shall not exceed 25 percent of the area of the wall between the projection room and the auditorium. Openings shall be provided with glass or other approved material, so as to close completely the opening.

409.3 Projection room and equipment ventilation. Ventilation shall be provided in accordance with the *New York City Mechanical Code*.

409.3.1 Supply air. Each projection room shall be provided with adequate air supply inlets so arranged as to provide well-distributed air throughout the room. Air inlet ducts shall provide an amount of air equivalent to the amount of air being exhausted by projection equipment. Air is permitted to be taken from the outside; from adjacent spaces within the building, provided the volume and infiltration rate is sufficient; or from the building air-conditioning system, provided it is so arranged as to provide sufficient air when other systems are not in operation.

409.3.2 Exhaust air. Projection rooms are permitted to be exhausted through the lamp exhaust system. The lamp exhaust system shall be positively interconnected with the lamp so that the lamp will not operate unless there is the required airflow. Exhaust air ducts shall terminate at the exterior of the building in such a location that the exhaust air cannot be readily recirculated into any air supply system. The projection room ventilation system is permitted to also serve appurtenant rooms, such as the generator and rewind rooms.

409.3.3 Projection machines. Each projection machine shall be provided with an exhaust duct that will draw air from each lamp and exhaust it directly to the outside of the building. The lamp exhaust is permitted to serve to exhaust air from the projection room to provide room air circulation. Such ducts shall be of rigid materials, except

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for a flexible connector approved for the purpose. The projection lamp or projection room exhaust system, or both, is permitted to be combined but shall not be interconnected with any other exhaust or return system, or both, within the building.

409.4 Lighting control. Provisions shall be made for control of the auditorium lighting and the means of egress lighting systems of theaters from inside the projection room and from not less than one other convenient point in the building.

409.5 Miscellaneous equipment. Each projection room shall be provided with rewind and film storage facilities.

SECTION BC 410 STAGES, PLATFORMS AND TECHNICAL PRODUCTION AREAS

410.1 Applicability. The provisions of Sections 410.1 through 410.11 shall apply to all parts of buildings and structures that contain stages or platforms and similar appurtenances as herein defined.

410.2 Definitions. This section contains terms defined elsewhere in the code, and terms with definitions that are specific to this section.

410.2.1 Terms defined elsewhere in the code. The following terms are defined in Chapter 2:

FLY GALLERY.

GRIDIRON.

PINRAIL.

PROSCENIUM WALL.

STAGE.

TECHNICAL PRODUCTION AREA.

410.2.2 Definitions specific to this section. The following term shall, for the purposes of this section, have the meaning shown herein:

PLATFORM. A raised area within a building used for worship, the presentation of music, plays or other entertainment; the head table for special guests; the raised area for lecturers and speakers; boxing and wrestling rings; theater-in-the-round stages; and similar purposes wherein, other than horizontal sliding curtains, there are no overhead hanging curtains, drops, scenery or stage effects other than lighting and sound. A temporary platform is one installed for not more than 30 days.

410.3 Stages. Stage construction shall comply with Sections 410.3.1 through 410.3.7.

410.3.1 Stage construction. Stage floors shall be constructed of materials as required for floors for the type of construction of the building in which such stages are located. Where areas below the stage are used for other occupancies, such stage floor shall be constructed in accordance with the requirements for separated occupancies as per Chapter 5.

Exceptions:

1. Stages of Type IIB or IV construction with a nominal 2-inch (51 mm) wood deck, provided that the stage is separated from other areas in accordance with Section 410.3.4.
2. In all types of construction, the finished floor shall be constructed of wood or approved noncombustible materials. Openings through stage floors shall be equipped with tight-fitting, solid wood trap doors with approved safety locks. The room or space below the stage into which the traps or lifts open shall be completely enclosed by construction having at least the fire-resistance rating required for the stage floor, and such room or space shall not be used as a workshop or storage area. Storage shall not be deemed to include the location in this area of scenery or scenic elements used during a performance. However, no combustible material that has a flame spread rating greater than 25 or that has not been rendered flameproof in accordance with the rules of the Fire Commissioner may be stored in this location at any time.

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410.3.1.1 Stage height and area. Stage areas shall be measured to include the entire performance area and adjacent backstage and support areas not separated from the performance area by fire-resistance-rated construction. Stage height shall be measured from the lowest point on the stage floor to the highest point of the roof or floor deck above the stage.

410.3.2 Technical production areas: galleries, gridirons, catwalks and pinrails. Beams designed only for the attachment of portable or fixed theater equipment, gridirons, galleries and catwalks shall be constructed of non-combustible materials. Floors of fly galleries and catwalks shall be constructed of noncombustible material. These areas shall not be considered to be floors, stories, mezzanines or levels in applying this code.

410.3.3 Exterior stage doors. Where protection of openings is required, exterior exit doors shall be protected with fire door assemblies that comply with Section 716. Exterior openings that are located on the stage for means of egress or loading and unloading purposes, and that are likely to be open during occupancy of the theater, shall be constructed with vestibules to prevent air drafts into the auditorium.

410.3.4 Proscenium wall. Where the stage height is greater than 40 feet (12 192 mm), all portions of the stage shall be completely separated from the seating area by a proscenium wall with not less than a 2-hour fire-resistance rating extending continuously from the foundation to the roof.

††† **410.3.5 Proscenium curtain.** Where a proscenium wall is required to have a fire-resistance rating, the stage opening shall be provided with one of the following:

1. A fire curtain complying with NFPA 80; or
2. An approved stage water curtain and sprinklers complying with Section 410.7 and Section 903.3.1.1 of this code, and the following:
 - 2.1. A deluge valve actuated by a “rate of rise system” and “fixed temperature system” shall control the water curtain system;
 - 2.2. The heat actuating devices shall be located on not more than 10-foot (3048 mm) centers around the perimeter of the sprinklered area stage or as otherwise required for the type of device used to assure operation of the system;
 - 2.3. In addition to the automatic controls, manual-operating devices shall be located at the voice/alarm communication system required by Section 410.9, and adjacent to at least one exit from the stage. Such exit shall be remote from the voice/alarm communication system;
 - 2.4. All valves controlling deluge and sprinkler supplies on stages where the stage height is greater than 40 feet (12 192 mm) shall be provided with tamper switches wired to an annunciator panel located at the voice/alarm communication system required by Section 410.9;
 - 2.5. The operation of any section of the sprinkler system on stages where the stage height is greater than 40 feet (12 192 mm) or the operation of the deluge system shall activate the emergency ventilating equipment required in Section 410.3.7 and shall be provided with central station supervision in addition to the required local alarm; and
 - 2.6. The water flow alarm, tamper switches, the sprinkler system on stages where the stage height is greater than 40 feet (12 192 mm), and deluge system equipment shall be provided with central station supervision in addition to the required local alarm.

410.3.6 Scenery. All scenery or scenic elements shall be of noncombustible materials, or of materials having a flame-spread rating not exceeding 25, or of materials that have been rendered flameproof in compliance with the rules of the Fire Commissioner. Scenery and scenic elements not complying with the above requirements may be used only when expressly permitted by the Fire Department.

410.3.7 Stage ventilation. Emergency ventilation shall be provided for stages larger than 1,000 square feet (93 m²) in floor area, or with a stage height greater than 40 feet (12 192 mm). Such ventilation shall comply with Section 410.3.7.1 or 410.3.7.2.

410.3.7.1 Roof vents. Two or more vents constructed to open automatically by approved heat-activated devices and with an aggregate clear opening area of not less than 5 percent of the area of the stage shall be located near

the center and above the highest part of the stage area. Supplemental means shall be provided for manual operation of the ventilator. Curbs shall be provided as required for skylights in Section 2610.2. Vents shall be labeled.

410.3.7.2 Smoke control. Smoke control in accordance with Section 909 shall be provided to maintain the smoke layer interface not less than 6 feet (1829 mm) above the highest level of the assembly seating or above the top of the proscenium opening where a proscenium wall is provided in compliance with Section 410.3.4.

410.4 Platform construction. Permanent platforms shall be constructed of materials as required for the type of construction of the building in which the permanent platform is located. Permanent platforms are permitted to be constructed in accordance with the following:

1. The area below the platform shall be enclosed on all sides with solid construction.
2. Platforms constructed of untreated wood shall not exceed an area of 400 square feet (37 m²). Platforms constructed of fire-retardant-treated wood shall not exceed an area of 1,200 square feet (111 m²). Platforms constructed of noncombustible materials shall not be limited in area.
3. Where wood is used, the floor of the platform shall be at least 1-inch (25 mm) nominal thickness. Such floor shall be laid on a solid, noncombustible backing, or shall have all spaces between supporting members fire-blocked with noncombustible material.

Where the space beneath the permanent platform is used for storage or any other purpose other than equipment, wiring or plumbing, the floor construction shall not be less than 1-hour fire-resistant-rated construction. Where the space beneath the permanent platform is used only for equipment, wiring or plumbing, the underside of the permanent platform need not be protected.

410.4.1 Temporary platforms. Platforms installed for a period of not more than 30 days are permitted to be constructed of any materials permitted by this code. The space between the floor and the platform above shall only be used for plumbing and electrical wiring to platform equipment.

410.5 Dressing and appurtenant rooms. Dressing and appurtenant rooms shall comply with Sections 410.5.1 and 410.5.2.

410.5.1 Separation from stage. The stage shall be separated from dressing rooms, scene docks, property rooms, workshops, storerooms and compartments appurtenant to the stage and other parts of the building by fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both. The fire-resistance rating shall be not less than 2 hours for stage heights greater than 50 feet (15 240 mm) and not less than 1 hour for stage heights of 50 feet (15 240 mm) or less.

410.5.2 Separation from each other. Dressing rooms, scene docks, property rooms, workshops, storerooms and compartments appurtenant to the stage shall be separated from each other by not less than 1-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.

410.6 Means of egress. Except as modified or as provided for in this section, the provisions of Chapter 10 shall apply.

410.6.1 Stage exits. No fewer than one exit or exit access doorway shall be provided from each side of the stage and from each side of the space under the stage. Technical production areas shall comply with Section 410.6.3.

410.6.2 Stairway and ramp enclosure. Exit access stairways and ramps serving a stage or platform are not required to be enclosed. Exit access stairways and ramps serving technical production areas are not required to be enclosed.

410.6.3 Technical production areas. Technical production areas shall be provided with means of egress and means of escape in accordance with Sections 410.6.3.1 through 410.6.3.5.

410.6.3.1 Number of means of egress. No fewer than one means of egress shall be provided from technical production areas.

410.6.3.2 Exit access travel distance. The exit access travel distance shall be not greater than 300 feet (91 440 mm) for buildings without a sprinkler system and 400 feet (121 900 mm) for buildings equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1.

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410.6.3.3 Two means of egress. Where two means of egress are required, the common path of travel shall be not greater than 100 feet (30 480 mm).

Exception: A means of escape to a roof in place of a second means of egress is permitted.

410.6.3.4 Path of egress travel. The following exit access components are permitted where serving technical production areas:

1. Stairways.
2. Ramps.
3. Spiral stairways.
4. Catwalks.
5. Alternating tread devices.
6. Permanent ladders.

410.6.3.5 Width. The path of egress travel within and from technical support areas shall be not less than 22 inches (559 mm) in width.

410.7 Automatic sprinkler system. An automatic sprinkler system in accordance with Section 903.3.1.1 shall be provided in the following areas when located within a place of assembly:

1. Stages including those with stage height greater than 40 feet (12 192 mm);
2. Technical production areas over the stage including under the roof, above and below the gridiron, catwalks and galleries;
3. Dressing rooms, performer lounges, shops and storerooms accessory to stages or platforms, with live entertainment;
4. Platforms with live entertainment;
5. Cabarets; or
6. When openings are provided in the stage floor for stage lifts, trap doors or stairs, sprinklers spaced 5 feet (1524 mm) on centers shall be provided around the opening at the ceiling below the stage, and baffles at least 12 inches (305 mm) in depth shall be installed around the perimeter of the opening.

Exceptions:

1. Sprinklers are not required under stages or platforms less than 4 feet (1219 mm) in clear height that are utilized exclusively for storage of tables and chairs, provided the concealed space is separated from the adjacent spaces by Type X gypsum board not less than $\frac{5}{8}$ inch (15.9 mm) in thickness.
2. Sprinklers are not required within portable orchestra enclosures on stages.

410.8 Standpipes. Standpipe systems shall be provided in accordance with Section 905.

410.9 Voice/alarm communication system. Stages shall be provided with a voice/alarm communication system in accordance with Section 907.2.1.1.

410.10 Construction in seating areas. Construction in seating areas shall be in accordance with Sections 410.10.1 and 410.10.2.

410.10.1 Scenery and scenic elements in seating areas. Scenery or scenic elements may be placed in seating sections if such elements:

1. Are noncombustible, or of materials that have been rendered flameproof in accordance with the rules of the Fire Commissioner, or have a flame spread rating of 25 or less.
2. Are adequately braced or secured.
3. Do not obstruct the required visibility of, or paths of travel to, exit openings.

410.10.2 Platforms or runways in seating areas. Platforms or runways for performances, to accommodate the operation of cameras, electronic equipment, or motion picture projection machines not using carbon-arc or other

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light source that emits a gaseous discharge may be constructed in seating sections, provided such platforms or runways comply with the requirements of Section 410.4, except Item 1, and Section 410.9.

410.11 Stage lighting. Stage lights shall be placed such that they will not develop temperatures on the surface of any material that will cause that material to ignite, or smoke, or cause its flameproofing to deteriorate.

SECTION BC 411 SPECIAL AMUSEMENT BUILDINGS

411.1 General. Special amusement buildings having an occupant load of 75 or more shall comply with the requirements for the appropriate Group A occupancy and Sections 411.1 through 411.8. Special amusement buildings having an occupant load of less than 75 shall comply with the requirements for a Group B occupancy and Sections 411.1 through 411.8.

Exception: Special amusement buildings or portions thereof that are without walls or a roof and constructed to prevent the accumulation of smoke. All decorative material shall be noncombustible or composed of flame-retardant fabric.

411.2 Definition.‡ The following term is defined in Chapter 2:

SPECIAL AMUSEMENT BUILDING.

411.3 Automatic fire detection. Special amusement buildings shall be equipped with an automatic fire detection system in accordance with Section 907.

411.4 Automatic sprinkler system. Special amusement buildings shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1. Where the special amusement building is temporary, the sprinkler water supply shall be of an approved temporary means.

Exception: Automatic sprinklers are not required where the total floor area of a temporary special amusement building is less than 1,000 square feet (93 m²) and the exit access travel distance from any point to an exit is less than 50 feet (15 240 mm).

411.5 Alarm. Actuation of a single smoke detector, the automatic sprinkler system or other automatic fire detection device shall initiate a pre-signal system in accordance with NFPA 72 as modified by Appendix Q of this code at a constantly attended location from which the Fire Department shall be notified and live voice evacuation instructions shall be initiated using an emergency voice/alarm communications system in accordance with Section 907.5.2.2.

411.6 Emergency voice/alarm communications system. An emergency voice/alarm communications system shall be provided in accordance with Sections 907.2.12 and 907.5.2.2, which is also permitted to serve as a public address system and shall be audible throughout the entire special amusement building.

411.7 Exit marking. Exit signs shall be installed at the required exit or exit access doorways of amusement buildings in accordance with this section and Section 1013 of this code. Approved directional exit markings listed in accordance with UL 1994, shall also be provided. Where mirrors, mazes or other designs are utilized that disguise the path of egress travel such that they are not apparent, approved low-level exit signs that comply with Section 1013.5 of this code, and directional path markings shall be provided and located not more than 8 inches (203 mm) above the walking surface and on or near the path of egress travel. Such markings shall become visible in an emergency. The directional exit marking shall be activated by the automatic fire detection system and the automatic sprinkler system in accordance with Section 907.2.12.2.

411.8 Interior finish. The interior finish shall be Class A in accordance with Section 803.1.

SECTION BC 412 AIRCRAFT-RELATED OCCUPANCIES

412.1 General. Aircraft-related occupancies shall comply with the *New York City Fire Code* and Sections 412.1 through 412.8 of this code.

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412.2 Definitions. The following terms are defined in Chapter 2:

FIXED BASE OPERATOR (FBO).

HELIPORT.

HELISTOP.

RESIDENTIAL AIRCRAFT HANGAR.

TRANSIENT AIRCRAFT.

412.3 Airport traffic control towers. The provisions of Sections 412.3.1 through 412.3.9 shall apply to airport traffic control towers occupied only for the following uses:

1. Airport traffic control cab.
2. Electrical and mechanical equipment rooms.
3. Airport terminal radar and electronics rooms.
4. Office spaces incidental to the tower operation.
5. Lounges for employees, including sanitary facilities.

412.3.1 Type of construction. Airport traffic control towers shall be constructed to comply with the height limitations of Table 412.3.1.

**TABLE 412.3.1
HEIGHT LIMITATIONS FOR
AIRPORT TRAFFIC CONTROL TOWERS**

TYPE OF CONSTRUCTION	HEIGHT ^a (feet)
IA	Unlimited
IB	240
IIA	100
IIB	85
IIIA	65

For SI: 1 foot = 304.8 mm, 1 square foot = 0.0929 m².

a. Height to be measured from grade plane to cab floor.

412.3.2 Stairways. Stairways in airport traffic control towers shall be in accordance with Section 1011. Stairways in airport traffic control towers are not required to comply with Section 1011.12. Stairways shall be smokeproof enclosures complying with one of the alternatives provided in Section 909.20. The stairway shall be separated from elevators by a minimum distance of one-half of the diagonal of the area served measured in a straight line.

Exception: The stairway and elevator hoistway are permitted to be located in the same shaft enclosure without the minimum separation distance, provided they are separated from each other by a 4-hour fire barrier having no openings. Such stairway shall be pressurized to a minimum of 0.15 inch of water column (43 Pa) and a maximum of 0.35 inch of water column (101 Pa) in the shaft relative to the building with stairway doors closed.

412.3.3 Exit access. From observation levels, airport traffic control towers shall be permitted to have a single means of exit access for a distance of travel not greater than 100 feet (30 480 mm). Exit access stairways from the observation level need not be enclosed.

412.3.4 Number of exits. Not less than one exit stairway shall be permitted for airport traffic control towers of any height provided that the occupant load per floor is not greater than 15 and the area per floor does not exceed 1,500 square feet (140 m²).

412.3.4.1 Interior finish. Where an airport traffic control tower is provided with only one exit stairway, interior wall and ceiling finishes shall be either Class A or Class B.

412.3.5 Automatic fire detection systems. Airport traffic control towers shall be provided with an automatic fire detection system installed in accordance with Section 907.2.

412.3.6 Automatic sprinkler system. Where an occupied floor is located more than 35 feet (10 668 mm) above the lowest level of fire department vehicle access, airport traffic control towers shall be equipped with an automatic sprinkler system in accordance with Section 903.3.1.1.

412.3.7 Elevator protection. Wires or cables that provide normal or standby power, control signals, communication with the car, lighting, heating, air conditioning, ventilation and fire detecting systems to elevators shall be protected by construction having a fire-resistance rating of not less than 1 hour, or shall be circuit integrity cable having a fire-resistance rating of not less than 1 hour.

412.3.7.1 Elevators for occupant evacuation. Where provided in addition to an exit stairway, occupant evacuation elevators shall be in accordance with Section 3008.

412.3.8 Accessibility. Airport traffic control towers need not be accessible as specified in the provisions of Chapter 11.

412.3.9 Standby power and emergency power. A standby power system and an emergency power system that conform to Chapter 27 shall be provided in airport traffic control towers more than 65 feet (19 812 mm) in height. Standby power shall be provided to the following equipment:

1. Pressurization equipment and mechanical equipment provided in accordance with Section 412.3.2.
2. Elevator operating equipment.
3. Fire alarm and smoke detection systems.

412.3.9.1 Means of egress illumination. Emergency power shall be provided for means of egress illumination in accordance with Sections 1008.3.4 and 1008.3.5.

412.4 Aircraft hangars. Aircraft hangars shall be in accordance with Sections 412.4.1 through 412.4.6.

412.4.1 Exterior walls. Exterior walls located less than 30 feet (9144 mm) from lot lines or a public way shall have a fire-resistance rating not less than 2 hours.

412.4.2 Basements. Where hangars have basements, floors over basements shall be of Type IA construction and shall be made tight against seepage of water, oil or vapors. There shall be no opening or communication between basements and the hangar. Access to basements shall be from outside only.

412.4.3 Floor surface. Floors shall be graded and drained to prevent water or fuel from remaining on the floor. Floor drains shall discharge through an oil separator to the sewer or to an outside vented sump.

Exception: Aircraft hangars with individual lease spaces not exceeding 2,000 square feet (186 m²) each in which servicing, repairing or washing is not conducted and fuel is not dispensed shall have floors that are graded toward the door, but shall not require a separator.

412.4.4 Heating equipment. Heating equipment shall be placed in another room separated by 2-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both. Entrance shall be from the outside or by means of a vestibule providing a two-doorway separation.

Exceptions:

1. Unit heaters and vented infrared radiant heating equipment suspended not less than 10 feet (3048 mm) above the upper surface of wings or engine enclosures of the highest aircraft that are permitted to be housed in the hangar need not be located in a separate room provided they are mounted not less than 8 feet (2438 mm) above the floor in shops, offices and other sections of the hangar communicating with storage or service areas.
2. Entrance to the separated room shall be permitted by a single interior door provided the sources of ignition in the appliances are not less than 18 inches (457 mm) above the floor.

412.4.5 Finishing. The process of “doping,” involving use of a volatile flammable solvent, or of painting, shall be carried on in a separate detached building equipped with automatic fire-extinguishing equipment in accordance with Section 903.

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412.4.6 Fire suppression. Aircraft hangars shall be provided with a fire suppression system designed in accordance with NFPA 409, based upon the classification for the hangar given in Table 412.4.6 of this code.

Exception: Where a fixed base operator has separate repair facilities on site, Group II hangars operated by a fixed base operator used for storage of transient aircraft only shall have a fire suppression system, but the system is exempt from foam requirements.

TABLE 412.4.6
HANGAR FIRE SUPPRESSION REQUIREMENTS^{a, b, c}

MAXIMUM SINGLE FIRE AREA (square feet)	TYPE OF CONSTRUCTION								
	IA	IB	IIA	IIB	IIIA	IIIB	IV	VA ^d	VB ^d
> 40,001	Group I	Group I	Group I	Group I	Group I	Group I	Group I	Group I	Group I
40,000	Group II	Group II	Group II	Group II	Group II	Group II	Group II	Group II	Group II
30,000	Group III	Group II	Group II	Group II	Group II	Group II	Group II	Group II	Group II
20,000	Group III	Group III	Group II	Group II	Group II	Group II	Group II	Group II	Group II
15,000	Group III	Group III	Group III	Group II	Group III	Group II	Group III	Group II	Group II
12,000	Group III	Group III	Group III	Group III	Group III	Group III	Group III	Group II	Group II
8,000	Group III	Group III	Group III	Group III	Group III	Group III	Group III	Group III	Group II
5,000	Group III	Group III	Group III	Group III	Group III	Group III	Group III	Group II	Group II

For SI: 1 foot = 304.8 mm, 1 square foot = 0.0929m².

- a. Aircraft hangars with a door height greater than 28 feet shall be provided with fire suppression for a Group I hangar regardless of maximum fire area.
- b. Groups shall be as classified in accordance with NFPA 409.
- c. Membrane structures complying with Section 3102 shall be classified as a Group IV hangar.
- d. Construction Types VA and VB are not permitted in Fire Districts. See Tables 504.3, 504.4 and 506.2.

412.4.6.1 Hazardous operations. Any Group III aircraft hangar according to Table 412.4.6 that contains hazardous operations including, but not limited to, the following shall be provided with a Group I or II fire suppression system in accordance with NFPA 409 as applicable:

1. Doping.
2. Hot work including, but not limited to, welding, torch cutting and torch soldering.
3. Fuel transfer.
4. Fuel tank repair or maintenance not including defueled tanks in accordance with NFPA 409, inerted tanks or tanks that have never been fueled.
5. Spray finishing operations.
6. Total fuel capacity of all aircraft within the unsprinklered single fire area in excess of 1,600 gallons (6057 L).
7. Total fuel capacity of all aircraft within the maximum single fire area in excess of 7,500 gallons (28 390 L) for a hangar with an automatic sprinkler system in accordance with Section 903.3.1.1.

412.4.6.2 Separation of maximum single fire areas. Maximum single fire areas established in accordance with hangar classification and construction type in Table 412.4.6 shall be separated by 2-hour fire walls constructed in accordance with Section 706. In determining the maximum single fire area as set forth in Table 412.4.6,

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ancillary uses that are separated from aircraft servicing areas by a fire barrier of not less than 1 hour, constructed in accordance with Section 707, shall not be included in the area.

412.5 Residential aircraft hangars. Residential aircraft hangars shall comply with Sections 412.5.1 through 412.5.5.

412.5.1 Fire separation. A hangar shall not be attached to a dwelling unless separated by a fire barrier having a fire-resistance rating of not less than 1 hour. Such separation shall be continuous from the foundation to the underside of the roof and unpierced except for doors leading to the dwelling unit. Doors into the dwelling unit shall be equipped with self-closing devices and conform to the requirements of Section 716 with a noncombustible raised sill not less than 4 inches (102 mm) in height. Openings from a hangar directly into a room used for sleeping purposes shall not be permitted.

412.5.2 Egress. A hangar shall provide two means of egress. One of the doors into the dwelling shall be considered as meeting only one of the two means of egress.

412.5.3 Smoke alarms. Smoke alarms shall be provided within the hangar in accordance with Section 907.2.11.

412.5.4 Independent systems. Electrical, mechanical and plumbing drain, waste and vent (DWV) systems installed within the hangar shall be independent of the systems installed within the dwelling. Building sewer lines shall be permitted to be connected outside the structures.

Exception: Smoke detector wiring and feed for electrical subpanels in the hangar.

412.5.5 Height and area limits. Residential aircraft hangars shall be not greater than 2,000 square feet (186 m²) in area and 20 feet (6096 mm) in building height.

412.6 Aircraft paint hangars. Aircraft painting operations where flammable liquids are used in excess of the maximum allowable quantities per control area listed in Table 307.1(1) shall be conducted in an aircraft paint hangar that complies with the provisions of Sections 412.6.1 through 412.6.6.

412.6.1 Occupancy group. Aircraft paint hangars shall be classified as Group H-2. Aircraft paint hangars shall comply with the applicable requirements of the *New York City Fire Code* and this code for such occupancy.

412.6.2 Construction. The aircraft paint hangar shall be of Type I or II construction.

412.6.3 Operations. Only those flammable liquids necessary for painting operations shall be permitted in quantities less than the maximum allowable quantities per control area in Table 307.1(1). Spray equipment cleaning operations shall be conducted in a liquid use, dispensing and mixing room.

412.6.4 Storage. Storage of flammable liquids shall be in a liquid storage room.

412.6.5 Fire suppression. Aircraft paint hangars shall be provided with fire suppression as required by NFPA 409.

412.6.6 Ventilation. Aircraft paint hangars shall be provided with ventilation as required in the *New York City Mechanical Code*.

412.7 Aircraft manufacturing facilities. In buildings used for the manufacturing of aircraft, exit access travel distances indicated in Section 1017.1 shall be increased in accordance with the following:

1. The building shall be of Type I or II construction.
2. Exit access travel distance shall not exceed the distances given in Table 412.7.

412.7.1 Ancillary areas. Rooms, areas and spaces ancillary to the primary manufacturing area shall be permitted to egress through such area having a minimum height as indicated in Table 412.7. Exit access travel distance within the ancillary room, area or space shall not exceed that indicated in Table 1017.2 based on the occupancy classification of that ancillary area. Total exit access travel distance shall not exceed that indicated in Table 412.7.

**TABLE 412.7
AIRCRAFT MANUFACTURING EXIT ACCESS TRAVEL DISTANCE**

HEIGHT (feet) ^b	MANUFACTURING AREA (sq. ft.) ^a					
	> 150,000	> 200,000	> 250,000	> 500,000	> 750,000	>1,000,000

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> 25	400	450	500	500	500	500
> 50	400	500	600	700	700	700
> 75	400	500	700	850	1,000	1,000
> 100	400	500	750	1,000	1,250	1,500

For SI: 1 foot = 304.8 mm.

- Contiguous floor area of the aircraft manufacturing facility having the indicated height.
- Minimum height from finished floor to bottom of ceiling or roof slab or deck.

412.8 Heliports and helistops. Heliports and helistops shall be permitted to be erected on buildings or other locations where they are constructed in accordance with Sections 412.8.1 through 412.8.5.

412.8.1 Size. The landing area for helicopters less than 3,500 pounds (1588 kg) shall be not less than 20 feet (6096 mm) in length and width. The landing area shall be surrounded on all sides by a clear area having a minimum average width at roof level of 15 feet (4572 mm) but with no width less than 5 feet (1524 mm).

412.8.2 Design. Helicopter landing areas and the supports thereof on the roof of a building shall be noncombustible construction. Landing areas shall be designed to confine any flammable liquid spillage to the landing area itself and provisions shall be made to drain such spillage away from any exit or stairway serving the helicopter landing area or from a structure housing such exit or stairway. For structural design requirements, see Section 1607.6.

412.8.3 Means of egress. The means of egress from heliports and helistops shall comply with the provisions of Chapter 10. Landing areas located on buildings or structures shall have two or more means of egress. For landing areas less than 60 feet (18 288 mm) in length or less than 2,000 square feet (186 m²) in area, the second means of egress is permitted to be a fire escape, alternating tread device or ladder leading to the floor below.

412.8.4 Rooftop heliports and helistops. Rooftop heliports and helistops shall comply with NFPA 418 and the *New York City Fire Code*.

412.8.5 Standpipe system. In buildings equipped with a standpipe system, the standpipe shall extend to the roof level in accordance with Section 905.3.6.

SECTION BC 413 COMBUSTIBLE STORAGE

413.1 General. Storage of combustible materials in attics, under-floor spaces, concealed spaces, and below-grade storage rooms shall comply with this section.

413.2 Attic, under-floor and concealed spaces. Attic, under-floor and concealed spaces used for storage of combustible materials shall be protected on the storage side as required for 1-hour fire-resistance-rated construction. Openings shall be protected by assemblies that are self-closing and are of noncombustible construction or solid wood core not less than 1³/₄ inch (45 mm) in thickness.

Exception: Neither fire-resistance-rated construction nor opening protectives are required in any of the following locations:

- Areas protected by approved automatic sprinkler systems.
- Group R-3 and U occupancies.

413.3 High-piled stock or rack storage. High-piled stock or rack storage in any occupancy group shall comply with the *New York City Fire Code*, including the requirement for permit approval.

SECTION BC 414 HAZARDOUS MATERIALS

414.1 General. The provisions of Sections 414.1 through 414.6 shall apply to buildings and structures occupied for the manufacturing, processing, dispensing, use or storage of hazardous materials.

414.1.1 Other provisions. Buildings and structures with an occupancy in Group H shall comply with this section and the applicable provisions of the *New York City Fire Code* and Section 415 of this code, including, but not limited to, location, installation and mechanical provisions.

414.1.2 Materials. The safe design of hazardous material occupancies is material dependent. Individual material requirements are also found in the *New York City Mechanical Code*, the *New York City Fire Code*, and Sections 307 and 415 of this code.

414.1.2.1 Aerosols. Level 2 and 3 aerosol products shall be stored and displayed in accordance with the *New York City Fire Code*. See the *New York City Fire Code* and Section 311.2 of this code for occupancy group requirements.

414.1.3 Information required. A report shall be submitted to the Fire Department identifying the maximum expected quantities of hazardous materials to be stored, used in a closed system and used in an open system, and subdivided to separately address hazardous material classification categories based on Tables 307.1(1) and 307.1(2). The methods of protection from such hazards, including but not limited to control areas, fire protection systems and Group H occupancies shall be indicated in the report and on the construction documents. The opinion and report shall be prepared by a qualified person, firm or corporation approved by the Fire Commissioner and provided without charge to the Fire Department.

For buildings and structures with an occupancy in Group H, separate floor plans shall be submitted identifying the locations of anticipated contents and processes so as to reflect the nature of each occupied portion of every building and structure.

414.2 Control areas. Control areas shall comply with the *New York City Fire Code* and Sections 414.2.1 through 414.2.5 of this code.

414.2.1 Construction requirements. Control areas shall be separated from each other by fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.

414.2.2 Percentage of maximum allowable quantities. The percentage of maximum allowable quantities of hazardous materials per control area permitted at each floor level within a building shall be in accordance with Table 414.2.2.

**TABLE 414.2.2
DESIGN AND NUMBER OF CONTROL AREAS**

FLOOR LEVEL		PERCENTAGE OF THE MAXIMUM ALLOWABLE QUANTITY PER CONTROL AREA ^a	NUMBER OF CONTROL AREAS PER FLOOR ^b	FIRE-RESISTANCE RATING FOR FIRE BARRIERS IN HOURS ^b
Above grade	Higher than 9	5	1	2
	7-9	5	2	2
	6	12.5	2	2
	5	12.5	2	2
	4	12.5	2	2
	3	50	2	1
	2	75	3	1
	1	100	4	1
Below grade	1	75	3	1
	2	50	2	1
	Lower than 2	Not Allowed	Not Allowed	Not Allowed

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- a. Percentages shall be of the maximum allowable quantity per control area shown in Tables 307.1(1) and 307.1(2), with all increases allowed in the notes to those tables.
- b. Separation shall include fire barriers and horizontal assemblies as necessary to provide separation from other portions of the building.

414.2.3 Number. The maximum number of control areas within a building shall be in accordance with Table 414.2.2.

414.2.4 Fire-resistance-rating requirements. The required fire-resistance rating for fire barriers shall be in accordance with Table 414.2.2. The floor assembly of the control area and the construction supporting the floor of the control area shall have a fire-resistance rating of not less than 2 hours.

Exception: The floor assembly of the control area and the construction supporting the floor of the control area are allowed to be 1-hour fire-resistance rated in buildings of Types IIA, IIIA and VA construction, provided that both of the following conditions exist:

1. The building is equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1; and
2. The building is three or fewer stories above grade plane.

414.2.5 Hazardous material in Group M display and storage areas and in Group S storage areas. The aggregate quantity of nonflammable solid and nonflammable or noncombustible liquid hazardous materials permitted within a single control area of a Group M display and storage area, a Group S storage area or an outdoor control area is permitted to exceed the maximum allowable quantities per control area specified in Tables 307.1(1) and 307.1(2) without classifying the building or use as a Group H occupancy, provided that the materials are displayed and stored in accordance with the *New York City Fire Code* and quantities do not exceed the maximum allowable specified in Table 414.2.5(1) of this code.

In Group M occupancy wholesale and retail sales uses, indoor storage of flammable and combustible liquids shall not exceed the maximum allowable quantities per control area as indicated in Table 414.2.5(2) of this code, provided that the materials are displayed and stored in accordance with the *New York City Fire Code*.

The maximum quantity of aerosol products in Group M occupancy retail display areas, storage areas adjacent to retail display areas and retail storage areas shall be in accordance with the *New York City Fire Code*.

TABLE 414.2.5(1)
MAXIMUM ALLOWABLE QUANTITY PER INDOOR AND OUTDOOR CONTROL AREA IN GROUP M AND S OCCUPANCIES
NONFLAMMABLE SOLIDS AND NONFLAMMABLE AND NONCOMBUSTIBLE LIQUIDS^{d, e, f}

CONDITION		MAXIMUM ALLOWABLE QUANTITY PER CONTROL AREA	
Material ^a	Class	Solids pounds	Liquids gallons
A. Health-hazard materials—nonflammable and noncombustible solids and liquids			
Corrosives ^{b, c}	Not Applicable	9,750	975
Highly toxics	Not Applicable	20 ^{b, c}	2 ^{b, c}
Toxics ^{b, c}	Not Applicable	1,000	100
B. Physical-hazard materials—nonflammable and noncombustible solids and liquids			
1. Oxidizers ^{b, c}	4	Not Allowed	Not Allowed
	3	1,150 ^g	115
	2	2,250 ^h	225
	1	18,000 ^{i, j}	18,000 ^{i, j}
	4	Not Allowed	Not Allowed

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2. Unstable (reactives) ^{b, c}	3	550	55
	2	1,150	115
	1	Not Limited	Not Limited
3. Water reactives	3 ^{b, c}	550	55
	2 ^{b, c}	1,150	115
	1	Not Limited	Not Limited

For SI: 1 pound = 0.454 kg, 1 gallon = 3.785 L.

- a. Hazard categories are as specified in the *New York City Fire Code*.
- b. Maximum allowable quantities shall be increased 100 percent in buildings that are sprinklered in accordance with Section 903.3.1.1. When Note c also applies, the increase for both notes shall be applied cumulatively.
- c. Maximum allowable quantities shall be increased 100 percent when stored in approved storage cabinets, in accordance with the *New York City Fire Code*. When Note b also applies, the increase for both notes shall be applied cumulatively.
- d. See Table 414.2.2 for design and number of control areas.
- e. Allowable quantities for other hazardous material categories shall be in accordance with Section 307.
- f. Maximum quantities shall be increased 100 percent in outdoor control areas.
- g. Maximum amounts shall be increased to 2,250 pounds when individual packages are in the original sealed containers from the manufacturer or packager and do not exceed 10 pounds each.
- h. Maximum amounts shall be increased to 4,500 pounds when individual packages are in the original sealed containers from the manufacturer or packager and do not exceed 10 pounds each.
- i. The permitted quantities shall not be limited in a building equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1.
- j. Quantities are unlimited in an outdoor control area.

TABLE 414.2.5(2)
MAXIMUM ALLOWABLE QUANTITY OF FLAMMABLE AND COMBUSTIBLE LIQUIDS
IN WHOLESALE AND RETAIL SALES OCCUPANCIES PER CONTROL AREA^a

TYPE OF LIQUID	MAXIMUM ALLOWABLE QUANTITY PER CONTROL AREA (gallons)		
	Sprinklered in accordance with note b densities and arrangements	Sprinklered in accordance with Tables 5704.3.6.3(4) through 5704.3.6.3(8) and 5704.3.7.5.1 of the <i>New York City Fire Code</i>	Nonsprinklered
Class IA	60	60	30
Class IB, IC, II and IIIA	7,500 ^e	15,000 ^e	1,600
Class IIIB	Unlimited	Unlimited	13,200

For SI: 1 foot = 304.8 mm, 1 square foot = 0.0929m², 1 gallon = 3.785 L, 1 gallon per minute per square foot = 40.75 L/min/m².

- a. Control areas shall be separated from each other by not less than a 1-hour fire barrier wall.
- b. To be considered as sprinklered, a building shall be equipped throughout with an approved automatic sprinkler system with a design providing minimum densities as follows:
 1. For uncartoned commodities on shelves 6 feet or less in height where the ceiling height does not exceed 18 feet, quantities are those permitted with a minimum sprinkler design density of Ordinary Hazard Group 2.
 2. For cartoned, palletized or racked commodities where storage is 4 feet 6 inches or less in height and where the ceiling height does not exceed 18 feet, quantities are those permitted with a minimum sprinkler design density of 0.21 gallon per minute per square foot over the most remote 1,500-square-foot area.
- c. Where wholesale and retail sales or storage areas exceed 50,000 square feet in area, the maximum allowable quantities are allowed to be increased by 2 percent for each 1,000 square feet of area in excess of 50,000 square feet, up to a maximum of 100 percent of the table amounts. A control area separation is not required. The cumulative amounts, including amounts attained by having an additional control area, shall not exceed 30,000 gallons.

414.2.5.1 Below-grade storage of flammable or combustible mixtures accessory to Group M establishments. Class I liquids shall not be stored in below-grade locations. Storage of Class II and Class III liquids, accessory to retail stores, and partly or completely below grade shall be permitted, provided that such below-grade area is equipped throughout with an automatic sprinkler system in accordance with Section 903.3.1.1 and

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enclosed by fire barriers having a fire-resistance rating of at least 2 hours, constructed in accordance with the construction type of the building. Openings shall be protected by self-closing assemblies having a fire-resistance rating of at least 1½ hours.

Exception: Class IIIB liquids may be stored in below-grade locations that are not provided throughout with an automatic sprinkler system when such liquids are stored in a room that is separated, vertically and horizontally, from surrounding spaces by fire barriers of not less than 2-hour fire-resistance rating and such room is provided with an automatic sprinkler system.

414.3 Ventilation. Rooms, areas or spaces in which explosive, corrosive, combustible, flammable or highly toxic dusts, mists, fumes, vapors or gases are or may be emitted due to the processing, use, handling or storage of materials shall be mechanically ventilated where required by the *New York City Fire Code* or the *New York City Mechanical Code*.

Emissions generated at workstations shall be confined to the area in which they are generated as specified in the *New York City Fire Code* and the *New York City Mechanical Code*.

414.4 Hazardous material systems. Systems involving hazardous materials shall be suitable for the intended application. Controls shall be designed to prevent materials from entering or leaving process or reaction systems at other than the intended time, rate or path. Automatic controls, where provided, shall be designed to be fail safe.

414.5 Inside storage, dispensing and use. The inside storage, dispensing and use of hazardous materials shall be in accordance with the *New York City Fire Code* and Sections 414.5.1 through 414.5.4 of this code.

414.5.1 Explosion control. Explosion control shall be provided in accordance with the *New York City Fire Code* as required by Table 414.5.1 where quantities of hazardous materials specified in that table exceed the maximum allowable quantities in Table 307.1(1) or where a structure, room or space is occupied for purposes involving explosion hazards as required by the *New York City Fire Code* or Section 415 of this code.

TABLE 414.5.1
EXPLOSION CONTROL REQUIREMENTS^{a, h}

MATERIAL	CLASS	EXPLOSION CONTROL METHODS	
		Barricade construction	Explosion (deflagration) venting or explosion (deflagration) prevention systems ^b
HAZARD CATEGORY			
Combustible dusts ^c	—	Not Required	Required
Cryogenic flammables	—	Not Required	Required
Explosives	Division 1.1	Required	Not Required
	Division 1.2	Required	Not Required
	Division 1.3	Not Required	Required
	Division 1.4	Not Required	Required
	Division 1.5	Required	Not Required
	Division 1.6	Required	Not Required
Flammable gas	Gaseous	Not Required	Required
	Liquefied	Not Required	Required
Flammable liquid	IA ^d	Not Required	Required
	IB ^e	Not Required	Required
Organic peroxides	U	Required	Not Permitted
	I	Required	Not Permitted
Oxidizer liquids and solids	4	Required	Not Permitted
Pyrophoric gas	—	Not Required	Required

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Unstable (reactive)	4 3 Detonable 3 Nondetonable	Required Required Not Required	Not Permitted Not Permitted Required
Water-reactive liquids and solids	3 2 ^g	Not Required Not Required	Required Required
SPECIAL USES			
Grain processing	—	Not Required	Required
Where explosion hazards exist ^f	Detonation Deflagration	Required Not Required	Not Permitted Required

a. See Section 414.1.3.

b. See the *New York City Fire Code*.

c. As generated during manufacturing or processing.

d. Storage or use.

e. In open use or dispensing.

f. Rooms containing dispensing and use of hazardous materials when an explosive environment can occur because of the characteristics or nature of the hazardous materials or as a result of the dispensing or use process.

g. A method of explosion control shall be provided when Class 2 water-reactive materials can form potentially explosive mixtures.

h. Explosion venting is not required for Group H-5 fabrication areas complying with the *New York City Fire Code* and Section 415.11.1 of this code.

414.5.2 Emergency or standby power. Where required by the *New York City Fire Code* or this code, mechanical ventilation, treatment systems, temperature control, alarm, detection or other electrically operated systems shall be provided with emergency or standby power in accordance with Section 2702 of this code. For storage and use areas for highly toxic or toxic materials, see the *New York City Fire Code*.

414.5.2.1 Exempt applications. Emergency or standby power is not required for the mechanical ventilation systems provided for any of the following:

1. Storage of Class IB and IC flammable and combustible liquids in closed containers not exceeding 6.5 gallons (25 L) capacity.
2. Storage of Class 1 and 2 oxidizers.
3. Storage of Class II, III, IV and V organic peroxides.
4. Storage of asphyxiant, irritant and radioactive gases.

414.5.2.2 Fail-safe engineered systems. Standby power for mechanical ventilation, treatment systems and temperature control systems shall not be required where an approved fail-safe engineered system is installed.

414.5.3 Spill control, drainage and containment. Rooms, buildings or areas occupied for the storage of solid and liquid hazardous materials shall be provided with a means to control spillage and to contain or drain off spillage and fire protection water discharged in the storage area where required in the *New York City Fire Code*. The methods of spill control shall be in accordance with the *New York City Fire Code*.

414.5.4 Storage of medical gases. When required by the *New York City Fire Code*, a one-hour room shall be provided for the storage of medical gases inside of buildings, with the protected openings and ventilation provisions as required by the *New York City Fire Code*.

414.6 Outdoor storage, dispensing and use. The outdoor storage, dispensing and use of hazardous materials shall be in accordance with the *New York City Fire Code*.

414.6.1 Weather protection. Where weather protection is provided for sheltering outdoor hazardous material storage or use areas, such areas shall be considered outdoor storage or use when the weather protection structure complies with Sections 414.6.1.1 through 414.6.1.3.

414.6.1.1 Walls. Walls shall not obstruct more than one side of the structure.

Exception: Walls shall be permitted to obstruct portions of multiple sides of the structure, provided that the obstructed area is not greater than 25 percent of the structure's perimeter.

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414.6.1.2 Separation distance. The distance from the structure to buildings, lot lines, public ways or means of egress to a public way shall be not less than the distance required for an outside hazardous material storage or use area without weather protection.

414.6.1.3 Noncombustible construction. The overhead structure shall be of approved noncombustible construction with a maximum area of 1,500 square feet (140 m²).

Exception: The maximum area is permitted to be increased as provided by Section 506.

SECTION BC 415 GROUPS H-1, H-2, H-3, H-4 AND H-5

415.1 Scope. The provisions of Sections 415.1 through 415.11 shall apply to the storage and use of hazardous materials in excess of the maximum allowable quantities per control area listed in Section 307.1. Buildings and structures with an occupancy in Group H shall also comply with the applicable provisions of the *New York City Fire Code* and Section 414 of this code including, but not limited to, location and installation provisions.

415.2 Definitions. The following terms are defined in Chapter 2:

CONTINUOUS GAS DETECTION SYSTEM.

DETACHED BUILDING.

EMERGENCY CONTROL STATION.

EXHAUSTED ENCLOSURE.

FABRICATION AREA.

FLAMMABLE VAPORS OR FUMES.

GAS CABINET.

GAS ROOM.

HAZARDOUS PRODUCTION MATERIAL (HPM).

HPM FLAMMABLE LIQUID.

HPM ROOM.

IMMEDIATELY DANGEROUS TO LIFE AND HEALTH (IDLH).

LIQUID.

LIQUID STORAGE ROOM.

LIQUID USE, DISPENSING AND MIXING ROOM.

LOWER FLAMMABLE LIMIT (LFL).

NORMAL TEMPERATURE AND PRESSURE (NTP).

**PHYSIOLOGICAL WARNING
THRESHOLD LEVEL.**

SERVICE CORRIDOR.

SOLID.

STORAGE, HAZARDOUS MATERIALS.

USE (MATERIAL).

WORKSTATION.

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415.3 Automatic fire detection systems. Group H occupancies shall be provided with an automatic fire detection system in accordance with Section 907.2.

415.4 Automatic sprinkler system. Group H occupancies shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.2.5.

415.5 Emergency alarms. Emergency alarms for the detection and notification of an emergency condition in Group H occupancies shall be provided as set forth herein.

415.5.1 Storage. An approved manual emergency alarm system shall be provided in buildings, rooms or areas used for storage of hazardous materials. Emergency alarm-initiating devices shall be installed outside of each interior exit or exit access door of storage buildings, rooms or areas. Activation of an emergency alarm-initiating device shall sound a local alarm to alert occupants of an emergency situation involving hazardous materials.

415.5.2 Dispensing, use and handling. Where hazardous materials having a hazard ranking of 3 or 4 in accordance with NFPA 704 are transported through corridors, interior exit stairways or ramps, or exit passageways, there shall be an emergency telephone system, a local manual alarm station or an approved alarm-initiating device at not more than 150-foot (45 720 mm) intervals and at each exit and exit access doorway throughout the transport route. The signal shall be relayed to an approved central, proprietary or remote station service or constantly attended on-site location and shall initiate a local audible alarm.

415.5.3 Supervision. Emergency alarm systems shall be supervised by an approved central, proprietary or remote station service or shall initiate an audible and visual signal at a constantly attended on-site location.

415.5.4 Emergency alarm systems. Emergency alarm systems shall be provided with emergency power in accordance with Section 2702.

415.6 Fire separation distance. Group H occupancies shall be located on property in accordance with the other provisions of this chapter. In Groups H-2 and H-3, not less than 25 percent of the perimeter wall of the occupancy shall be an exterior wall.

Exceptions:

1. Liquid use, dispensing and mixing rooms having a floor area of not more than 500 square feet (46.5 m²) need not be located on the outer perimeter of the building where they are in accordance with the *New York City Fire Code* and NFPA 30.
2. Liquid storage rooms having a floor area of not more than 1,000 square feet (93 m²) need not be located on the outer perimeter where they are in accordance with the *New York City Fire Code* and NFPA 30.
3. Spray paint booths that comply with the *New York City Fire Code* need not be located on the outer perimeter.

415.6.1 Group H occupancy minimum fire separation distance. Regardless of any other provisions, buildings containing Group H occupancies shall be set back to the minimum fire separation distance as set forth in Sections 415.6.1.1 through 415.6.1.4. Distances shall be measured from the walls enclosing the occupancy to lot lines, including those on a public way. Distances to assumed lot lines established for the purpose of determining exterior wall and opening protection are not to be used to establish the minimum fire separation distance for buildings on sites where explosives are manufactured or used when separation is provided in accordance with the quantity distance tables specified for explosive materials in the *New York City Fire Code*.

**TABLE 415.6.1
MINIMUM SEPARATION DISTANCES FOR BUILDINGS CONTAINING EXPLOSIVE MATERIALS^f**

QUANTITY OF EXPLOSIVE MATERIAL ^{a, d}		MINIMUM DISTANCE (feet)		
		Lot lines ^b and inhabited buildings ^c		Separation of magazines ^e
Pounds over	Pounds not over	Barricaded	Unbarricaded	
0	5	70	140	12
5	10	90	180	16
10	20	110	220	20
20	30	125	250	22

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30	40	140	280	24
40	50	150	300	28
50	75	170	340	30
75	100	190	380	32
100	125	200	400	36
125	150	215	430	38
150	200	235	470	42
200	250	255	510	46
250	300	270	540	48
300	400	295	590	54
400	500	320	640	58
500	600	340	680	62
600	700	355	710	64
700	800	375	750	66
800	900	390	780	70
900	1,000	400	800	72

For SI: 1 pound = 0.454 kg, 1 foot = 304.8 mm.

- The number of pounds of explosives listed is the number of pounds of trinitrotoluene (TNT) or the equivalent pounds of other explosive.
- The distance listed is the distance to lot line, including lot lines at public ways.
- For the purpose of this table, an inhabited building is any building on the same tax lot that is regularly occupied by people. Where two or more buildings containing explosives or magazines are located on the tax lot, each building or magazine shall comply with the minimum distances specified from inhabited buildings and, in addition, they shall be separated from each other by not less than the distance shown for "Separation of magazines," except that the quantity of explosive materials contained in detonator buildings or magazines shall govern in regard to the spacing of said detonator buildings or magazines from buildings or magazines containing other explosive materials. If any two or more buildings or magazines are separated from each other by less than the specified "Separation of magazines" distances, then such two or more buildings or magazines, as a group, shall be considered as one building or magazine, and the total quantity of explosive materials stored in such group shall be treated as if the explosive were in a single building or magazine located on the site of any building or magazine of the group, and shall comply with the minimum distance specified from other magazines or inhabited buildings.
- Storage of explosives in a quantity exceeding 1,000 pounds in one magazine is prohibited.
- Magazine is a building or structure, other than an operating building, approved for storage of explosive materials. Portable or mobile magazines not exceeding 120 square feet (11 m²) in area need not comply with the requirements of this code, however, all magazines shall comply with the *New York City Fire Code*.
- This table shall not apply to the temporary storage incidental to transportation.

415.6.1.1 Group H-1. Group H-1 occupancies shall be set back not less than 75 feet (22 860 mm) and not less than required by the *New York City Fire Code*.

Exceptions:

- Fireworks manufacturing buildings separated in accordance with NFPA 1124.
- Buildings containing the following materials when separated in accordance with Table 415.6.1 of this code:
 - Organic peroxides, unclassified detonable.
 - Unstable reactive materials Class 4.
 - Unstable reactive materials, Class 3 detonable.
 - Detonable pyrophoric materials.

415.6.1.2 Group H-2. Group H-2 occupancies shall be set back not less than 30 feet (9144 mm) where the area of the occupancy exceeds 1,000 square feet (93 m²), and it is not required to be located in a detached building.

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415.6.1.3 Groups H-2 and H-3. Group H-2 and H-3 occupancies shall be set back not less than 50 feet (15 240 mm) where a detached building is required (see Table 415.6.2).

415.6.1.4 Explosive materials. Group H-2 and H-3 occupancies containing materials with explosive characteristics shall be separated as required by the *New York City Fire Code*. Where separations are not specified, the distances required shall not be less than the distances required by Table 415.6.1.

415.6.2 Detached buildings for Group H-1, H-2 or H-3 occupancy. The storage or use of hazardous materials in excess of those amounts listed in Table 415.6.2 shall be in accordance with the applicable provisions of Sections 415.7 and 415.8.

**TABLE 415.6.2
DETACHED BUILDING REQUIRED**

A DETACHED BUILDING IS REQUIRED WHEN THE QUANTITY OF MATERIAL EXCEEDS THAT LISTED HEREIN			
Material	Class	Solids and Liquids (tons) ^{a,b}	Gases (cubic feet) ^{a,b}
Explosives	Division 1.1 Division 1.2 Division 1.3 Division 1.4 Division 1.4 ^c Division 1.5 Division 1.6	Maximum Allowable Quantity Maximum Allowable Quantity Maximum Allowable Quantity Maximum Allowable Quantity 1 Maximum Allowable Quantity Maximum Allowable Quantity	Not Applicable
Oxidizers	Class 4	Maximum Allowable Quantity	Maximum Allowable Quantity
Unstable (reactives) detonable	Class 3 or 4	Maximum Allowable Quantity	Maximum Allowable Quantity
Oxidizer, liquids and solids	Class 3 Class 2	1,200 2,000	Not Applicable Not Applicable
Organic peroxides	Detonable Class I Class II Class III	Maximum Allowable Quantity Maximum Allowable Quantity 25 50	Not Applicable Not Applicable Not Applicable Not Applicable
Unstable (reactives) nondetonable	Class 3 Class 2	1 25	2,000 10,000
Water reactives	Class 3 Class 2	1 25	Not Applicable Not Applicable
Pyrophoric gases	Not Applicable	Not Applicable	2,000

For SI: 1 ton = 906 kg, 1 cubic foot = 0.02832 m³, 1 pound = 0.454 kg.

- a. For materials that are detonable, the distance to other buildings or lot lines shall be as specified in Table 415.6.1 based on trinitrotoluene (TNT) equivalence of the material. For materials classified as explosives, see the *New York City Fire Code*. Unclassified detonable organic peroxides, detonable pyrophoric materials, detonable unstable (reactive) materials and detonable water-reactive materials shall be considered as explosives for the purposes of storage.
- b. "Maximum Allowable Quantity" means the maximum allowable quantity per control area set forth in Table 307.1(1).
- c. Limited to Division 1.4 materials and articles, including articles packaged for shipment, that are not regulated as an explosive under Bureau of Alcohol, Tobacco Firearms and Explosives (BATF) regulations or unpackaged articles used in process operations that do not propagate a detonation or deflagration between articles, provided the net explosive weight of individual articles does not exceed 1 pound.

415.6.2.1 Wall and opening protection. Where a detached building is required by Table 415.6.2, there are no requirements for wall and opening protection based on fire separation distance.

415.7 Special provisions for Group H-1 occupancies. Group H-1 occupancies shall be in detached buildings used for no other purpose, shall not exceed one story in height and be without basements, crawl spaces or other under-floor spaces. Roofs shall be of lightweight construction with suitable thermal insulation to prevent sensitive material from reaching its decomposition temperature. Group H-1 occupancies containing materials that are in themselves both

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physical and health hazards in quantities exceeding the maximum allowable quantities per control area in Table 307.1(2) shall comply with requirements for both Group H-1 and H-4 occupancies.

415.7.1 Floors in storage rooms. Floors in storage areas for organic peroxides, pyrophoric materials and unstable (reactive) materials shall be of liquid-tight, noncombustible construction.

415.8 Special provisions for Group H-2 and H-3 occupancies. Group H-2 and H-3 occupancies containing quantities of hazardous materials in excess of those set forth in Table 415.6.2 shall be in detached buildings used for manufacturing, processing, dispensing, use or storage of hazardous materials. Materials listed for Group H-1 occupancies in Section 307.3 are permitted to be located within Group H-2 or H-3 detached buildings provided the amount of materials per control area do not exceed the maximum allowed quantity specified in Table 307.1(1).

415.8.1 Multiple hazards. Group H-2 or H-3 occupancies containing materials that are in themselves both physical and health hazards in quantities exceeding the maximum allowable quantities per control area in Table 307.1(2) shall comply with requirements for Group H-2, H-3 or H-4 occupancies as applicable.

415.8.2 Separation of incompatible materials. Hazardous materials other than those listed in Table 415.6.2 shall be allowed in manufacturing, processing, dispensing, use or storage areas when separated from incompatible materials in accordance with the provisions of the *New York City Fire Code*.

415.8.3 Water reactives. Group H-2 and H-3 occupancies containing water-reactive materials shall be resistant to water penetration. Piping for conveying liquids shall not be over or through areas containing water reactives, unless isolated by approved liquid-tight construction.

Exception: Fire protection piping shall be permitted over or through areas containing water reactives without isolating it with liquid-tight construction.

415.8.4 Floors in storage rooms. Floors in storage areas for organic peroxides, oxidizers, pyrophoric materials, unstable (reactive) materials and water-reactive solids and liquids shall be of liquid-tight, noncombustible construction.

415.8.5 Waterproof room. Rooms or areas used for the storage of water-reactive solids and liquids shall be constructed in a manner that resists the penetration of water through the use of waterproof materials. Piping carrying water for other than approved automatic sprinkler systems shall not be within such rooms or areas.

415.9 Group H-2. Occupancies in Group H-2 shall be constructed in accordance with the *New York City Fire Code* and Sections 415.9.1 through 415.9.3 of this code.

415.9.1 Flammable and combustible liquids. The storage, handling, processing and transporting of flammable and combustible liquids in Group H-2 and H-3 occupancies shall be in accordance with the *New York City Mechanical Code*, the *New York City Fire Code*, and Sections 415.9.1.1 through 415.9.1.9 of this code.

415.9.1.1 Mixed occupancies. Where the storage tank area is located in a building of two or more occupancies and the quantity of liquid exceeds the maximum allowable quantity for one control area, the use shall be completely separated from adjacent fire areas in accordance with the requirements of Section 508.4.

415.9.1.1.1 Height exception. Where storage tanks are located within a building no more than one story above grade plane, the height limitation of Section 504 shall not apply for Group H.

415.9.1.2 Tank protection. Storage tanks shall be noncombustible and protected from physical damage. Fire barriers or horizontal assemblies or both around the storage tanks shall be permitted as the method of protection from physical damage.

415.9.1.2.1 Tanks. Storage tanks shall be approved tanks conforming to the requirements of the *New York City Fire Code*.

415.9.1.3 Suppression. Group H shall be equipped throughout with an approved automatic sprinkler system, installed in accordance with Section 903.

415.9.1.4 Leakage containment. A liquid-tight containment area compatible with the stored liquid shall be provided. The method of spill control, drainage control and secondary containment shall be in accordance with the *New York City Fire Code*.

Exception: Rooms where only double-wall storage tanks conforming to Section 415.9.1.3 are used to store Class I, II and IIIA flammable and combustible liquids shall not be required to have a leakage containment area.

415.9.1.5 Leakage alarm. An approved automatic alarm shall be provided to indicate a leak in a storage tank and room. The alarm shall sound an audible signal, 15 dBA above the ambient sound level, at every point of entry into the room in which the leaking storage tank is located. An approved sign shall be posted on every entry door to the tank storage room indicating the potential hazard of the interior room environment, or the sign shall state: WARNING, WHEN ALARM SOUNDS, THE ENVIRONMENT WITHIN THE ROOM MAY BE HAZARDOUS. The leakage alarm shall also be supervised in accordance with Chapter 9 to transmit a trouble signal.

415.9.1.6 Tank vent. Storage tank vents for Class I, II or IIIA liquids shall terminate to the outdoor air in accordance with the *New York City Fire Code*.

415.9.1.7 Room ventilation. Storage tank areas storing Class I, II or IIIA liquids shall be provided with mechanical ventilation. The mechanical ventilation system shall be in accordance with the *New York City Mechanical Code* and the *New York City Fire Code*.

415.9.1.8 Explosion venting. Where Class I liquids are being stored, explosion venting shall be provided in accordance with the *New York City Fire Code*.

415.9.1.9 Tank openings other than vents. Tank openings other than vents from tanks inside buildings shall be designed to ensure that liquids or vapor concentrations are not released inside the building.

415.9.2 Liquefied petroleum gas-distribution facilities. The construction and installation of liquefied petroleum gas facilities shall be in accordance with the requirements of this code, the *New York City Fire Code*, the *New York City Mechanical Code*, the *New York City Fuel Gas Code* and NFPA 58.

415.9.3 Dry cleaning plants. The construction and installation of dry cleaning plants shall be in accordance with the requirements of this code, the *New York City Mechanical Code*, the *New York City Plumbing Code* and NFPA 32. Dry cleaning solvents and systems shall be classified in accordance with the *New York City Fire Code*.

415.10 Groups H-3 and H-4. Groups H-3 and H-4 shall be constructed in accordance with the applicable provisions of this code and the *New York City Fire Code*.

415.10.1 Flammable and combustible liquids. The storage, handling, processing and transporting of flammable and combustible liquids in Group H-3 occupancies shall be in accordance with Section 415.9.1.

415.10.2 Gas rooms. Where gas rooms are provided, such rooms shall be separated from other areas by not less than 1-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.

415.10.3 Floors in storage rooms. Floors in storage areas for corrosive liquids and highly toxic or toxic materials shall be of liquid-tight, noncombustible construction.

415.10.4 Separation-highly toxic solids and liquids. Highly toxic solids and liquids not stored in approved hazardous materials storage cabinets shall be isolated from other hazardous materials storage by not less than 1-hour fire-resistance rating.

415.11 Group H-5. In addition to the requirements set forth elsewhere in this code, Group H-5 shall comply with the provisions of the *New York City Fire Code* and Section 415.9 of this code, including, but not limited to, automatic sprinkler protection for workstations, gas cabinets, exhausted enclosures, pass-throughs in exit access corridors and exhaust ducts.

415.11.1 Fabrication areas. Fabrication areas shall comply with Sections 415.11.1.1 through 415.11.1.8.

415.11.1.1 Hazardous materials. Hazardous materials and hazardous production materials (HPM) shall comply with Sections 415.11.1.1.1 through 415.11.1.1.2.

415.11.1.1.1 Aggregate quantities. The aggregate quantities of hazardous materials stored and used in a single fabrication area shall not exceed the quantities set forth in Table 415.11.1.1.1.

Exception: The quantity limitations for any hazard category in Table 415.11.1.1.1 shall not apply where the fabrication area contains quantities of hazardous materials not exceeding the maximum allowable quantities per control area established by Tables 307.1(1) and 307.1(2).

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**TABLE 415.11.1.1.1
QUANTITY LIMITS FOR HAZARDOUS MATERIALS IN A SINGLE FABRICATION AREA IN GROUP H-5^a**

HAZARD CATEGORY		SOLIDS (pounds per square foot)	LIQUIDS (gallons per square foot)	GAS (cubic feet @ NTP/square foot)
PHYSICAL-HAZARD MATERIALS				
Combustible dust		Note b	Not Applicable	Not Applicable
Combustible fiber	Loose Baled	Note b Notes b, c	Not Applicable	Not Applicable
Combustible liquid	II IIIA IIIB	Not Applicable	0.01 0.02 Not Limited	Not Applicable
Combination Class	I, II and IIIA		0.04	
Cryogenic gas	Flammable Oxidizing	Not Applicable	Not Applicable	Note d 1.25
Explosives		Note b	Note b	Note b
Flammable gas	Gaseous Liquefied	Not Applicable	Not Applicable	Note d Note d
Flammable liquid	IA IB IC	Not Applicable	0.0025 0.025 0.025	Not Applicable
Combination Class	IA, IB and IC		0.025	
Combination Class	I, II and IIIA		0.04	
Flammable solid			0.001	
Organic peroxide	Unclassified detonable Class I Class II Class III Class IV Class V	Note b Note b 0.025 0.1 Not Limited Not limited	Not Applicable	Not Applicable
Oxidizing gas	Gaseous Liquefied	Not Applicable	Not Applicable	1.25 1.25 1.25
Combination of gaseous and liquefied				
Oxidizer	Class 4 Class 3 Class 2 Class 1	Note b 0.003 0.003 0.003	Note b 0.03 0.03 0.03	Not Applicable
Combination	Class 1, 2, 3	0.003	0.03	
Pyrophoric material		0.01	0.00125	Notes d and e
Unstable reactive	Class 4 Class 3 Class 2 Class 1	Note b 0.025 0.1 Not Limited	Note b 0.0025 0.01 Not Limited	Note b Note b Note b Not Limited
Water reactive	Class 3 Class 2 Class 1	Note b 0.25 Not Limited	0.00125 0.025 Not Limited	Not Applicable
HEALTH-HAZARD MATERIALS				

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Corrosives	Not Limited	Not Limited	Not Limited
Highly toxic	Not Limited	Not Limited	Note d
Toxics	Not Limited	Not Limited	Note d

For SI: 1 pound per square foot = 4.882 kg/m², 1 gallon per square foot = 40.7 L/m²,
1 cubic foot @ NTP/square foot = 0.305 m³ @ NTP/m² 1 cubic foot = 0.02832 m³.

- a. Hazardous materials within piping shall not be included in the calculated quantities.
- b. Quantity of hazardous materials in a single fabrication shall not exceed the maximum allowable quantities per control area in Tables 307.1(1) and 307.1(2).
- c. Densely packed baled cotton that complies with the packing requirements of ISO 8115 shall not be included in this material class.
- d. The aggregate quantity of flammable, pyrophoric, toxic and highly toxic gases shall not exceed 9,000 cubic feet at NTP.
- e. The aggregate quantity of pyrophoric gases in the building shall not exceed the amounts set forth in Table 415.6.2.

415.11.1.1.2 Hazardous production materials. The maximum quantities of hazardous production materials (HPM) stored in a single fabrication area shall not exceed the maximum allowable quantities per control area established by Tables 307.1(1) and 307.1(2).

415.11.1.2 Separation. Fabrication areas, whose sizes are limited by the quantity of hazardous materials allowed by Table 415.11.1.1.1, shall be separated from each other, from corridors and from other parts of the building by not less than 1-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.

Exceptions:

1. Doors within such fire barrier walls, including doors to corridors, shall be only self-closing fire door assemblies having a fire protection rating of not less than $\frac{3}{4}$ hour.
2. Windows between fabrication areas and exit access corridors are permitted to be fixed glazing listed and labeled for a fire protection rating of not less than $\frac{3}{4}$ hour in accordance with Section 716.

415.11.1.3 Location of occupied levels. Occupied levels of fabrication areas shall be located at or above the first story above grade plane.

415.11.1.4 Floors. Except for surfacing, floors within fabrication areas shall be of noncombustible construction.

Openings through floors of fabrication areas are permitted to be unprotected where the interconnected levels are used solely for mechanical equipment directly related to such fabrication areas (see also Section 415.11.1.5).

Floors forming a part of an occupancy separation shall be liquid tight.

415.11.1.5 Shafts and openings through floors. Elevator hoistways, vent shafts and other openings through floors shall be enclosed where required by Sections 712 and 713. Mechanical, duct and piping penetrations within a fabrication area shall not extend through more than two floors. The annular space around penetrations for cables, cable trays, tubing, piping, conduit or ducts shall be sealed at the floor level to restrict the movement of air. The fabrication area, including the areas through which the ductwork and piping extend, shall be considered a single conditioned environment.

415.11.1.6 Ventilation. Mechanical exhaust ventilation at the rate of not less than 1 cubic foot per minute per square foot ($0.0051 \text{ m}^3/(\text{s} \cdot \text{m}^2)$) of floor area shall be provided throughout portions of the fabrication area where HPM are used or stored. The exhaust air duct system of one fabrication area shall not connect to another duct system outside that fabrication area within the building.

A ventilation system shall be provided to capture and exhaust fumes and vapors at workstations.

Two or more operations at a workstation shall not be connected to the same exhaust system where either one or the combination of the substances removed could constitute a fire, explosion or hazardous chemical reaction within the exhaust duct system.

Exhaust ducts penetrating fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711 shall be contained in a shaft of equivalent fire-resistance-rated construction. Exhaust ducts shall not penetrate fire walls.

Fire dampers shall not be installed in exhaust ducts.

415.11.1.7 Transporting hazardous production materials to fabrication areas. HPM shall be transported to fabrication areas through enclosed piping or tubing systems that comply with Section 415.11.6, through service

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corridors complying with Section 415.11.3, or in exit access corridors as permitted in the exception to Section 415.11.2. The handling or transporting of HPM within service corridors shall comply with the *New York City Fire Code*.

415.11.1.8 Electrical. Electrical equipment and devices within the fabrication area shall comply with the *New York City Electrical Code*. The requirements for hazardous locations need not be applied where the average air change is at least four times that set forth in Section 415.11.1.6 and where the number of air changes at any location is not less than three times that required by Section 415.11.1.6. The use of recirculated air shall be permitted.

415.11.1.8.1 Workstations. Workstations shall not be energized without adequate exhaust ventilation. See Section 415.11.1.6 for workstation exhaust ventilation requirements.

415.11.2 Corridors. Corridors shall comply with Chapter 10 and shall be separated from fabrication areas as specified in Section 415.11.1.2. Corridors shall not contain HPM and shall not be used for transporting such materials except through closed piping systems as provided in Section 415.11.6.

Exception: Where existing fabrication areas are altered or modified, HPM is allowed to be transported in existing corridors, subject to the following conditions:

1. Nonproduction HPM is allowed to be transported in corridors if utilized for maintenance, lab work and testing.
2. Corridors. Corridors adjacent to the fabrication area where the alteration work is to be done shall comply with Section 1020 for a length determined as follows:
 - 2.1. The length of the common wall of the corridor and the fabrication area; and
 - 2.2. For the distance along the corridor to the point of entry of HPM into the corridor serving that fabrication area.
3. Emergency alarm system. There shall be an emergency telephone system, a local manual alarm station or other approved alarm-initiating device within corridors at not more than 150-foot (45 720 mm) intervals and at each exit and doorway. The signal shall be relayed to an approved central, proprietary or remote station service or the emergency control station and shall also initiate a locale audible alarm.
4. Pass-throughs. Self-closing doors having a fire-protection rating of not less than 1 hour shall separate pass-throughs from existing corridors. Pass-throughs shall be constructed as required for the corridors, and protected by an approved automatic fire-extinguishing system.

415.11.3 Service corridors. Service corridors within a Group H-5 occupancy shall comply with Sections 415.11.3.1 through 415.11.3.4.

415.11.3.1 Use conditions. Service corridors shall be separated from exit access corridors as required by Section 415.11.1.2. Service corridors shall not be used as a required exit access corridor.

415.11.3.2 Mechanical ventilation. Service corridors shall be mechanically ventilated as required by Section 415.11.1.6 or at not less than six air changes per hour.

415.11.3.3 Means of egress. The distance of travel from any point in a service corridor to an exit, exit access corridor or door into a fabrication area shall be not greater than 75 feet (22 860 mm). Dead ends shall be not greater than 4 feet (1219 mm) in length. There shall be not less than two exits, and not more than one-half of the required means of egress shall require travel into a fabrication area. Doors from service corridors shall swing in the direction of egress travel and shall be self-closing.

415.11.3.4 Minimum width. The clear width of a service corridor shall be not less than 5 feet (1524 mm), or 33 inches (838 mm) wider than the widest cart or truck used in the service corridor, whichever is greater.

415.11.3.5 Emergency alarm system. Emergency alarm systems shall be provided in accordance with this section and Sections 415.5.1 and 415.5.2. The maximum allowable quantity per control area provisions shall not apply to emergency alarm systems required for HPM.

415.11.3.5.1 Service corridors. An emergency alarm system shall be provided in service corridors, with no fewer than one alarm device in each service corridor.

415.11.3.5.2 Exit access corridors and exit enclosures. Emergency alarms for exit access corridors and exit enclosures shall comply with Section 415.5.2.

415.11.3.5.3 Liquid storage rooms, HPM rooms and gas rooms. Emergency alarms for liquid storage rooms, HPM rooms and gas rooms shall comply with Section 415.5.1.

415.11.3.5.4 Alarm-initiating devices. An approved emergency telephone system, local alarm manual pull stations, or other approved alarm-initiating devices are allowed to be used as emergency alarm-initiating devices.

415.11.3.5.5 Alarm signals. Activation of the emergency alarm system shall sound a local alarm and transmit a signal to the emergency control station.

415.11.4 Storage of hazardous production materials. Storage of HPM in fabrication areas shall be within approved or listed storage cabinets or gas cabinets or within a workstation. The storage of HPM in quantities greater than those listed in the *New York City Fire Code* shall be in liquid storage rooms, HPM rooms or gas rooms as appropriate for the materials stored. The storage of other hazardous materials shall be in accordance with other applicable provisions of this code and the *New York City Fire Code*.

415.11.5 HPM rooms, gas rooms, liquid storage room construction. HPM rooms, gas rooms and liquid shall be constructed in accordance with Sections 415.11.5.1 through 415.11.5.9.

415.11.5.1 HPM rooms and gas rooms. HPM rooms and gas rooms shall be separated from other areas by fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both. The fire-resistance rating shall be not less than 2 hours where the area is 300 square feet (27.9 m²) or more and not less than 1 hour where the area is less than 300 square feet (27.9 m²).

415.11.5.2 Liquid storage rooms. Liquid storage rooms shall be constructed in accordance with the following requirements:

1. Rooms greater than 500 square feet (46.5 m²) in area, shall have no fewer than one exterior door approved for Fire Department access.
2. Rooms shall be separated from other areas by fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both. The fire-resistance rating shall be not less than 1 hour for rooms up to 150 square feet (13.9 m²) in area and not less than 2 hours where the room is more than 150 square feet (13.9 m²) in area.
3. Shelving, racks and wainscoting in such areas shall be of noncombustible construction or wood of not less than 1-inch (25 mm) nominal thickness or fire-retardant-treated wood complying with Section 2303.2.
4. Rooms used for the storage of Class I flammable liquids shall not be located in a basement.

415.11.5.3 Floors. Except for surfacing, floors of HPM rooms and liquid storage rooms shall be of noncombustible liquid-tight construction. Raised grating over floors shall be of noncombustible materials.

415.11.5.4 Location. Where HPM rooms, liquid storage rooms and gas rooms are provided, they shall have no fewer than one exterior wall and such wall shall be not less than 30 feet (9144 mm) from tax-lot lines, including tax-lot lines adjacent to public ways.

415.11.5.5 Explosion control. Explosion control shall be provided where required by Section 414.5.1.

415.11.5.6 Exits. Where two exits are required from HPM rooms, liquid storage rooms and gas rooms, one shall be directly to the outside of the building.

415.11.5.7 Doors. Doors in a fire barrier wall, including doors to corridors, shall be self-closing fire door assemblies having a fire-protection rating of not less than ³/₄ hour.

415.11.5.8 Ventilation. Mechanical exhaust ventilation shall be provided in liquid storage rooms, HPM rooms and gas rooms at the rate of not less than 1 cubic foot per minute per square foot (0.044 L/s/m²) of floor area or six air changes per hour.

Exhaust ventilation for gas rooms shall be designed to operate at a negative pressure in relation to the surrounding areas and direct the exhaust ventilation to an exhaust system.

415.11.5.9 Emergency alarm system. An approved emergency alarm system shall be provided for HPM rooms, liquid storage rooms and gas rooms.

Emergency alarm-initiating devices shall be installed outside of each interior exit door of such rooms.

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Activation of an emergency alarm-initiating device shall sound a local alarm and transmit a signal to the emergency control station.

An approved emergency telephone system, local alarm manual pull stations or other approved alarm-initiating devices are allowed to be used as emergency alarm-initiating devices.

415.11.6 Piping and tubing. Hazardous production materials piping and tubing shall comply with this section and ASME B31.3.

415.11.6.1 HPM having a health-hazard ranking of 3 or 4. Systems supplying HPM liquids or gases having a health-hazard ranking of 3 or 4 shall be welded throughout, except for connections, to the systems that are within a ventilated enclosure if the material is a gas, or an approved method of drainage or containment is provided for the connections if the material is a liquid.

415.11.6.2 Location in service corridors. Hazardous production materials supply piping or tubing in service corridors shall be exposed to view.

415.11.6.3 Excess flow control. Where HPM gases or liquids are carried in pressurized piping above 15 pounds per square inch gauge (psig) (103.4 kPa), excess flow control shall be provided. Where the piping originates from within a liquid storage room, HPM room or gas room, the excess flow control shall be located within the liquid storage room, HPM room or gas room. Where the piping originates from a bulk source, the excess flow control shall be located as close to the bulk source as practical.

415.11.6.4 Installations in corridors and above other occupancies. The installation of HPM piping and tubing within the space defined by the walls of exit access corridors and the floor or roof above, or in concealed spaces above other occupancies, shall be in accordance with Sections 415.11.6.1 through 415.11.6.3 and the following conditions:

1. Automatic sprinklers shall be installed within the space unless the space is less than 6 inches (152 mm) in the least dimension.
2. Ventilation not less than six air changes per hour shall be provided. The space shall not be used to convey air from any other area.
3. Where the piping or tubing is used to transport HPM liquids, a receptor shall be installed below such piping or tubing. The receptor shall be designed to collect any discharge or leakage and drain it to an approved location. The 1-hour enclosure shall not be used as part of the receptor.
4. HPM supply piping and tubing and nonmetallic waste lines shall be separated from the corridor and from occupancies other than Group H-5 by fire barriers or by an approved method or assembly that has a fire-resistance rating of not less than 1 hour. Access openings into the enclosure shall be protected by approved fire-resistance-rated assemblies.
5. Readily accessible manual or automatic remotely activated fail-safe emergency shutoff valves shall be installed on piping and tubing other than waste lines at the following locations:
 - 5.1. At branch connections into the fabrication area.
 - 5.2. At entries into exit access corridors.

Exception: Transverse crossings of the corridors by supply piping that is enclosed within a ferrous pipe or tube for the width of the corridor need not comply with Items 1 through 5.

415.11.6.5 Identification. Piping, tubing and HPM waste lines shall be identified in accordance with ANSI A13.1 to indicate the material being transported.

415.11.7 Continuous gas detection systems. A continuous gas detection system shall be provided for HPM gases where the physiological warning threshold level of the gas is at a higher level than the accepted permissible exposure limit (PEL) for the gas and for flammable gases in accordance with Sections 415.11.7.1 and 415.11.7.2.

415.11.7.1 Where required. A continuous gas detection system shall be provided in the areas identified in Sections 415.11.7.1.1 through 415.11.7.1.4.

415.11.7.1.1 Fabrication areas. A continuous gas detection system shall be provided in fabrication areas where gas is used in the fabrication area.

415.11.7.1.2 HPM rooms. A continuous gas detection system shall be provided in HPM rooms where gas is used in the room.

415.11.7.1.3 Gas cabinets, exhausted enclosures and gas rooms. A continuous gas detection system shall be provided in gas cabinets and exhausted enclosures. A continuous gas detection system shall be provided in gas rooms where gases are not located in gas cabinets or exhausted enclosures.

415.11.7.1.4 Corridors. Where gases are transported in piping placed within the space defined by the walls of a corridor and the floor or roof above the corridor, a continuous gas detection system shall be provided where piping is located and in the corridor.

Exception: A continuous gas detection system is not required for occasional transverse crossings of the corridors by supply piping that is enclosed in a ferrous pipe or tube for the width of the corridor.

415.11.7.2 Gas detection system operation. The continuous gas detection system shall be capable of monitoring the room, area or equipment in which the gas is located at or below all the following gas concentrations:

1. Immediately dangerous to life and health (IDLH) values where the monitoring point is within an exhausted enclosure, ventilated enclosure or gas cabinet.
2. Permissible exposure limit (PEL) levels where the monitoring point is in an area outside an exhausted enclosure, ventilated enclosure or gas cabinet.
3. For flammable gases, the monitoring detection threshold level shall be vapor concentrations in excess of 25 percent of the lower flammable limit (LFL) where the monitoring is within or outside an exhausted enclosure, ventilated enclosure or gas cabinet.
4. Except as noted in this section, monitoring for highly toxic and toxic gases shall also comply with the *New York City Fire Code*.

415.11.7.2.1 Alarms. The gas detection system shall initiate a local alarm and transmit a signal to the emergency control station when a short-term hazard condition is detected. The alarm shall be both visual and audible and shall provide warning both inside and outside the area where the gas is detected. The audible alarm shall be distinct from all other alarms.

415.11.7.2.2 Shutoff of gas supply. The gas detection system shall automatically close the shutoff valve at the source on gas supply piping and tubing related to the system being monitored for which gas is detected when a short-term hazard condition is detected. Automatic closure of shutoff valves shall comply with the following:

1. Where the gas detection sampling point initiating the gas detection system alarm is within a gas cabinet or exhausted enclosure, the shutoff valve in the gas cabinet or exhausted enclosure for the specific gas detected shall automatically close.
2. Where the gas detection sampling point initiating the gas detection system alarm is within a room and compressed gas containers are not in gas cabinets or an exhausted enclosure, the shutoff valves on all gas lines for the specific gas detected shall automatically close.
3. Where the gas detection sampling point initiating the gas detection system alarm is within a piping distribution manifold enclosure, the shutoff valve supplying the manifold for the compressed gas container of the specific gas detected shall automatically close.

Exception: Where the gas detection sampling point initiating the gas detection system alarm is at the use location or within a gas valve enclosure of a branch line downstream of a piping distribution manifold, the shutoff valve for the branch line located in the piping distribution manifold enclosure shall automatically close.

415.11.8 Manual fire alarm system. An approved manual fire alarm system shall be provided throughout buildings containing Group H-5. Activation of the alarm system shall initiate a local alarm and transmit a signal to the emergency control station. The fire alarm system shall be designed and installed in accordance with Section 907.

415.11.9 Emergency control station. An emergency control station shall be provided in accordance with Sections 415.11.9.1 through 415.11.9.3.

415.11.9.1 Location. The emergency control station shall be located on the premises at an approved location outside the fabrication area.

415.11.9.2 Staffing. Trained personnel shall continuously staff the emergency control station.

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415.11.9.3 Signals. The emergency control station shall monitor signals from emergency equipment and alarm and detection systems. Such emergency equipment and alarm and detection systems shall include the following systems, whether required by this code or the *New York City Construction Codes* or voluntarily installed:

1. Automatic sprinkler system alarm and monitoring systems.
2. Manual fire alarm systems.
3. Emergency alarm systems.
4. Continuous gas detection systems.
5. Smoke detection systems.
6. Emergency power system.
7. Automatic detection and alarm systems for pyrophoric liquids and Class 3 water-reactive liquids required in the *New York City Fire Code*.
8. Exhaust ventilation flow alarm devices for pyrophoric liquids and Class 3 water-reactive liquids cabinet exhaust ventilation systems required in the *New York City Fire Code*.

415.11.10 Emergency power system. An emergency power system shall be provided in Group H-5 occupancies in accordance with Section 2702. The emergency power system shall supply power automatically to the electrical systems specified in Section 415.11.10.1 when the normal electrical supply system is interrupted.

415.11.10.1 Required electrical systems. Emergency power shall be provided for electrically operated equipment and connected control circuits for the following systems:

1. HPM exhaust ventilation systems.
2. HPM gas cabinet ventilation systems.
3. HPM exhausted enclosure ventilation systems.
4. HPM gas room ventilation systems.
5. HPM gas detection systems.
6. Emergency alarm systems.
7. Manual and automatic fire alarm systems.
8. Automatic sprinkler system monitoring and alarm systems.
9. Automatic alarm and detection systems for pyrophoric liquids and Class 3 water-reactive liquids required in the *New York City Fire Code*.
10. Flow alarm switches for pyrophoric liquids and Class 3 water-reactive liquids cabinet exhaust ventilation systems required in the *New York City Fire Code*.
11. Electrically operated systems required elsewhere in this code applicable to the use, storage or handling of HPM.

415.11.10.2 Exhaust ventilation systems. Exhaust ventilation systems are allowed to be designed to operate at not less than one-half the normal fan speed on the emergency power system where it is demonstrated that the level of exhaust will maintain a safe atmosphere.

415.11.11 Automatic sprinkler system protection in exhaust ducts for HPM. An approved automatic fire sprinkler system shall be provided in exhaust ducts conveying vapors, fumes, mists or dusts generated from HPM in accordance with the *New York City Mechanical Code* and Sections 415.11.11.1 through 415.11.11.3 of this code.

415.11.11.1 Metallic and noncombustible nonmetallic exhaust ducts. An approved automatic sprinkler system shall be provided in metallic and noncombustible nonmetallic exhaust ducts where all of the following conditions apply:

1. Where the largest cross-sectional diameter is equal to or greater than 10 inches (254 mm).
2. The ducts are within the building.
3. The ducts are conveying flammable gases, vapors or fumes.

415.11.11.2 Combustible nonmetallic exhaust ducts. Automatic fire sprinkler system protection shall be provided in combustible nonmetallic exhaust ducts where the largest cross-sectional diameter of the duct is equal to or greater than 10 inches (254 mm).

Exception: Ducts need not be provided with automatic sprinkler protection as follows:

1. Ducts listed or approved for applications without automatic sprinkler system protection.
2. Ducts not more than 12 feet (3658 mm) in length installed below ceiling level.

415.11.11.3 Automatic sprinkler locations. Sprinkler systems shall be installed at 12-foot (3658 mm) intervals in horizontal ducts and at changes in direction. In vertical ducts, sprinklers shall be installed at the top and at alternate floor levels.

SECTION BC 416 APPLICATION OF FLAMMABLE FINISHES

416.1 General. The provisions of this section shall apply to the construction, installation and use of buildings and structures, or parts thereof, for the application of flammable paints, varnishes and lacquers or other flammable materials or mixtures or compounds used for painting, varnishing, staining or similar purposes. Such construction and equipment shall comply with the *New York City Fire Code*.

416.2 Spray rooms. Spray rooms shall be enclosed with not less than 2-hour fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both. Floors shall be water-proofed and drained in an approved manner.

416.2.1 Surfaces. The interior surfaces of spray rooms shall be smooth and shall be so constructed to permit the free passage of exhaust air from all parts of the interior and to facilitate washing and cleaning, and shall be so designed to confine residues within the room. Aluminum shall not be used.

416.2.2 Ventilation. Mechanical ventilation and interlocks with the spraying operation shall be in accordance with the *New York City Mechanical Code*.

416.3 Spraying spaces. Spraying spaces shall be ventilated with an exhaust system capable of at least six air changes per hour to prevent the accumulation of flammable mist or vapors in accordance with the *New York City Mechanical Code*. Where such spaces are not separately enclosed, noncombustible spray curtains shall be provided to restrict the spread of flammable vapors.

416.3.1 Surfaces. The interior surfaces of spraying spaces shall be smooth and continuous without edges; shall be so constructed to permit the free passage of exhaust air from all parts of the interior and to facilitate washing and cleaning; and shall be so designed to confine residues within the spraying space. Aluminum shall not be used.

416.4 Spray booths. Spray booths shall be designed, constructed and operated in accordance with the *New York City Fire Code*.

416.5 Fire protection. An automatic sprinkler system or fire-extinguishing system shall be provided in all spray, dip and immersing spaces and storage rooms and shall be installed in accordance with Chapter 9.

SECTION BC 417 DRYING ROOMS

417.1 General. A drying room or dry kiln installed within a building shall be constructed entirely of approved non-combustible materials or assemblies of such materials regulated by the approved rules or as required in the general and specific sections of this chapter for special occupancies, where applicable to the general requirements of Chapter 28 of this code, and in accordance with the *New York City Fire Code*.

417.2 Piping clearance. Overhead heating pipes shall have a clearance of not less than 2 inches (51 mm) from combustible contents in the dryer.

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417.3 Insulation. Where the operating temperature of the dryer is 175°F (79°C) or more, metal enclosures shall be insulated from adjacent combustible materials by not less than 12 inches (305 mm) of airspace, or the metal walls shall be lined with 1/4-inch (6.4 mm) insulating mill board or other approved equivalent insulation.

417.4 Fire protection. Drying rooms designed for high-hazard materials and processes, including special occupancies as provided for in this chapter, shall be protected by an approved automatic fire-extinguishing system complying with the provisions of Chapter 9.

SECTION BC 418 ORGANIC COATINGS

418.1 Building features. Manufacturing of organic coatings shall be done only in buildings that do not have pits or basements and shall comply with the *New York City Fire Code*. Such buildings shall be protected by an approved automatic sprinkler system in accordance with Chapter 9 of this code.

418.2 Location. Organic coating manufacturing operations and operations incidental to or connected therewith shall not be located in buildings having other occupancies.

418.3 Process mills. Mills operating with close clearances and that process flammable and heat-sensitive materials, such as nitrocellulose, shall be classified as Group H-1 occupancy.

418.4 Tank storage. Storage areas for flammable and combustible liquid tanks inside of structures shall be located at or above grade and shall be separated from the processing area by not less than 2-hour fire-resistance-rated fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both.

418.5 Nitrocellulose storage. Nitrocellulose storage shall be located on a detached pad or in a separate structure or a room enclosed with no less than 2-hour fire-resistance-rated fire barriers.

418.6 Finished products. Storage rooms for finished products that are flammable or combustible liquids shall be separated from the processing area by fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both having a fire-resistance rating of at least 2 hours, and openings in the walls shall be protected with approved opening protectives.

SECTION BC 419 RESERVED

SECTION BC 420 GROUPS I-1, R-1, R-2, R-3

420.1 General. Occupancies in Groups I-1, R-1, R-2 and R-3 shall comply with the provisions of Sections 420.1 through 420.6 and other applicable provisions of this code.

420.2 Separation walls. Walls separating dwelling units in the same building, walls separating sleeping units in the same building and walls separating dwelling or sleeping units from other occupancies contiguous to them in the same building shall be constructed as fire barriers constructed in accordance with Section 707 and shall be at least 1-hour fire-resistance-rated.

420.3 Horizontal separation. Floor assemblies separating dwelling units in the same buildings, floor assemblies separating sleeping units in the same building and floor assemblies separating dwelling or sleeping units from other occupancies contiguous to them in the same building shall be constructed as horizontal assemblies in accordance with Section 711.

420.4 Reserved.

420.5 Automatic sprinkler system. Group R occupancies shall be equipped throughout with an automatic sprinkler system in accordance with Section 903.2.8. Group I-1 occupancies shall be equipped throughout with an automatic

sprinkler system in accordance with Section 903.2.6. Quick response or residential automatic sprinklers shall be installed in accordance with Section 903.3.2.

420.6 Fire alarm systems and smoke alarms. Fire alarm systems and smoke alarms shall be provided in Group I-1, R-1, and R-2 occupancies in accordance with Sections 907.2.6, 907.2.8 and 907.2.9, respectively. Single or multiple station smoke alarms shall be provided in Groups I-1, R-2, and R-3 in accordance with Section 907.2.11.

SECTION BC 421 HYDROGEN FUEL GAS ROOMS

421.1 General. Where required by the *New York City Fire Code*, hydrogen fuel gas rooms shall be designed and constructed in accordance with Sections 421.1 through 421.7 of this code.

421.1.1 Scope. Hydrogen fuel gas rooms shall be separately ventilated, fully enclosed, and shall contain all equipment and piping for the immediate on-premises use of hydrogen in indoor fuel cells or other energy production process, and incidental indoor storage of gaseous hydrogen in accordance with the *New York City Fire Code*. A hydrogen fuel gas room shall not be used to house production or dispensing of hydrogen motor fuel.

421.1.2 Storage. Hydrogen fuel gas rooms shall not be used for storage of gaseous hydrogen in quantities exceeding the maximum allowable quantity per control area set forth in Table 307.1(1). Where approved by the Fire Department, such hydrogen fuel gas rooms permitted to store greater quantities shall be classified as Occupancy Group H.

421.1.3 Definition.‡ The following term is defined in Chapter 2:

HYDROGEN FUEL GAS ROOM.

421.2 Location. Hydrogen fuel gas rooms shall not be located below grade.

421.3 Design and construction. Hydrogen fuel gas rooms not classified as Occupancy Group H shall be separated from other areas of the building in accordance with Section 509.1. Such separation shall have a fire-resistance rating of not less than 2 hours for all occupancy groups.

421.3.1 Pressure control. Hydrogen fuel gas rooms shall be provided with a ventilation system designed to maintain the room at a negative pressure in relation to surrounding area.

421.3.2 Windows. Operable windows in interior walls shall not be permitted. Fixed windows shall be permitted where in accordance with Section 716.

421.4 Exhaust ventilation. Hydrogen fuel gas rooms shall be separately ventilated and provided with mechanical exhaust ventilation in accordance with the applicable provisions of NFPA 2.

421.5 Gas detection system. Hydrogen fuel gas rooms shall be provided with a gas detection system that complies with Sections 421.5.1, 421.5.2, and 918.

421.5.1 System activation. Activation of a gas detection alarm shall initiate the following notification and action:

1. Initiation of distinct audible and visible alarm signals both inside and outside of the hydrogen fuel gas room.
2. Automatic activation of the mechanical exhaust ventilation system.

421.5.2 Failure of the gas detection system. Failure of the gas detection system shall automatically discontinue hydrogen generation, activate the mechanical exhaust ventilation system and transmit a supervisory signal to the control panel and a central station.

421.6 Explosion control. Explosion control shall be provided where required by Section 414.5.1.

421.7 Emergency power. Mechanical exhaust ventilation and gas detection systems shall be provided with an emergency power system in accordance with Section 2702.

SECTION BC 422 AMBULATORY CARE FACILITIES

422.1 General. Ambulatory care facilities shall comply with the provisions of Sections 422.1 through 422.5 and other applicable provisions of this code.

422.2 Separation. Ambulatory care facilities, where there is the potential for four or more care recipients who will be incapable of self-preservation at any time, whether rendered incapable by staff or staff accepted responsibility for a care recipient already incapable, shall be separated from adjacent spaces, corridors or tenants with a fire partition installed in accordance with Section 708.

422.3 Smoke compartments. Where the aggregate area of one or more ambulatory care facilities is greater than 10,000 square feet (929 m²) on one story, the story shall be provided with a smoke barrier to subdivide the story into no fewer than two smoke compartments. The area of any one such smoke compartment shall be not greater than 22,500 square feet (2092 m²). The distance of travel from any point in a smoke compartment to a smoke barrier door shall be not greater than 200 feet (60 960 mm). The smoke barrier shall be installed in accordance with Section 709 with the exception that smoke barriers shall be continuous from outside wall to an outside wall, a floor to a floor, or from a smoke barrier to a smoke barrier or a combination thereof.

422.3.1 Means of egress. Where ambulatory care facilities require smoke compartmentation in accordance with Section 422.3 of this code, the fire protection and emergency preparedness plans shall comply with the applicable provisions of the *New York City Fire Code* and shall be in accordance with Section 1001.4 of this code.

422.3.2 Refuge area. Not less than 30 net square feet (2.8 m²) for each nonambulatory care recipient shall be provided within the aggregate area of corridors, care recipient rooms, treatment rooms, lounge or dining areas and other low-hazard areas within each smoke compartment. Each occupant of an ambulatory care facility shall be provided with access to a refuge area without passing through or utilizing adjacent tenant spaces.

422.3.3 Independent egress. A means of egress shall be provided from each smoke compartment created by smoke barriers without having to return through the smoke compartment from which means of egress originated.

422.4 Automatic sprinkler systems. Automatic sprinkler systems shall be provided for ambulatory care facilities in accordance with Section 903.2.2.

422.5 Fire alarm systems. A fire alarm system shall be provided for ambulatory care facilities in accordance with Section 907.2.2.

SECTION BC 423 STORM SHELTERS

423.1 General. In addition to other applicable requirements in this code, storm shelters shall be constructed in accordance with ICC 500.

423.1.1 Scope. This section applies to the construction of storm shelters constructed as separate detached buildings or constructed as safe rooms within buildings for the purpose of providing safe refuge from storms that produce high winds, such as tornados and hurricanes. Such structures shall be designated to be hurricane shelters, tornado shelters, or combined hurricane and tornado shelters.

423.2 Definitions specific to this section. The following terms shall, for the purposes of this section, have the meanings shown herein:

STORM SHELTER. A building, structure or portion(s) thereof, constructed in accordance with ICC 500 and designated for use during a severe wind storm event, such as a hurricane or tornado.

Community storm shelter. A storm shelter not defined as a “Residential storm shelter.”

Residential storm shelter. A storm shelter serving occupants of dwelling units and having an occupant load not exceeding 16 persons.

SECTION BC 424 CHILDREN'S PLAY STRUCTURES

424.1 Children's play structures. Children's play structures installed inside all occupancies covered by this code that exceed 10 feet (3048 mm) in height and 150 square feet (14 m²) in area shall comply with Sections 424.2 through 424.5.

424.2 Materials. Children's play structures shall be constructed of noncombustible materials or of combustible materials that comply with the following:

1. Fire-retardant-treated wood complying with Section 2303.2 of this code.
2. Light-transmitting plastics complying with Section 2606 of this code.
3. Foam plastics (including the pipe foam used in soft-contained play equipment structures) having a maximum heat-release rate not greater than 100 kilowatts when tested in accordance with UL 1975 or when tested in accordance with NFPA 289, using the 20 kW ignition source.
4. Aluminum composite material (ACM) meeting the requirements of Class A interior finish in accordance with Chapter 8 of this code when tested as an assembly in the maximum thickness intended for use.
5. Textiles and films complying with the fire propagation performance criteria contained in Test Method 1 or Test Method 2, as appropriate, of NFPA 701.
6. Plastic materials used to construct rigid components of soft-contained play equipment structures (such as tubes, windows, panels, junction boxes, pipes, slides and decks) exhibiting a peak rate of heat release not exceeding 400 kW/m² when tested in accordance with ASTM E 1354 at an incident heat flux of 50 kW/m² in the horizontal orientation at a thickness of 6 mm.
7. Ball pool balls, used in soft-contained play equipment structures, having a maximum heat-release rate not greater than 100 kilowatts when tested in accordance with UL 1975 or when tested in accordance with NFPA 289, using the 20 kW ignition source. The minimum specimen test size shall be 36 inches by 36 inches (914 mm by 914 mm) by an average of 21 inches (533 mm) deep, and the balls shall be held in a box constructed of galvanized steel poultry netting wire mesh.
8. Foam plastics shall be covered by a fabric, coating or film meeting the fire propagation performance criteria contained in Test Method 1 or Test Method 2, as appropriate, of NFPA 701.
9. The floor covering placed under the children's play structure shall exhibit a Class I interior floor finish classification, as described in Section 804, when tested in accordance with NFPA 253.

424.3 Fire protection. Children's play structures shall be provided with the same level of approved fire protection system required for other structures in the same occupancy.

424.4 Separation. Children's play structures shall have a horizontal separation from building walls, partitions and from elements of the means of egress of not less than 5 feet (1524 mm). Children's playground structures shall have a horizontal separation from other children's play structures of not less than 20 feet (6090 mm).

424.5 Area limits. Children's play structures shall be not greater than 300 square feet (28 m²) in area.

SECTION BC 425 HYPERBARIC FACILITIES

425.1 Hyperbaric facilities. Hyperbaric facilities shall meet the requirements contained in Chapter 14 of NFPA 99.

SECTION BC 426 COMBUSTIBLE DUSTS, GRAIN PROCESSING AND STORAGE

426.1 Combustible dusts, grain processing and storage. The provisions of Sections 426.1.1 through 426.1.7 shall apply to buildings in which materials that produce combustible dusts are stored or handled. Buildings that store or

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handle combustible dusts shall comply with the applicable provisions of NFPA 61, NFPA 69, NFPA 85, NFPA 120, NFPA 484, NFPA 654, NFPA 655 and NFPA 664 and the *New York City Fire Code*.

426.1.1 Type of construction and height exceptions. Buildings shall be constructed in compliance with the height, number of stories and area limitations specified in Sections 504 and 506; except that where erected of Type I or II construction, the heights and areas of grain elevators and similar structures shall be unlimited, and where of Type IV construction, the maximum building height shall be 65 feet (19 812 mm) and except further that, in isolated areas, the maximum building height of Type IV structures shall be increased to 85 feet (25 908 mm).

426.1.2 Grinding rooms. Every room or space occupied for grinding or other operations that produce combustible dusts in such a manner that the room or space is classified as a Group H-2 occupancy shall be enclosed with fire barriers constructed in accordance with Section 707 or horizontal assemblies constructed in accordance with Section 711, or both. The fire-resistance rating of the enclosure shall be not less than 2 hours where the area is not more than 3,000 square feet (279 m²), and not less than 4 hours where the area is greater than 3,000 square feet (279 m²).

426.1.3 Conveyors. Conveyors, chutes, piping and similar equipment passing through the enclosures of rooms or spaces shall be constructed dirt tight and vapor tight, and be of approved noncombustible materials complying with Chapter 30.

426.1.4 Explosion control. Explosion control shall be provided as specified in the *New York City Fire Code*, or spaces shall be equipped with the equivalent mechanical ventilation complying with the *New York City Mechanical Code*.

426.1.5 Grain elevators. Grain elevators, malt houses and buildings for similar occupancies shall not be located within 30 feet (9144 mm) of interior lot lines or structures on the same lot, except where erected along a railroad right-of-way.

426.1.6 Coal pockets. Coal pockets located less than 30 feet (9144 mm) from interior lot lines or from structures on the same lot shall be constructed of not less than Type IB construction. Where more than 30 feet (9144 mm) from interior lot lines, or where erected along a railroad right-of-way, the minimum type of construction of such structures not more than 65 feet (19 812 mm) in building height shall be Type IV.

426.1.7 Tire rebuilding. Buffing operations shall be located in a room separated from the remainder of the building housing the tire rebuilding or tire recapping operation by a 1-hour fire barrier.

Exception: Buffing operations are not required to be separated where all of the following conditions are met:

1. Buffing operations are equipped with an approved continuous automatic water-spray system directed at the point of cutting action;
2. Buffing machines are connected to particle-collecting systems providing a minimum air movement of 1,500 cubic feet per minute (cfm) (0.71 m³/s) in volume and 4,500 feet per minute (fpm) (23 m/s) in-line velocity; and
3. The collecting system shall discharge the rubber particles to an approved outdoor noncombustible or fire-resistant container, which is emptied at frequent intervals to prevent overflow.

SECTION BC 427 NONPRODUCTION CHEMICAL LABORATORIES

427.1 General. The provisions of this section shall govern the design, construction and chemical quantity limitations of buildings and portions thereof occupied or designed to be occupied as a nonproduction laboratory. Nonproduction laboratories shall be classified into occupancy groups in accordance with Section 427.5. The provisions of this section shall not apply to production laboratories classified in Occupancy Group F or H.

427.2 Standards. Except as otherwise provided in this section, nonproduction laboratories shall meet the design, construction and chemical quantity limitation requirements for laboratory unit fire hazard Class D, as set forth in NFPA 45 as modified by the *New York City Fire Code*, and the *New York City Mechanical Code*. The storage arrangement, handling and use of chemicals within nonproduction laboratories and accessory storage rooms, and the operation

and maintenance of nonproduction laboratories shall comply with the requirements of the *New York City Fire Code*, and the rules of the Fire Commissioner.

427.3 Prohibitions. It shall be unlawful in any nonproduction laboratory to:

1. Store or use any explosive.
2. Store or use any unclassified detonable organic peroxide, detonable pyrophoric material, detonable unstable (reactive) material or detonable water-reactive material.
3. Store or use any Class 4 unstable (reactive) material.
4. Store or use any Class 4 oxidizing material.
5. Store or use any flammable gas below grade.

427.4 Definitions specific to this section. The following terms shall, for the purposes of this section, have the meanings shown herein:

LABORATORY BUILDING. A structure consisting wholly or principally of one or more laboratory units.

LABORATORY CHEMICAL. A material with a health, flammability, or instability hazard rating of 2, 3, or 4 as defined in NFPA 704.

LABORATORY, NONPRODUCTION. A building or portion thereof wherein chemicals or gases are used or synthesized on a nonproduction basis for testing, research, experimental, instructional or educational purposes.

LABORATORY UNIT. An enclosed space of a minimum 1-hour fire-rated construction, designed or used as a nonproduction laboratory. Laboratory units may include one or more separate laboratory work areas, and accessory storage rooms or spaces within or contiguous with the laboratory unit, such as offices and lavatories.

STORAGE CABINET. A cabinet designed and constructed in accordance with UL 1275, and used for the storage of not more than 60 gallons (227 L) of flammable and combustible liquids.

STORAGE ROOM. A room where laboratory chemicals or gases are stored and not otherwise used or synthesized.

427.5 Classification. Buildings or portions thereof occupied as a nonproduction laboratory may be classified as a Group B occupancy provided they comply with the provisions of Section 427. Nonproduction laboratories not in compliance with the provisions of Section 427.7 for laboratory chemical quantity limitations shall be classified as Group H occupancy.

427.5.1 Accessory use nonproduction laboratories. Accessory use nonproduction laboratories occupying an area not more than 10 percent of the area of the story in which such laboratories are located and not exceeding the tabular values in Sections 504 and 506, for the allowable height or area for such use shall be classified according to the main occupancy. Such accessory nonproduction laboratories shall comply with the provisions of Section 427.

427.6 Fire protection. Fire protection systems in nonproduction laboratories shall be provided in accordance with Section 427.6.1 through 427.6.3.

427.6.1 Sprinkler system. Laboratory units shall be provided throughout with an automatic sprinkler system in accordance with Section 903.3.1.1. The entire building shall be provided throughout with an automatic sprinkler system when the aggregate floor area of all laboratory units within any building exceeds 20,000 square feet (1858 m²).

427.6.2 Standpipe and hose system. In all nonproduction laboratory buildings that are two or more stories above or below the grade level (level of exit discharge), standpipes shall be installed in accordance with Section 905.2.

427.6.3 Fire alarm system. A manual fire alarm system shall be installed in accordance with Section 907.2.2, 907.2.3, or 907.2.6, as applicable.

427.7 Quantity limitations. Quantity limitations in nonproduction laboratories shall be provided in accordance with Section 427.7.1 through 427.7.10.

SPECIAL DETAILED REQUIREMENTS BASED ON USE AND OCCUPANCY

427.7.1 Flammable and combustible liquids. The density and total quantity of flammable and combustible liquids allowed within a laboratory unit, outside of storage rooms, shall comply with Table 9.1.1(A) of NFPA 45, as modified by the *New York City Fire Code*, for laboratory unit fire hazard Class D.

Exceptions: The following exceptions apply to other than educational or instructional laboratories:

1. The density of flammable and combustible liquids allowed within a laboratory unit may be increased to those set forth in Table 9.1.1(A) of NFPA 45 as modified by the *New York City Fire Code*, for laboratory unit fire hazard Class B provided the total quantity of flammable and combustible liquid, including any in storage cabinets or safety cans, does not exceed 25 gallons (95 L).
2. The density of flammable and combustible liquids allowed within a laboratory unit may be increased to those set forth in Table 9.1.1(A) of NFPA 45 as modified by the *New York City Fire Code*, for laboratory unit fire hazard Class B provided the total quantity of flammable and combustible liquid, including any in storage cabinets or safety cans, does not exceed 30 gallons (114 L) and the walls, floors and ceilings of the laboratory unit are separated from all adjoining areas by 2-hour fire-rated construction.
3. Based on a density of 1 gallon per 100 square feet (3.785 L per 9.3 m²) of area, the quantity of flammable and combustible liquid allowed within a laboratory unit, excluding quantities in storage cabinets or safety cans, may be increased to 100 gallons (379 L), and the total quantities of flammable and combustible liquids, including quantities in storage cabinets or safety cans, may be increased to 200 gallons (757 L) provided the walls, floors and ceilings of the laboratory unit are separated from all adjoining areas by 2-hour fire-rated construction.

427.7.2 Flammable solids. The total quantity of flammable solids allowed within a laboratory unit, excluding any quantities in a storage room, shall not exceed 10 pounds (4.5 kg).

Exception: The total quantity of flammable solids allowed within a laboratory unit that is provided with walls, floors and ceilings that separate the laboratory unit from all adjoining areas by 2-hour fire-rated construction shall not exceed 15 pounds (6.8 kg).

††† **427.7.3 Oxidizers and organic peroxides.** The total quantity of solid and liquid oxidizers and organic peroxides combined allowed within a laboratory unit, excluding any quantities in a storage room, shall not exceed 40 pounds (18 kg), provided not more than 2 pounds (0.908 kg) of which are Class 3 oxidizers and 1 pound (0.454 kg) of which is Class I organic peroxides.

Exception: The total quantity of solid and liquid oxidizers and organic peroxides combined allowed within a laboratory unit that is provided with walls, floors and ceilings that separate the laboratory unit from all adjoining areas by 2-hour fire-rated construction shall not exceed 50 pounds (23 kg), provided not more than 2 pounds (0.908 kg) of which is Class 3 oxidizers and 1 pound (0.454 kg) of which are Class I organic peroxides.

427.7.4 Unstable (reactive). The total quantity of unstable (reactive) material allowed within a laboratory unit, excluding any quantities in a storage room, shall not exceed 6 pounds (2.7 kg), provided not more than 1 pound (0.454 kg) of which is Class 3 unstable (reactive).

Exception: The total quantity of unstable (reactive) material allowed within a laboratory unit that is provided with walls, floors and ceilings that separate the laboratory unit from all adjoining areas by 2-hour fire-rated construction shall not exceed 12 pounds (5.4 kg), provided not more than 1 pound (0.454 kg) of which is Class 3 unstable (reactive).

427.7.5 Water reactive material. The total quantity of water reactive material allowed within a laboratory unit, excluding any quantities in a storage room, shall not exceed 2.5 pounds (1.1 kg).

Exception: The total quantity of water-reactive material allowed within a laboratory unit that is provided with walls, floors and ceilings that separate the laboratory unit from all adjoining areas by 2-hour fire-rated construction shall not exceed 5 pounds (2.3 kg).

††† **427.7.6 Pyrophoric material.** The total quantity of solid or liquid pyrophoric material allowed within a laboratory unit, excluding any quantities in a storage room, shall not exceed 0.5 pounds (0.227 kg).

Exception: The total quantity of pyrophoric material allowed within a laboratory unit that is provided with walls, floors and ceilings that separate the laboratory unit from all adjoining areas by 2-hour fire-rated construction shall not exceed 1 pound (0.454 kg).

427.7.7 Highly toxic material. The total quantity of solid or liquid highly toxic material allowed within a laboratory unit, excluding any quantities in a storage room, shall not exceed 5 pounds (2.27 kg).

427.7.8 Toxic material. The total quantity of solid or liquid toxic material allowed within a laboratory unit, excluding any quantities in a storage room, shall not exceed 250 pounds (113.4 kg).

427.7.9 Corrosive material. The total quantity of solid or liquid corrosive material allowed within a laboratory unit, excluding any quantity in a storage room, shall not exceed 250 gallons (946 L).

427.7.10 Hazardous gases. The storage, handling and use of hazardous gases, including highly toxic, toxic, corrosive, and asphyxiant gases, within a laboratory unit shall be in accordance the *New York City Fire Code*, including quantities permitted and oxygen sensors.

427.8 Storage room classification. Storage rooms for laboratory chemicals accessory to a laboratory unit shall be classified as Occupancy Group S-1 provided they comply with the provisions of Section 427.9. Storage rooms not in compliance with the provisions of Section 427.9 for laboratory chemical quantity limitations shall be classified as a Group H occupancy.

427.9 Storage rooms. In addition to the quantities allowed within a laboratory unit by Section 427.7, laboratory chemicals that are accessory to a laboratory unit may be stored, whether outside of or entirely within a laboratory unit, in dedicated storage rooms complying with all of the following:

1. Storage room capacity shall be limited to a maximum of 300 gallons (1136 L) of laboratory chemicals, not to exceed 5 gallons per square foot (0.125 L/m²) of floor area.
2. Flammable gas storage rooms shall be limited to 2,500 standard cubic feet (71 standard cubic m) of flammable gas.
3. Storage rooms shall be enclosed by a minimum of 2-hour rated fire barriers.
4. Storage rooms shall be equipped with a continuously operated ventilation system that provides at least 6 changes of air per hour and vents to the outdoors.
5. Each entrance to the storage room shall be provided with a minimum 1½-hour rated self-closing fire door.
6. Each entrance to the storage room shall be provided with a sill at the doorway, except that no sill shall be required in storage rooms containing only flammable gases.
7. Storage rooms shall be equipped with an automatic sprinkler system providing at least one sprinkler for each 90 square feet (8.4 m²) or portion thereof.
8. Laboratory chemicals that are incompatible with each other shall not be stored in the same storage room unless in compliance with the *New York City Fire Code* for separation of incompatible materials.
9. Storage rooms shall not open directly to an exit or any enclosed exit access component.
10. The floor of any storage room storing flammable gases shall be located at or above grade.

427.10 Ventilation. Nonproduction laboratories shall be ventilated in accordance with Section 408 of the *New York City Mechanical Code*.

SECTION BC 428 USES AND OCCUPANCIES INVOLVING RADIOACTIVE MATERIALS AND RADIATION-PRODUCING EQUIPMENT

428.1 Scope. This section shall apply to the construction, alteration, and use of buildings or spaces for radioactive materials and radiation-producing equipment.

428.2 City, state and federal regulations. In addition to the requirements of this section, occupancies involving radioactive materials and radiation-producing equipment shall also comply with applicable requirements of the *New York City Health Code*, and applicable state and federal regulations.

428.3 Laboratories. All laboratories utilizing radioactive materials or radiation-producing equipment required to register under the requirements of the *New York City Health Code* shall comply with the requirements of Sections 428.3.1 through 428.3.7.

SPECIAL DETAILED REQUIREMENTS BASED ON USE AND OCCUPANCY

428.3.1 Construction. All buildings in which such laboratories occur shall be of Type I or II construction.

428.3.2 Floors. All floors shall comply with the fire resistance requirements for the class of construction, and provide the degree of radioactive resistance required by applicable city, state, and federal regulations. A finished material shall be applied to provide a continuous nonporous surface, which may be readily removed.

428.3.3 Interior finish. All insulation of acoustical treatments and interior partitions shall be of noncombustible material. Walls and ceilings shall have nonporous finishes of Class A rating.

428.3.4 Sprinkler protection. Automatic sprinkler protection complying with the construction provisions of Chapter 9 of the *New York City Building Code* shall be provided, and such protection shall be designed for the type of combustible materials wherever such material is used, and for the radioactive material that may be expected to melt, vaporize, or oxidize under fire conditions. Laboratory equipment susceptible to damage from water or other materials used in the sprinkler system may be shielded by hoods except when the equipment provides a source of combustion. Where sprinkler protection uses water, or small water-spray installations are used to fight small isolated fires, floors shall be provided with drainage so that water may be carried to retention tanks for later disposal as required by the *New York City Health Code* when contamination of the water is to be anticipated.

428.3.5 Electrical controls. Electrical controls and equipment shall be installed in accordance with the requirements of the *New York City Electrical Code*.

428.3.6 Ventilation. Exhaust air from areas in which radioactive materials are used or stored shall be exhausted to the outdoors in such manner as not to create a health hazard, and shall not be recirculated to other areas of the building. Air pressure in rooms in which radioactive materials are used or stored shall be maintained below the air pressure of adjoining rooms, so that there is no flow of radioactive gases or dusts into adjoining rooms.

428.3.6.1 Ducts. Ducts shall be of sheet steel of not less than No. 16 manufacturers' standard gauge or of other equivalent noncombustible material having a melting point above 1800°F (982°C). Exhaust ducts within the building, on the discharge side of the fan, shall be welded airtight. Exhaust ducts within the building, on the suction side of the fan, shall have laps in the direction of airflow with smoke-tight joints, and shall be subjected to a smoke test in accordance with the requirements for chimneys in the *New York City Mechanical Code*. Access hatches with tight-closing covers shall be provided for cleaning and for fire-fighting in the exhaust system ducts.

428.3.6.2 Fume hoods. Fume hoods shall be exhausted to the outdoors. Controls for hood fans shall be interlocked so that contaminated air cannot be drawn into any space from a hood where the exhaust fan is not in operation.

428.3.6.3 Fans. Fan equipment other than the impeller and impeller housing shall be located outside the exhaust stream.

428.3.6.4 Exhaust. When the degree of contamination of the exhaust stream exceeds the concentration limits permitted by the health code, the duct system shall be equipped with devices to decontaminate the air to a safe level before discharging to the outdoor air.

428.3.7 Plumbing. Drainage lines from sinks used for radioactive wastes shall be without traps, and shall lead to retention tanks when required by the provisions of the *New York City Health Code*.

428.4 Radiation machines. Radiation machines or particle accelerators, linear accelerators, cyclotrons, synchrotrons‡, betatrons, or bevatrons shall be located only in buildings of Type I or II construction; however, this requirement shall not apply to conventional medical, dental, research, or industrial x-ray machines of less than one million-volt capacity.

428.5 Storage. Radioactive materials shall be stored in sealed containers. When required by the commissioner to avoid too concentrated an exposure within any one space, radioactive materials shall be stored in vaults designed in accordance with the radiation shielding or other requirements for the materials to be stored, and with the requirements of Sections 428.5.1 through 428.5.3.

428.5.1 Fire protection. When any materials are subject to melting, vaporization, or oxidation under fire conditions, the storage vaults shall be constructed of walls having a fire-resistance rating of at least 4 hours, and the vaults shall be equipped with automatic sprinklers complying with the construction requirements of Chapter 9 and shall be vented through devices to decontaminate the air to a safe level.

SPECIAL DETAILED REQUIREMENTS BASED ON USE AND OCCUPANCY

428.5.2 Doors. Doors opening into storage vaults shall meet shielding requirements and have a fire-protection rating of not less than 3 hours.

428.5.3 Bins, shelving, partitions and pallets. All bins, shelving, partitions, and pallets in storage vaults shall be of noncombustible materials. Other methods of storage permitted by the health department or applicable state and federal regulations, such as storage under water, may be used.